Thoughts And Rambles On Probability And Statistics

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To my future self. I hope you can look back on this and think this was all worth it.

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Preface

These notes started as my attempt to make sense of a first-year graduate course in probability, and they have slowly grown into something that looks like a book. The title is honest. These are thoughts, and a lot of them are rambles, and I've kept it that way on purpose because I think the rambling is where most of the understanding actually happens.

The book is roughly split into two halves. Chapters 1 through 5 build the machinery: measure theory and the probability triple, random variables and their distributions, expectation and the convergence theorems, the standard inequalities, and the characteristic function. Chapter 6 talks about what it means for a sequence of random variables to converge, and then the last two chapters use everything to get at the two results that make statistics possible: the law of large numbers in Chapter 7 and the central limit theorem in Chapter 8.

My bias is that of a physicist. I care more about why a result is true and where it comes from than about stating it in its most general form, so you will see a lot more derivations and pictures than theorem-proof blocks. When a proof is genuinely illuminating, I will go through it carefully. When it's just bookkeeping, I will say so and point you somewhere else. The worked problems are in grey boxes, and I'd recommend actually trying them before reading the solutions. The problems at the end of each chapter are left for you.

A lot of the structure and many of the problems are borrowed from the books that taught me: Rosenthal's wonderful little book [5], Gut's graduate course [4], and Wasserman's *All of Statistics* [6]. Durrett [2], Billingsley [1] and Feller [3] are where you should go if you want to see everything done properly.

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1

Just A Bit Of Measure Theory

"Erm, what the σ"

– Unknown

I've been postponing this for a bit now, and a big part of it is because I've been finding it difficult to motivate myself to think about measure theory. If I can't even convince myself to do measure theory, then it would be impossible to convince someone else to do it. However, it is a responsibility that anyone who does probability must take on. As such, we will grit our teeth, hold our tongue, and push forward! My promise is that every definition in this chapter will show up because something breaks without it, and I'll try to show you the thing that breaks first.

1.1 The Need For Measure Theory

It's not immediately obvious to someone who is a novice in the arts of probability why we need measure theory. There are a lot of examples one could give, but I think Jeff Rosenthal's exposition of this [5] is, quite frankly, goated. I will work through it here. In basic probability and statistics, we are familiar with the basic random variables like the Poisson distribution or the uniform distribution, and we say that the distinction between them is that one is a discrete R.V. while the other is continuous. If we wanted to calculate the expectation value or variance or whatever, then we know exactly how to calculate these quantities: sums for the discrete ones, integrals for the continuous ones. What if we wanted to say: let Z be a random variable such that with probability $\frac{1}{2}$, it is a Poisson R.V. with $\lambda = 5$, and

with probability $\frac{1}{2}$, it is a uniform R.V. on $[0,1]$. It is *not* immediately obvious how to work with a random variable like this. It has no density (it puts probability $\frac{1}{2}e^{-5}$ on the single point 0, and densities can't do that), and it has no probability mass function either (it puts probability $\frac{1}{2}$ on an interval made of uncountably many points, each of which individually gets probability zero). So is $\mathbb{E}[Z]$ a sum or an integral? It's also not immediately obvious how to build a theory of probability that describes an object like this. That's the first reason for measure theory: we want one language in which "probability" means the same thing for every random variable, and "expectation" is one operation instead of a sum in some cases and an integral in others.

1.1.1 Probabilities On The Uniform

When we think of probability on an intuitive level, we think about it as a fraction of {objects of interest} divided by {all possible relevant objects}. For example, if Alice, Bob, and Charlie are your potential lab partners, what's the probability that Alice is your partner? Well, in our heads we can just say $\frac{1}{3}$, since there is only 1 Alice out of 3 possible partners. In the language of set theory, we can say that a probability is the ratio of the cardinalities of sets. But what happens if you want to find the probability that I will be in the office between 12pm and 3pm knowing that I have to be in the office at some time between 12pm and 5pm? Time is a continuous parameter in this case, and the usual notion of probabilities being ratios of cardinalities doesn't make sense anymore. Both sets have uncountably many points, so their "ratio of counts" is ∞/∞. What does it mean to count every possible time between 12pm and 3pm relative to 12pm and 5pm? We need a notion of length to compare these events, and the object that we will use to describe lengths (and areas, volumes, and probabilities) of sets is called a **measure**. If we can have a general notion of size, then we can generalize the probability of an event as just the size of the set of desired outcomes compared with the size of the whole sample space, 3/5 in this case.

1.1.2 Not Every Set Can Be Measured

Here is the second reason, and it's the one that I wish someone had shown me first, because it explains the weirdest part of the theory. Once we have length, it's tempting to want to assign a length to *every* subset of $[0,1]$. What properties should "length" have? At the very least, we'd want

(1) The length of an interval $[a,b]$ is $b-a$.
(2) Shifting a set doesn't change its length (if we wrap around the circle, so that $[0,1)$ is the circle and shifting by r means adding r mod 1).
(3) The length of a countable disjoint union is the sum of the lengths.

These all sound completely innocent. As it turns out, no function on all subsets of $[0,1)$ can satisfy them simultaneously, and the counterexample is due to Vitali. Call two numbers in $[0,1)$ equivalent if their difference is rational. This chops $[0,1)$ up into equivalence classes (each class looks like $x+\mathbb{Q}$, wrapped around). Now build a set V by picking exactly one number from each class (this uses the axiom of choice). Then look at the shifted copies $V_q = V + q \pmod 1$ for each rational $q \in [0,1)$. Two things are true:

- The copies are disjoint. If x were in both V_q and $V_{q'}$, then $x-q$ and $x-q'$ would be two different elements of V that differ by a rational, but V only has one element from each class.
- The copies cover everything. Any $x \in [0,1)$ is equivalent to the representative $v \in V$ from its class, so $x = v + q$ for some rational q, which means $x \in V_q$.

So $[0,1)$ is a countable disjoint union of the shifted copies V_q. If V had a length ℓ, then every copy has length ℓ by property 2, and by property 3

$$1 = \lambda([0,1)) = \sum_{q \in \mathbb{Q} \cap [0,1)} \lambda(V_q) = \sum_q \ell = \ell + \ell + \ell + \cdots$$

If $\ell = 0$ the right hand side is 0, and if $\ell > 0$ it's ∞. Neither is 1! So the set V cannot be given a length at all. This isn't a failure of cleverness. It's a theorem: there are sets that are too pathological to measure. The resolution is to give up on measuring *every* set, and only measure the "reasonable" ones. The collection of reasonable sets is what a σ-algebra is, and it's the reason that σ-algebras show up everywhere in probability. In practice, you will never write down a non-measurable set by accident (you need the axiom of choice to build one), so the σ-algebra is more like a legal disclaimer than something you fight with day to day. But it has to be there.

1.2 Sets And De Morgan's Law

The only real thing that we will take from set theory is De Morgan's law. Everything else from set theory is pretty straightforward. I will assume that you know things like complements, intersections, unions, and symmetric differences. Even if you are a bit rusty, I will leave some problems at the end of the chapter for practice. The thing that we want to discuss is the relationship between unions of sets and their complements

$$(A \cup B)^c = A^c \cap B^c \tag{1.1}$$

And the proof is really straightforward. We can show this by the definition of what it means for $x \in A \cup B$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \tag{1.2}$$

Then if we take the complement, what we are really saying is that it's not true that "x is in A or x is in B". The negation of an "or" is an "and": x is not in A, and x is not in B either

$$(A \cup B)^c = \{x \mid x \notin A \text{ and } x \notin B\} \tag{1.3}$$

But if it's not in A, then it must be in A^c, similarly with B. So

$$(A \cup B)^c = \{x \mid x \in A^c \text{ and } x \in B^c\} = A^c \cap B^c \tag{1.4}$$

The reason that it is not an "or" relation is that being outside of just one of the sets isn't enough. A point in A^c could still be in B, and then it's in $A \cup B$. The relation for intersections of sets takes a similar form

$$(A \cap B)^c = A^c \cup B^c \tag{1.5}$$

and both are drawn in figure 1.1. The proof will be left as an exercise for the reader, but

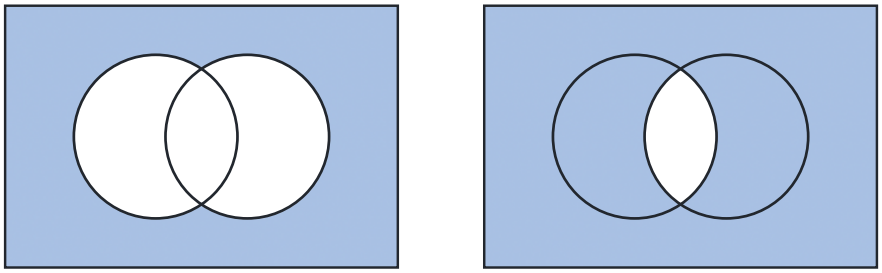


Figure 1.1: De Morgan's laws, with A the left circle, B the right circle and Ω the rectangle. Left: the shaded region is $(A \cup B)^c = A^c \cap B^c$, everything outside of both circles. Right: the shaded region is $(A \cap B)^c = A^c \cup B^c$, everything except the overlap.

it's literally identical to the proof of the first De Morgan's law. But this is only really good for 2 sets. In probability theory in general, we will deal with countable unions or countable intersections, in which case we need a way to generalize the results that we have found. For finitely many sets, we can use induction. We already have the base case proven. Let's assume that

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c$$

to be true. Then $\left(\bigcup_{i=1}^{n+1} A_i\right)^c$ is just the same as

$$\left(\bigcup_{i=1}^{n} A_i \cup A_{n+1}\right)^c$$

which is a union of 2 sets, so from the basic De Morgan's law and the induction hypothesis, we have that

$$\left(\bigcup_{i=1}^{n} A_i\right)^c \cap A_{n+1}^c = \bigcap_{i=1}^{n} A_i^c \cap A_{n+1}^c = \bigcap_{i=1}^{n+1} A_i^c$$

Thus we have shown the generalized De Morgan's laws

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \qquad \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c \tag{1.6}$$

Induction only ever gets us to finite n, though. For infinitely many sets we need a different argument, so let's do a problem.

Problem 1.

Let $\{A_i : i \in I\}$ be a collection of events where I is an arbitrary index set (finite, countable, or even uncountable). Show that

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c \quad \text{and} \quad \left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$$

Solution: The trick is to not use induction at all and to go back to the definitions, which never cared how many sets there were. A point x is in $\bigcup_{i\in I} A_i$ if there *exists* some i with $x \in A_i$. So x is in the complement if there does *not* exist such an i, which is the same thing as saying that *for every* i, $x \notin A_i$, i.e. $x \in A_i^c$ for all i. That's exactly the statement $x \in \bigcap_{i\in I} A_i^c$. For the second one, apply the first law to the sets A_i^c and take complements of both sides:

$$\left(\bigcup_{i\in I} A_i^c\right)^c = \bigcap_{i\in I} A_i \quad \Longrightarrow \quad \bigcup_{i\in I} A_i^c = \left(\bigcap_{i\in I} A_i\right)^c$$

De Morgan's law is really just the logical fact that "not (there exists)" is "for all (not)".

There's one more tool from set theory that is going to make our lives a lot easier, and that's the **indicator function**

$$\mathbb{I}_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

It turns set operations into arithmetic: $\mathbb{I}_{A\cap B} = \mathbb{I}_A\mathbb{I}_B$, $\mathbb{I}_{A^c} = 1 - \mathbb{I}_A$, and for disjoint sets $\mathbb{I}_{A\cup B} = \mathbb{I}_A + \mathbb{I}_B$. De Morgan's law, for example, is $1 - \mathbb{I}_A\mathbb{I}_B = 1 - (1 - \mathbb{I}_{A^c})(1 - \mathbb{I}_{B^c})$ expanded out. Indicators will turn out to be the simplest random variables there are, and their expectations are probabilities.

1.3 σ-Algebras

In the first section, we discussed why we need measure theory for a theory of probability, and the Vitali set told us that we can't measure everything. In this section, we will discuss which sets in a space we are going to be allowed to measure, and then from there we can talk about the axioms of probability. A collection $\mathcal{F}$ of subsets of Ω is called a **σ-algebra** if it satisfies the following 3 properties:

(1) The entire set Ω is contained in $\mathcal{F}$.
(2) It is closed under complements: if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
(3) It is closed under countable unions: if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

From these, the empty set is in there too ($\emptyset = \Omega^c$), and so are countable intersections, by De Morgan's law, so saying both parts is a bit redundant. (Some books list them anyway.) Elements of $\mathcal{F}$ are called **measurable sets**, or in probability, **events**.

Problem 2.

Show that closure under complements and countable unions leads to closure under countable intersections.

Solution: Suppose that $A_1, A_2, \ldots \in \mathcal{F}$. By property 2, each complement A_i^c is also in $\mathcal{F}$. By property 3, the union of the complements is in $\mathcal{F}$

$$\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{F}$$

and then by property 2 again, its complement is in $\mathcal{F}$. Using De Morgan,

$$\left(\bigcup_{i=1}^{\infty} A_i^c\right)^c = \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$$

which completes the proof.

When I was learning this for the first time, I had no idea what to make of σ-algebras and their significance. It also didn't really help that later on, we didn't really use them explicitly. So let me try to give you the two ways I think about them now.

The first way is the one from the Vitali set: a σ-algebra is the collection of sets that we are allowed to assign probabilities to. The three properties are exactly what we need to make probability work. We need complements because if we know the probability that it rains, we had better know the probability that it doesn't rain. And we need countable unions (not just finite ones) because probability is all about limits. Consider flipping a coin forever. The event "we eventually see heads" is $\bigcup_{n=1}^{\infty}\{\text{the } n\text{-th flip is heads}\}$, a countable union of simple events, and we certainly want to know its probability. With only finite unions, it wouldn't even be an event! On the other hand, we can't allow arbitrary (uncountable) unions, because every subset of $[0,1]$ is a union of single points, and single points are as reasonable as sets get, so we'd be forced to measure the Vitali set. Countable is the sweet spot.

The second way, which becomes really important when we talk about conditional expectation and stochastic processes, is that a σ-algebra represents *information*. It's the collection of yes/no questions whose answers you know. If you've been told everything, then you can answer whether $\omega \in A$ for every A, and your σ-algebra is everything. If you've been told nothing, the only questions you can answer are "did something happen?" (yes, Ω) and "did nothing happen?" (no, $\emptyset$), so your σ-algebra is $\{\emptyset, \Omega\}$. Let us give some examples of what are or are not σ-algebras.

Problem 3.

Let $\Omega = \{a, b, c\}$ be a sample space. Define the following collections of subsets of Ω:

$$\mathcal{A}_1 = \{\emptyset, \{a\}, \{b, c\}, \Omega\}, \qquad \mathcal{A}_2 = \{\emptyset, \{a, b\}, \{c\}, \Omega\}$$

a). Verify whether $\mathcal{A}_1$ and $\mathcal{A}_2$ are σ-algebras.

b). Find the smallest σ-algebra $\mathcal{A}$ that contains both $\mathcal{A}_1$ and $\mathcal{A}_2$.
Solution: Pretty simple, we just have to check whether or not they satisfy the above properties. It is clear that $\mathcal{A}_1$ is a σ-algebra because it contains Ω, and it's closed under complements (the complement of $\{a\}$ is $\{b, c\}$ and vice versa, and $\emptyset$ and Ω swap). Unions are also fine: any union of these sets is again one of them. The exact same argument is made for $\mathcal{A}_2$. Thus both $\mathcal{A}_1$ and $\mathcal{A}_2$ are σ-algebras. Pretty straightforward. In the information language, $\mathcal{A}_1$ is what you know if someone tells you whether the outcome was a, and $\mathcal{A}_2$ is what you know if someone tells you whether it was c.
Now, the question of interest is: what is the smallest σ-algebra that contains both? Just taking $\mathcal{A}_1 \cup \mathcal{A}_2$ is not enough, because it's not closed under unions: $\{a\} \cup \{c\} = \{a, c\}$ isn't in there. We are forced to keep adding unions, intersections and complements until nothing new appears. For instance $\{b\} = \{b, c\} \cap \{a, b\}$ has to be in there. Once we have all three singletons, every subset is a union of them, so the smallest σ-algebra is everything

$$\boxed{\mathcal{A} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \Omega\}}$$

This also makes sense with information: if you know whether it was a and whether it was c, then you know exactly which outcome happened.

We learned a few things from this exercise: unions of σ-algebras are not necessarily themselves σ-algebras, and (implicitly) that the power set (all possible subsets) of Ω is a σ-algebra. But that comes as no surprise, because it's every possible subset. It *has* to contain complements and unions. For a finite (or countable) Ω, we'll basically always use the power set. It's only for uncountable spaces like $\mathbb{R}$ that we have to be choosy.

1.3.1 Generated σ-Algebras

Problem 3 had us find "the smallest σ-algebra containing" some sets. This idea is so important that it gets its own notation. Given any collection $\mathcal{C}$ of subsets of Ω, we define $\sigma(\mathcal{C})$, the **σ-algebra generated by $\mathcal{C}$**, as the smallest σ-algebra containing every set in $\mathcal{C}$. Why should such a thing exist? Here's the slick argument. Take *every* σ-algebra that contains $\mathcal{C}$ (there's at least one, the power set) and intersect all of them

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{F} \ : \ \mathcal{F} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{F}\}$$

An intersection of σ-algebras is a σ-algebra (if A is in all of them, so is A^c, because each one is closed under complements, and the same goes for countable unions). It contains $\mathcal{C}$,

and it's inside every other σ-algebra containing $\mathcal{C}$, so it's the smallest. In practice we never construct it this way; we describe a few sets we care about, and say "and whatever else it takes".

In the information picture, $\sigma(\mathcal{C})$ is everything you can figure out if you're told the answers to the questions in $\mathcal{C}$. For example, flip a coin twice, $\Omega = \{HH, HT, TH, TT\}$. If you only get to see the first flip, the events whose occurrence you can decide are

$$\mathcal{F}_1 = \sigma\left(\{HH, HT\}\right) = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}$$

After the second flip, you know everything, $\mathcal{F}_2 =$ all 16 subsets. The fact that $\mathcal{F}_1 \subseteq \mathcal{F}_2$ is the statement that information only accumulates. A growing sequence of σ-algebras like this is called a **filtration**, and it's the backbone of the theory of martingales and stochastic processes.

1.3.2 Borel Sets

For the real line, the σ-algebra that everyone uses is the one generated by the intervals,

$$\mathcal{B}(\mathbb{R}) = \sigma\left(\{(a, b) : a < b\}\right)$$

called the **Borel σ-algebra**, and its sets are the **Borel sets**. Because we close under countable operations, it contains a lot more than intervals. Single points are Borel, since $\{x\} = \bigcap_n (x - \frac{1}{n}, x + \frac{1}{n})$. Then every countable set is Borel (a countable union of points), so the rationals $\mathbb{Q}$ are Borel, and so are the irrationals (the complement). Closed sets, open sets, the Cantor set that we'll see in a moment, they're all Borel. In fact, it's hard to describe a subset of $\mathbb{R}$ that isn't Borel without invoking the axiom of choice. There's nothing special about open intervals either: you get the exact same σ-algebra from the half lines $(-\infty, x]$, since $(a, b] = (-\infty, b] \setminus (-\infty, a]$ and $(a, b) = \bigcup_n (a, b - \frac{1}{n}]$. That last fact is the reason that the cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$ will completely determine a distribution in chapter 2.

Problem 4.

Let $\Omega = \mathbb{R}$, set $I_k = [k - 1, k)$ for $k \geq 1$, and let

$$\mathcal{A}_n = \sigma\{I_k : k = 1, 2, \ldots, n\}.$$

Show that $\bigcup_{n=1}^{\infty} \mathcal{A}_n$ is not a σ-algebra.
Solution: This problem is not at all obvious, but it's a nontrivial example of the fact that unions of σ-algebras are not necessarily σ-algebras themselves. To give some intuition as to what $\mathcal{A}_n$ looks like, we will write out the first one explicitly

$$\mathcal{A}_1 = \{\emptyset, [0,1), (-\infty, 0) \cup [1, \infty), \mathbb{R}\}$$

For $\mathcal{A}_2$, the generators $[0,1)$ and $[1,2)$ chop the line into three pieces, $[0,1)$, $[1,2)$, and the rest $R_2 = (-\infty, 0) \cup [2, \infty)$. Every set in $\mathcal{A}_2$ is a union of some of these pieces, and every such union is in it, so $\mathcal{A}_2$ has $2^3 = 8$ sets. In general, $\mathcal{A}_n$ consists of all unions of some of the pieces $I_1, \ldots, I_n$ and $R_n = (-\infty, 0) \cup [n, \infty)$, which is 2^{n+1} sets. The $\mathcal{A}_n$ are increasing, $\mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \cdots$.
What about $\bigcup_{n=1}^{\infty} \mathcal{A}_n$? Every single I_k is in it (it's in $\mathcal{A}_k$). If it were a σ-algebra, the countable union

$$\bigcup_{k=1}^{\infty} I_k = [0, \infty)$$

would have to be in it, which means it would have to be in some $\mathcal{A}_n$. But every set in $\mathcal{A}_n$ either contains all of R_n or none of it, and $[0, \infty)$ contains $[n, \infty)$ but not $(-\infty, 0)$. So $[0, \infty)$ is in none of the $\mathcal{A}_n$. Thus, $\bigcup_{n=1}^{\infty} \mathcal{A}_n$ is not a σ-algebra, since it is not closed under countable unions. (It is closed under *finite* unions and complements, so it's what's called an **algebra**. The σ is the countable part, and this is exactly the gap between the two.)

Notice that σ-algebras need not be collections of singleton points like in problem 3. They can, in fact, be built out of intervals, and on $\mathbb{R}$ they usually are. Now that we have a collection of subsets of Ω that are measurable, we have to define a **measure**.

1.4 Measures And Probability

We want a way to generalize the idea of a length or size or volume of a set. Remember that when we think about elementary probabilities, we think of them as the size of some set divided by the size of the sample space. Here we will define what a measure is. Let Ω be a set and $\mathcal{F}$ a σ-algebra of subsets of Ω. A **measure** is a function $\mu : \mathcal{F} \to [0, \infty]$ that satisfies the following properties:

(1) Non-negativity: $\mu(A) \geq 0$ for all $A \in \mathcal{F}$.
(2) Null empty set: $\mu(\emptyset) = 0$.

(3) Countable additivity (or σ-additivity): if $\{A_i\}_{i=1}^{\infty}$ is a countable collection of disjoint sets in $\mathcal{F}$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$), then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

The triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**. My claim is that similar to how a metric is a generalization of distance in a space, a measure is a generalization of the size of sets in a space, and just like with metrics, different measures will give different results. Notice that the whole thing hinges on the third property. It says that size is additive, which is the one thing everybody agrees "size" should mean, and the σ is what lets it survive limits.

1.4.1 What Additivity Buys Us

Everything else that we want from "size" follows from those three properties, and it's worth deriving them carefully once, because we will use them constantly.

Monotonicity. If $A \subseteq B$, then B is the disjoint union of A and $B \setminus A$, so $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$. Bigger sets have bigger measure. If $\mu(A) < \infty$, this also gives $\mu(B \setminus A) = \mu(B) - \mu(A)$.

Inclusion-exclusion. For two sets that overlap, $A \cup B$ is the disjoint union of A and $B \setminus A$, and B is the disjoint union of $B \setminus A$ and $A \cap B$. Putting these together,

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \tag{1.7}$$

when everything is finite. The overlap got counted twice, so we subtract it once.

Countable subadditivity. For sets that aren't disjoint, we can make them disjoint. Let $B_1 = A_1$, $B_2 = A_2 \setminus A_1$, $B_3 = A_3 \setminus (A_1 \cup A_2)$, and so on: B_n is the part of A_n that's new. The B_n are disjoint, $B_n \subseteq A_n$, and they have the same union as the A_n. So

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(B_n) \leq \sum_{n=1}^{\infty} \mu(A_n) \tag{1.8}$$

This is the **union bound**, and it's probably the most used inequality in all of probability. The probability that at least one bad thing happens is at most the sum of the probabilities of each bad thing.

Continuity from below. If $A_1 \subseteq A_2 \subseteq \cdots$ increase to $A = \bigcup_n A_n$, then

$$\mu(A) = \lim_{n \to \infty} \mu(A_n) \tag{1.9}$$

To see this, use the same disjointification: here $B_n = A_n \setminus A_{n-1}$ are the rings, $A_n = B_1 \cup \cdots \cup B_n$, and

$$\mu(A) = \sum_{k=1}^{\infty} \mu(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mu(B_k) = \lim_{n\to\infty} \mu(A_n)$$

An infinite sum is the limit of its partial sums. That's all this is.

Continuity from above. If $A_1 \supseteq A_2 \supseteq \cdots$ decrease to $A = \bigcap_n A_n$ and $\mu(A_1) < \infty$, then

$$\mu(A) = \lim_{n\to\infty} \mu(A_n) \tag{1.10}$$

Take complements inside A_1: the sets $A_1 \setminus A_n$ increase to $A_1 \setminus A$, so by continuity from below $\mu(A_1) - \mu(A_n) \to \mu(A_1) - \mu(A)$, and we can cancel $\mu(A_1)$ because it's finite. The finiteness really is needed. With the counting measure on $\mathbb{N}$ (coming up in a second), the sets $A_n = \{n, n+1, \ldots\}$ decrease to $\emptyset$, but each one has infinitely many elements. For probability measures everything is finite, so we never have to worry about it.

These continuity properties are the reason we insisted on countable additivity. They're what let us take limits inside of probabilities, and that's what we'll be doing for the rest of the book.

1.4.2 Examples Of Measures

Let's look at an example of a measure on a discrete set.

Problem 5.

Let $\mathbb{N}$ be the set of natural numbers with the power set as the σ-algebra. Define the **counting measure** $C(A)$ to be the number of elements in the set A. What is the measure of $\{1, 2, 3\}$ and $\{1, 10, 200\}$? What is $C(\mathbb{N})$? Check that C is a measure.
Solution: This one is really easy. We just have to find the cardinality of each of those sets. We get $C(\{1, 2, 3\}) = 3$, $C(\{1, 10, 200\}) = 3$, and $C(\mathbb{N}) = \infty$ to no surprise. It's a measure because counting is additive: if the A_i are disjoint, every element of the union is in exactly one A_i, so it's counted exactly once on both sides. Notice that the counting measure doesn't care at all *where* the points are, which is exactly what makes it different from length.

Another one that you will see all the time is the **Dirac measure** (or point mass) at a point x: $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise. It's the measure version of the Dirac delta function,

and it's how a measure can put weight on a single point. The counting measure on $\mathbb{N}$ is just $\sum_n \delta_n$, and the Poisson distribution is $\sum_k \frac{e^{-\lambda}\lambda^k}{k!}\delta_k$.

The measure that we really care about is length. On $\Omega = [0,1]$, we want a measure λ, called the **Lebesgue measure**, with $\lambda((a,b)) = b - a$. The trouble is that we've only said what it does to intervals, and intervals aren't a σ-algebra (the union of two disjoint intervals isn't an interval). So how do we extend it to all Borel sets? Here's the idea, due to Lebesgue and Carathéodory. For any set A, cover it by countably many intervals, add up their lengths, and take the best (smallest) such cover

$$\lambda^*(A) = \inf\left\{ \sum_{k=1}^{\infty} (b_k - a_k) \;:\; A \subseteq \bigcup_{k=1}^{\infty} (a_k, b_k) \right\} \tag{1.11}$$

This is called the **outer measure**, and it's defined for every subset, even the Vitali set. The catch is that λ^* is only subadditive, not additive, on all subsets (the Vitali argument shows it can't be). So we keep only the sets on which it behaves. Carathéodory's criterion says a set E is measurable if it splits every other set additively

$$\lambda^*(A) = \lambda^*(A \cap E) + \lambda^*(A \cap E^c) \quad \text{for every } A \subseteq \Omega$$

The **Carathéodory extension theorem** then tells us that the measurable sets form a σ-algebra, that it contains all the Borel sets, and that λ^* restricted to it is a measure that agrees with $b-a$ on intervals. It's also the only measure on the Borel sets that does. I'm not going to go through the proof, since it's a few pages of careful bookkeeping (Rosenthal [5] does it beautifully), but the picture is the right one to keep in mind: the length of a complicated set is the total length of the cheapest pile of intervals that covers it.

Problem 6.

Show that every countable subset of $[0,1]$ has Lebesgue measure 0. Conclude that if you pick a number uniformly at random from $[0,1]$, the probability that it is rational is 0.

Solution: Let $A = \{x_1, x_2, \ldots\}$ be countable and pick any $\epsilon > 0$. Cover the k-th point by an interval of length $\epsilon/2^k$ centered on it. Then A is covered, and the total length of the cover is

$$\sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon$$

So $\lambda^*(A) \leq \epsilon$ for every $\epsilon > 0$, which forces $\lambda(A) = 0$. (Alternatively: each point

has measure $\lambda(\{x\}) = \lim_n \lambda((x - \frac{1}{n}, x + \frac{1}{n})) = 0$ by continuity from above, and then countable additivity adds up countably many zeros.) The rationals in $[0, 1]$ are countable, so the probability is $\boxed{0}$. That's pretty wild: the rationals are dense (there's one in every interval, no matter how small), and yet a random number is "never" rational.

Countable sets have measure zero, but the converse is false: there are uncountable sets with measure zero too, and the most famous one is the **Cantor set**. Start with $C_0 = [0, 1]$. Remove the open middle third to get $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Remove the middle third of each remaining piece to get C_2, and keep going (figure 1.2). The Cantor set is what's left, $C = \bigcap_n C_n$. At step n there are 2^n pieces of length 3^{-n}, so $\lambda(C_n) = (2/3)^n$, and by continuity from above

$$\lambda(C) = \lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0$$

But it's uncountable: a number is in C exactly when it can be written in base 3 using only the digits 0 and 2 (at each step, you choose the left third, digit 0, or the right third, digit 2). Replacing every 2 with a 1 turns that into a binary expansion, and every number in $[0, 1]$ has a binary expansion, so C has at least as many points as $[0, 1]$ itself. Measure and counting really are different notions of size.

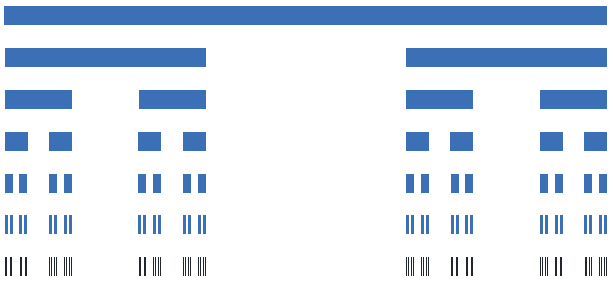


Figure 1.2: The first steps of the Cantor set construction, C_0 (top) through C_6 (bottom). Each step removes the open middle third of every remaining piece and keeps $2/3$ of the length, so C_n has length $(2/3)^n$. The limit has length 0, but it still has as many points as $[0, 1]$.

1.5 The Probability Triple

After all this, we can finally arrive at the probability triple. We say that $(\Omega, \mathcal{F}, \mathbb{P})$ is a **probability space**, where Ω is the sample space, $\mathcal{F}$ is the σ-algebra of events, and $\mathbb{P}$ is the

probability measure. The probability measure follows the same properties of a measure with 1 addition: normalization to unity. Formally, a **probability measure** is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ that satisfies

(1) Non-negativity: $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$.
(2) Countable additivity: if $\{A_i\}_{i=1}^{\infty}$ are disjoint events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

(3) Normalization: $\mathbb{P}(\Omega) = 1$.

These are the **Kolmogorov axioms**. Everything we derived for measures comes along for free: $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, monotonicity, inclusion-exclusion, the union bound $\mathbb{P}(\bigcup A_i) \leq \sum \mathbb{P}(A_i)$ for events that aren't disjoint, and both continuity properties without any finiteness worries.

The oversimplified summary is that a probability is just a measure with total mass 1. Functionally, it behaves no differently than how one is used to in introductory probability. However, this construction allows us to discuss probability measures and random variables that were otherwise not possible in probability 101. Another way to say this is that if μ is any measure with $0 < \mu(\Omega) < \infty$ (whether that be the Lebesgue or counting measure), then dividing by the total size of the sample space gives a probability measure

$$\mathbb{P}(A) = \frac{\mu(A)}{\mu(\Omega)} \tag{1.12}$$

With counting measure, this is the "favorable outcomes over total outcomes" of introductory probability. With length on $[12, 17]$, it's the office hours problem: $\mathbb{P}(\text{between 12 and 3}) = \frac{3}{5}$. And now we can finally write down Rosenthal's random variable from the beginning of the chapter. Its distribution is the probability measure on the Borel sets of $\mathbb{R}$

$$\mathbb{P}(Z \in A) = \frac{1}{2} \sum_{k=0}^{\infty} \frac{e^{-5} 5^k}{k!} \delta_k(A) + \frac{1}{2} \lambda\left(A \cap [0, 1]\right)$$

A sum of point masses plus a multiple of length. Both pieces are measures, a mix of measures is a measure, and the total mass is $\frac{1}{2} + \frac{1}{2} = 1$. There's nothing strange about it at all once we stop insisting that probabilities come from either a mass function or a density. Let's do an illustrative problem.

Problem 7.

In a deck of 52 cards, what is the probability of being dealt a royal flush in a 5 card hand?

Solution: For those that are not gambling addicts, a royal flush is a hand of 10, jack, queen, king, and ace all of the same suit. Every 5 card hand is equally likely, so we use the counting measure and (1.12). How many royal flushes are there? Pick a suit. How many suits can you pick? 4. Once the suit is chosen, the 5 cards are completely determined, and since order doesn't matter there's only 1 way to hold them. Thus, the measure of the event is 4. What about the measure of the sample space? Since order doesn't matter, this is just $\binom{52}{5} = 2{,}598{,}960$. The probability of a royal flush is

$$\boxed{\mathbb{P}(\text{royal flush}) = \frac{4}{\binom{52}{5}} \approx 1.5 \times 10^{-6}}$$

or about 1 in 650,000 hands.

1.5.1 Almost Surely

Problem 6 told us something that's worth dwelling on. If ω is uniform on $[0, 1]$, then the event "ω is irrational" has probability 1, but it's not the whole sample space, and the event "ω is rational" has probability 0, but it's not empty. In fact, *every* single outcome has probability 0, and yet one of them happens. So probability 0 doesn't mean impossible, and probability 1 doesn't mean certain. We say that an event with probability 1 happens **almost surely** (a.s.), and sets of probability 0 are called **null sets**. A lot of theorems in probability are of the form "this happens almost surely", and the "almost" is doing real work: the strong law of large numbers in chapter 7 says that the fraction of heads in infinitely many fair coin flips goes to 1/2 almost surely, even though the sequence HHHH... is a perfectly valid outcome. It's just one of a null set of exceptions. In the end, probability theory doesn't care about null sets, and we'll happily say two random variables are "equal" if they agree almost surely.

1.6 Conditioning And Independence

We have a working machine for calculating probabilities of a single event; now we will extend it to probabilities of multiple events. Before we begin, we have to make a distinction that confuses a lot of people: what is the difference between mutual exclusion and statistical

independence? If you ask two people what their favorite color is, their answers, generally speaking, are independent of one another. But if you ask one person their favorite color and they say red, then having yellow as a favorite color would be a mutually exclusive outcome (because you can't have 2 favorite colors). Mutually exclusive events are about as *dependent* as it gets: if one happens, you know for sure the other didn't.

How should we update probabilities when we learn something? If we are told that we are in the set B, then B effectively becomes our new sample space (everything outside of it is ruled out). The part of A that's still possible is $A \cap B$. And we have to renormalize so that the new sample space has probability 1, which by the same logic as (1.12) means dividing by $\mathbb{P}(B)$. This gives the definition of the **conditional probability**

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0 \tag{1.13}$$

You can check that for fixed B, $A \mapsto \mathbb{P}(A|B)$ satisfies the Kolmogorov axioms, so conditioning just gives us a new probability measure. Now we say events A and B are **independent** if knowing B has no bearing on A, $\mathbb{P}(A|B) = \mathbb{P}(A)$. An example is if you toss a coin twice: the first toss being tails does NOT change the probability of the second toss. Multiplying through by $\mathbb{P}(B)$, this is

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \tag{1.14}$$

and this symmetric form is the one we take as the official definition, since it also makes sense when $\mathbb{P}(B) = 0$ (and it shows independence is symmetric). For more than two events we have to be a bit careful. We say that events $A_1, A_2, \ldots$ (finitely or infinitely many) are **independent** if for *every* finite subcollection $A_{i_1}, \ldots, A_{i_k}$ of distinct events,

$$\mathbb{P}\left(A_{i_1} \cap \cdots \cap A_{i_k}\right) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}) \tag{1.15}$$

It's not enough to just check the whole collection at once, or just pairs, as we'll see in a moment. For infinitely many independent events, continuity from above then gives $\mathbb{P}(\bigcap_{i=1}^{\infty} A_i) = \prod_{i=1}^{\infty} \mathbb{P}(A_i)$ as well.

Let's look in general what would happen if we flip the condition

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \tag{1.16}$$

Since the numerator on both conditions is $\mathbb{P}(A \cap B)$, we can isolate the numerator and set the 2 equations equal to each other to get

$$\mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) \tag{1.17}$$

which is the celebrated **Bayes' theorem**. It's how we turn "the probability of the evidence given the hypothesis" into "the probability of the hypothesis given the evidence", and it's the whole foundation of Bayesian statistics.

But there's still more we can do when we are given information. Suppose that we partition our sample space into pieces B_i, and we want to find the probability of an event A, like in figure 1.3. By inspection, it's pretty obvious what the answer should be, but apparently proof by Venn diagram is not valid. Thus, let B_i be a partition of the sample space Ω, finite

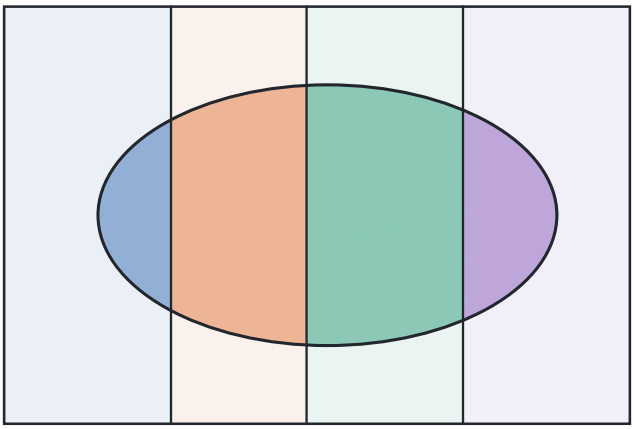


Figure 1.3: The law of total probability. The sample space Ω is partitioned into 4 vertical strips $B_1, \ldots, B_4$ (left to right, one color each), and the ellipse is the event A. The darker shaded pieces are the disjoint slices $A \cap B_i$.

or countable, so that

$$\bigcup_{i=1}^{\infty} B_i = \Omega \quad \text{and} \quad B_i \cap B_j = \emptyset \quad \forall i \neq j$$

We also use the trick that

$$A = A \cap \Omega = A \cap \bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} (A \cap B_i)$$

where the last step is just distributing the intersection over the union. The pieces $A \cap B_i$ are disjoint, because the B_i are. Then we can find the probability of A by using the σ-additivity property

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} (A \cap B_i)\right) = \sum_{i=1}^{\infty} \mathbb{P}(A \cap B_i)$$

and then rewriting each term with the definition of conditional probability (1.13)

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A|B_i)\mathbb{P}(B_i) \tag{1.18}$$

This has the very famous name of the **Law of Total Probability**. Its job is to break a hard problem into easier cases: if you don't know which B_i happened, consider each one separately and then average, weighted by how likely each case is.

Problem 8.

Suppose we toss a coin, with probability p of heads, until we get exactly two heads. Describe the sample space Ω. What is the probability that exactly k tosses are required?

Solution: This problem is actually not very intuitive. I lied (well, if you understand what the negative binomial is, then I guess it's super intuitive). But we will go through it systematically. The sample space is the set of all possible sequences we could see. Every sequence ends at its second head, so it's some number of tails, a head, some number of tails, and a final head. The size of the sample space is infinite. I will write the first few possibilities

$$\Omega = \{HH,\ THH,\ HTH,\ TTHH,\ THTH,\ HTTH,\ \ldots\}$$

(plus, technically, the null set of sequences that never get a second head). For exactly k tosses, how many heads and tails do we see? Every sequence has 2 heads and $k-2$ tails, so each one has probability $p^2(1-p)^{k-2}$ by independence. How many such sequences are there? The last toss must be the second head (by definition), so the only freedom we have is where to put the first head among the first $k-1$ tosses. The rest must be tails. That's $\binom{k-1}{1} = k-1$ choices. Thus, using the multiplication rule, the probability is

$$\binom{k-1}{1} p^2(1-p)^{k-2}$$

For a fair coin, $p = \frac{1}{2}$, and this is

$$\boxed{\mathbb{P}(k \text{ tosses}) = (k-1)\left(\frac{1}{2}\right)^k, \qquad k \geq 2}$$

As a sanity check, $\sum_{k\geq 2}(k-1)2^{-k} = 1$, so the second head does arrive almost surely.

1.6.1 Pairwise Independence

As it turns out, if we have a collection of events, it could be that every pair of them is independent but the whole collection is not. We say events are **pairwise independent** if

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \qquad \text{for all} \quad i \neq j$$

This is a weaker form of independence. Independence of the whole collection implies pairwise independence (obviously), but not the other way around. Here's the classic example. Flip two fair coins, and let $A = \{\text{first is heads}\}$, $B = \{\text{second is heads}\}$, $C = \{\text{the two coins match}\}$. Each has probability $\frac{1}{2}$, and each pair intersects in exactly one outcome out of four ($A \cap B = A \cap C = B \cap C = \{HH\}$), so every pair is independent. But if you know A and B, you know C for sure: $\mathbb{P}(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$. Any two of the events tell you nothing about the third, but all three together are linked. Quite frankly, I've never really had to use pairwise independence for anything other than this cautionary tale, but it does sometimes come up in the literature (pairwise independence is enough to make variances add, for example, which is why some versions of the law of large numbers only need it).

1.7 Limits Of Sets

We are close to closing the story on measure theory. There is still one last piece that we need to connect it all, and that is the limits of sets. Ideally, we want to analyze the asymptotic behavior of sequences of random variables, or convergence properties of probabilities. To do this, we must develop the tools of the limit supremum and the limit infimum.

Let us suppose that we have a sequence of sets A_k. Define the tail unions and tail intersections

$$S_n = \bigcup_{k=n}^{\infty} A_k, \qquad B_n = \bigcap_{k=n}^{\infty} A_k$$

Written out, we can see that the first few terms are

$$S_1 = A_1 \cup A_2 \cup A_3 \cup \cdots, \qquad B_1 = A_1 \cap A_2 \cap A_3 \cap \cdots$$

$$S_2 = A_2 \cup A_3 \cup A_4 \cup \cdots, \qquad B_2 = A_2 \cap A_3 \cap A_4 \cap \cdots$$

Clearly $B_n \subseteq A_n \subseteq S_n$. We note (whether it be clear or not) that the S_n are decreasing and the B_n are increasing. Why is this true? S_{n+1} is a union of fewer sets than S_n, so it's smaller, $S_{n+1} \subseteq S_n$, while B_{n+1} is an intersection of fewer sets than B_n, so it's bigger, $B_n \subseteq B_{n+1}$. Monotone sequences of sets always have a limit (the intersection for decreasing

ones, the union for increasing ones), so let's take them. We define the **limit supremum** and **limit infimum**

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} S_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \tag{1.19}$$

$$\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \tag{1.20}$$

These look like alphabet soup, so let's read them out loud. If $x \in \limsup A_k$, what does it mean? It means that x is an element of *every* S_n. So no matter how far out we start looking (n), x is in some A_k further down the line. That is, x is in infinitely many of the A_k, and we say that A_k happens **infinitely often** (i.o.) for that x. What about $\liminf A_k$? Now x is in *some* B_n, which means that from some index n onwards, x is in *all* of the A_k. Thus x is in all but finitely many of the A_k, and we say that A_k happens **eventually** (or almost always). For example, if x is in B_5, then it may or may not be in $A_1, \ldots, A_4$, but it's in every A_k from $k = 5$ onwards. Since "all but finitely many" is a stronger demand than "infinitely many", we always have

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$$

We say that the limit of the sequence exists if the liminf and the limsup coincide

$$\lim_{n\to\infty} A_n = \liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$$

For example, if $A_n = [0, 1]$ for even n and $A_n = [1, 2]$ for odd n, then every point of $[0, 2]$ is in infinitely many A_n, but only the point 1 is in all of them from some point on, so $\limsup A_n = [0, 2]$ and $\liminf A_n = \{1\}$, and the limit doesn't exist. If you like indicators, there's a nice way to say all of this: $\mathbb{I}_{\limsup A_n} = \limsup \mathbb{I}_{A_n}$ and $\mathbb{I}_{\liminf A_n} = \liminf \mathbb{I}_{A_n}$, as sequences of 0's and 1's.

Like I said before, these will often be used to characterize long term behaviors and convergence properties of probabilities and random variables. Let's work through one consequence of the probability axioms.

Problem 9.

Show that

$$\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n) \leq \limsup \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n)$$

Solution: Let's work through the first one, $\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n)$. To see this, expand the definition. The $B_n = \bigcap_{k\geq n} A_k$ increase to $\liminf A_n$, so by continuity from below

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k\right)$$

Now, the intersection of all the sets from n on is contained in each one of those sets, so by monotonicity $\mathbb{P}(\bigcap_{k\geq n} A_k) \leq \mathbb{P}(A_k)$ for every $k \geq n$, i.e. $\mathbb{P}(B_n) \leq \inf_{k\geq n} \mathbb{P}(A_k)$. Taking $n \to \infty$ on both sides,

$$\mathbb{P}(\liminf A_n) = \lim_{n\to\infty} \mathbb{P}(B_n) \leq \lim_{n\to\infty} \inf_{k\geq n} \mathbb{P}(A_k) = \liminf_{n\to\infty} \mathbb{P}(A_n)$$

The middle inequality $\liminf \mathbb{P}(A_n) \leq \limsup \mathbb{P}(A_n)$ is true by construction (lower limits are less than upper limits). All that is left is the last inequality, which is the mirror image of the first. The $S_n = \bigcup_{k\geq n} A_k$ decrease to $\limsup A_n$, so by continuity from above (fine, since probabilities are finite), $\mathbb{P}(S_n) \to \mathbb{P}(\limsup A_n)$. And each A_k with $k \geq n$ is contained in the union S_n, so $\sup_{k\geq n} \mathbb{P}(A_k) \leq \mathbb{P}(S_n)$. Taking $n \to \infty$,

$$\limsup_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty} \sup_{k\geq n} \mathbb{P}(A_k) \leq \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{k=n}^{\infty} A_k\right) = \mathbb{P}(\limsup A_n)$$

Putting it all together we get

$$\boxed{\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n) \leq \limsup \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n)}$$

In particular, if $\lim A_n$ exists, then all four are equal, so $\mathbb{P}(\lim A_n) = \lim \mathbb{P}(A_n)$. Probability is continuous along any convergent sequence of events.

1.8 The Borel-Cantelli Lemmas

We will end this chapter with the most useful application of the limits of sets. The question is: if we have a sequence of events A_n, will infinitely many of them happen? The Borel-Cantelli lemmas answer this in terms of the single number $\sum_n \mathbb{P}(A_n)$, and they're the workhorse behind every "almost surely" statement in this book.

1.8.1 The First Lemma

The first lemma states that, given a sequence of events $\{A_n\}$ (no independence needed!), if the sum of probabilities is finite

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then the probability that infinitely many of the events A_n occur is zero

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 0. \tag{1.21}$$

Here's the physicist's reason first. Let $N = \sum_n \mathbb{I}_{A_n}$ be the number of events that happen. Its expected value is (using linearity, which we'll justify for infinite sums in chapter 3)

$$\mathbb{E}[N] = \sum_{n=1}^{\infty} \mathbb{E}\left[\mathbb{I}_{A_n}\right] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$$

A random variable with a finite average can't be infinite with positive probability. So $N < \infty$ almost surely: only finitely many of the A_n happen. To show this directly, start with the definition of the lim sup

$$\limsup_{n\to\infty} A_n = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n$$

The lim sup is inside every one of the tail unions, so by monotonicity and then the union bound (1.8),

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) \leq \mathbb{P}\left(\bigcup_{n=m}^{\infty} A_n\right) \leq \sum_{n=m}^{\infty} \mathbb{P}(A_n) \quad \text{for every } m$$

The right hand side is the tail of a convergent series, so it goes to 0 as $m \to \infty$. Thus, we have calculated that the probability of the lim sup must be 0.

1.8.2 The Second Lemma

Now what if the sum is not finite but instead infinite? Is it then guaranteed that infinitely many of the events happen? Not in general. Take a single event A with $\mathbb{P}(A) = \frac{1}{2}$ and let $A_n = A$ for every n. The sum is infinite, but with probability $\frac{1}{2}$ none of them happen. The events have to be independent, so that one of them not happening doesn't stop the others. With independence, we will have to be a bit more slick. Begin by supposing that the A_n are

independent and the sum is infinite

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$$

Then

$$1 - \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = \mathbb{P}\left(\left(\limsup_{n\to\infty} A_n\right)^c\right) = \mathbb{P}\left(\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n\right)^c\right)$$

Using De Morgan's (twice), the complement of "infinitely often" is "eventually never"

$$= \mathbb{P}\left(\bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} A_n^c\right) \leq \sum_{m=1}^{\infty} \mathbb{P}\left(\bigcap_{n=m}^{\infty} A_n^c\right)$$

So it's enough to show that each of these intersections has probability 0. Complements of independent events are independent, so

$$\mathbb{P}\left(\bigcap_{n=m}^{\infty} A_n^c\right) = \prod_{n=m}^{\infty} \mathbb{P}(A_n^c) = \prod_{n=m}^{\infty} \left(1 - \mathbb{P}\left(A_n\right)\right)$$

What do we do now? In true physics fashion, we treat $1-\mathbb{P}(A_n)$ like the first order expansion of $e^{-\mathbb{P}(A_n)}$. But we don't even need to approximate: the inequality $1 - x \leq e^{-x}$ holds for every real x (the exponential is convex, and $1 - x$ is its tangent line at 0). So

$$\prod_{n=m}^{\infty} \left(1 - \mathbb{P}\left(A_n\right)\right) \leq \prod_{n=m}^{\infty} e^{-\mathbb{P}(A_n)} = \exp\left[-\sum_{n=m}^{\infty} \mathbb{P}(A_n)\right]$$

If the sum $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$ diverges, then the tail $\sum_{n=m}^{\infty} \mathbb{P}(A_n)$ also diverges for every m (we've only thrown away finitely many terms). The exponential gives $e^{-\infty}$, which is 0, so every term in the sum over m is 0, and therefore

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1 \tag{1.22}$$

So for independent events, there's no middle ground: either $\sum \mathbb{P}(A_n) < \infty$ and only finitely many happen (almost surely), or $\sum \mathbb{P}(A_n) = \infty$ and infinitely many happen (almost surely). This is an example of a **zero-one law**, and you can see the dichotomy in figure 1.4. The borderline is sharp: $1/n^2$ is summable and $1/n$ is not, and the difference between them is the difference between "the events stop" and "the events never stop" (though for $1/n$ they get rarer and rarer, with about $\log n$ of them by time n).

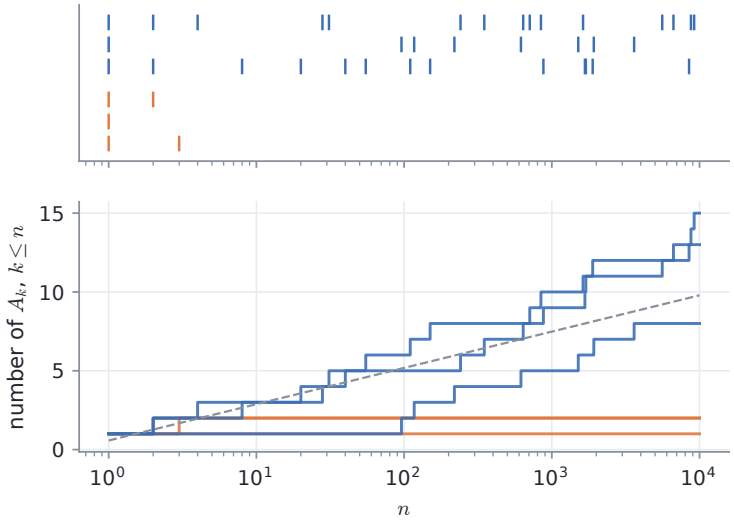


Figure 1.4: Independent events with $\mathbb{P}(A_n) = 1/n^2$ (orange) and $\mathbb{P}(A_n) = 1/n$ (blue), 3 runs each. Top: tick marks at the times n at which the events happen. Bottom: how many have happened by time n, with the dashed curve $\sum_{k \le n} 1/k \approx \log n$. With $1/n^2$ the count freezes after a handful of events; with $1/n$ it keeps growing like $\log n$ forever.

Problem 10.

Suppose you are flipping a fair coin infinitely many times.

a). Let A_n be the event that the n-th flip results in heads. Determine whether infinitely many heads occur with probability 1 or 0.

b). Let R_n be the event that flips $n, n+1, \ldots, n+9$ are all heads (a run of 10 heads starting at n). Do runs of 10 heads happen infinitely often?

Solution: a). Assuming that the coin is fair, the probability of getting heads is $\mathbb{P}(A_n) = 0.5$, and the flips are independent. The sum of all the probabilities is

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 0.5 = \infty$$

Since this sum diverges, applying the second Borel-Cantelli lemma tells us that the probability that infinitely many heads occur is 1.

b). Each R_n has probability 2^{-10}, so the sum diverges again. But careful, the R_n are NOT independent (R_1 and R_2 share 9 flips). The fix is to only look at the runs that start at $n = 1, 11, 21, \ldots$, which use disjoint blocks of flips and are therefore

independent. Their probabilities still sum to infinity, so by the second lemma infinitely many of those happen, and so certainly infinitely many of the R_n do. Replace "10 heads" with "the complete works of Shakespeare typed in binary" and you get the infinite monkey theorem.

1.9 Problems

1. Let $\Omega = \{1, 2, 3, 4\}$. Determine whether or not each of the following is a σ-algebra.

(a) $\mathcal{F}_1 = \{\emptyset, \{1, 2\}, \{3\}, \{3, 4\}, \{1, 2, 3\}, \Omega\}$.
(b) $\mathcal{F}_2 = \{\emptyset, \{3\}, \{4\}, \{2, 3\}, \{1, 3\}, \{1\}, \{1, 2, 3\}, \{1, 2, 3, 4\}\}$.
(c) $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 2, 3\}, \{2\}, \{1, 3, 4\}, \{2, 3\}, \Omega\}$.

2. Let $(\Omega, \mathcal{M}, \lambda)$ be the Lebesgue measure on the interval $[0, 1]$. Let

$$\Omega' = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

Let $\mathcal{F}$ be the collection of all subsets of Ω' of the form $\{(x, y) : x \in A, 0 \leq y \leq 1\}$ for some $A \in \mathcal{M}$. Finally, define a probability P on $\mathcal{F}$ by $P(\{(x, y) : x \in A, 0 \leq y \leq 1\}) = \lambda(A)$.

(a) Prove that $(\Omega', \mathcal{F}, P)$ is a probability triple.
(b) Let P^* be the outer measure corresponding to P and $\mathcal{F}$. Define the subset $S = \{(x, y) : 0 \leq x \leq 1, y = 1/2\}$. (Note that $S \notin \mathcal{F}$.) Prove that $P^*(S) = 1$ and $P^*(S^c) = 1$. What does this say about why S can't be measurable?

3. You have probably heard it before. Now you can solve it rigorously. It is called the "Monty Hall Problem". A prize is placed at random behind one of three doors. You pick a door. To be concrete, let's suppose you always pick door 1. Now Monty Hall chooses one of the other two doors, opens it and shows you that it is empty (or has a donkey). He then gives you the opportunity to keep your door or switch to the other unopened door. Should you stay or switch? Intuition suggests it doesn't matter. The correct answer is that you should switch. Prove it. It will help to specify the sample space and the relevant events carefully. Thus write $\Omega = \{(\omega_1, \omega_2) : \omega_i \in \{1, 2, 3\}\}$ where ω_1 is where the prize is and ω_2 is the door Monty opens.

4. Suppose that a fair coin is tossed repeatedly until both a head and a tail have appeared at least once.

(a) Describe the sample space Ω.
(b) What is the probability that three tosses will be required?

5. Suppose that the events $\{A_n, n \geq 1\}$ are independent. Show that the inclusion-exclusion formula reduces to

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) = 1 - \prod_{k=1}^{n} \left(1 - \mathbb{P}(A_k)\right).$$

6. Show, in the setup of the previous problem, that

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) \geq 1 - \exp\left\{-\sum_{k=1}^{n} \mathbb{P}(A_k)\right\},$$

and that

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty \implies \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = 1.$$

7. Show that the intersection of any collection of σ-algebras on Ω is a σ-algebra, but give an example of two σ-algebras whose union is not.

8. Let λ be the Lebesgue measure on $\mathbb{R}$ and let $A_n = [n, \infty)$. Compute $\lim_n \lambda(A_n)$ and $\lambda(\bigcap_n A_n)$. Which hypothesis of continuity from above fails?

9. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_n = 1) = p_n$ and $\mathbb{P}(X_n = 0) = 1 - p_n$. Show that $X_n \to 0$ almost surely if $\sum_n p_n < \infty$, and that X_n fails to converge almost surely if $\sum_n p_n = \infty$ and $p_n \to 0$.

2

Random Variables

"This is probably correct"

– Hieu

The worst is finally over. In this chapter, we will spend a good chunk of time deriving the standard distributions and then spend another chunk of time discussing how random variables transform.

2.1 Measurable Functions

In the last chapter, we spent time motivating measures and measurable sets all to derive the probability triple. But having a static set is uninteresting; fun things only happen when we transform it. Let us define a function

$$f : (X, A) \to (Y, B) \tag{2.1}$$

where A is a σ-algebra on the space X and B is a σ-algebra on Y. We call this function measurable if it respects the structure of the σ-algebras. That is, the pre-image of every set $E \in B$ is an element of A

$$f^{-1}(E) = \{x \in X \mid f(x) \in E\} \in A \text{ for all } E \in B \tag{2.2}$$

which intuitively just means that pulling a measurable set back through f gives you a measurable set. Notice that it's the pre-image and not the image; images of measurable sets can be very badly behaved, but pre-images play nicely with unions, intersections and complements ($f^{-1}(E^c) = f^{-1}(E)^c$ and $f^{-1}(\bigcup E_i) = \bigcup f^{-1}(E_i)$), which is exactly the structure a σ-algebra has. We should note that the measurability of a function is heavily dependent on the chosen σ-algebra. A basic illustration of this is if f takes on 2 different values but the σ-algebra A is the trivial one $\{\emptyset, X\}$. Then the pre-image of one of the values is neither $\emptyset$ nor X, so with respect to this σ-algebra, the function f is NOT measurable. In the language of chapter 1, the trivial σ-algebra is the state of knowing nothing, and a function that takes two values carries information that we don't have. Thus oftentimes you will hear one say that f is $\mathcal{F}$ measurable, where $\mathcal{F}$ is the σ algebra of a space X.

For real valued functions (with the Borel σ-algebra on $\mathbb{R}$), it's easier to say that f is measurable if

$$\{\omega \mid f(\omega) < a\} \in \mathcal{F} \tag{2.3}$$

for all a. This is equivalent to (2.2), because the half lines $(-\infty, a)$ generate the Borel σ-algebra (we saw this in chapter 1), so if the pre-images of the generators are measurable, the pre-images of everything built out of them are measurable too. An example of this is shown in figure 2.1. Suppose we have a function $f(x) = x^2 + 1$ and we choose a horizontal line $y = a$. What is the set of all points x such that $f(x) < a$? For $a > 1$, that would just be the interval $(-\sqrt{a-1}, \sqrt{a-1})$, which indeed is a Borel set. What happens if we choose $a \leq 1$? Then that set is just the empty set, but we said that the empty set is an element of every σ-algebra. In the same way that a measure is a generalization of size, definition (2.3) is a generalization of "well-behavedness" of functions. We make an extension to the definition and say that this measurability still holds even if the inequality in (2.3) is $\leq$, $\geq$, or $>$ (each of these sets can be built from the others with countable unions, intersections and complements, for example $\{f \leq a\} = \bigcap_n \{f < a + 1/n\}$). It's immediately clear, then, that every continuous function is measurable, because $\{f < a\}$ is the pre-image of an open set, which is open, and open sets are Borel.

Problem 1.

Let f and g be measurable functions. Show that 1). $f + g$ is measurable, 2). $f(g)$ is measurable, and 3). fg is measurable.

Solution:

Let's use definition (2.3). Define a new function $h = f + g$. Then we'll need to

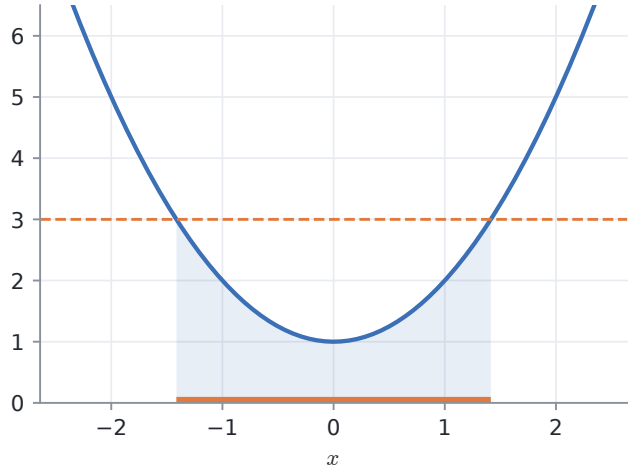


Figure 2.1: The function $f(x) = x^2 + 1$ (blue) and the level $y = a$ (dashed orange, here $a = 3$). The set of points where f lies below the line is the interval $(-\sqrt{a-1}, \sqrt{a-1})$, shaded and marked in orange on the x-axis, and intervals are Borel sets.

show that $\{\omega \mid h(\omega) < a\} \in \mathcal{F}$. The trick is that if $f(\omega) + g(\omega) < a$, then there is room between $f(\omega)$ and $a - g(\omega)$, and we can squeeze a rational number q in there: $f(\omega) < q < a - g(\omega)$. Since there are only countably many rationals,

$$\{f + g < a\} = \bigcup_{q \in \mathbb{Q}} \Big(\{f < q\} \cap \{g < a - q\} \Big)$$

which is a countable union of intersections of measurable sets, and therefore measurable.
What about compositions? Here we need $f : \mathbb{R} \to \mathbb{R}$ to be Borel measurable and g to be measurable. Then for any Borel set E, $(f \circ g)^{-1}(E) = g^{-1}\left(f^{-1}(E)\right)$. The inner set $f^{-1}(E)$ is Borel because f is measurable, and then g^{-1} of a Borel set is in $\mathcal{F}$ because g is measurable. Pre-images just stack.
Lastly, products. First, squares are measurable, because for $a > 0$, $\{f^2 < a\} = \{-\sqrt{a} < f < \sqrt{a}\} = \{f < \sqrt{a}\} \cap \{f > -\sqrt{a}\}$ (and it's empty for $a \leq 0$). Then we use the polarization trick

$$fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$$

which writes the product using only sums, scalar multiples and squares, all of which we just showed are measurable.

We've proven the basic arithmetic properties of measurable functions, but it would be nice to see actual examples of measurable functions. The most basic function is the indicator

function

$$f(x) = \mathbb{I}_A(x)$$

Which is 1 if $x \in A$ and 0 else. I think it's really funny that the most simple measurable function is not that intuitive. To show that f is measurable (when A is a measurable set), notice that a Borel set E can only contain 0, 1, both, or neither. If it contains only 1, the pre-image is A, which is in the σ-algebra. If it contains only 0, the pre-image is A^c, which is in the σ-algebra by closure under complements. Both gives Ω and neither gives $\emptyset$. At this point, you should have some intuition as to show how measurability of a function. We now move on to random variables.

2.2 Random Variables

A random variable is a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ on a probability space. That's basically it. It's a function that sends elements of the sample space Ω to $\mathbb{R}$, in such a way that questions like "is $X \leq 3$?" are events that we can assign probabilities to. For example, suppose we flip a coin 10 times. If X is the random variable counting the number of heads and ω is the sequence $HHHTTHTHHT$, then $X(\omega) = 6$. Notice that there is nothing random about the function X itself; all of the randomness lives in which ω nature picks. If we repeat the experiment a large number of times and plot a histogram of X, then it would follow some **distribution** (examine figure 2.2). In general, we can classify random

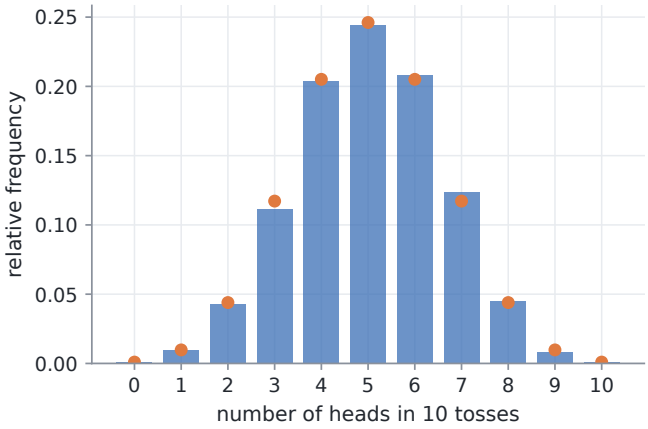


Figure 2.2: The frequency of seeing x heads in 10 coin tosses from 10,000 repetitions of the experiment (bars), together with the Binomial$(10, 1/2)$ probabilities we will derive later in this chapter (orange dots).

variables into different types (similar to how we can classify functions...because they are literally functions). The simplest random variable that we can make is just a sum of the

simplest measurable function we have: it's just the indicator function

$$X(\omega) = \sum_{i=1}^{n} x_i \mathbb{I}_{A_i}(\omega) \tag{2.4}$$

Obviously this is a valid random variable because we proved in the last section that the indicator function is a measurable function and sums of measurable functions are measurable. where the A_i are measurable sets (usually a partition of Ω). This has the appropriate name of being a **simple random variable**. Now, what if we extend the sum to infinity? Then that's still a valid random variable, it would just be an **elementary random variable**

$$X(\omega) = \sum_{i=1}^{\infty} x_i \mathbb{I}_{A_i}(\omega)$$

This is fine because a pointwise limit of measurable functions is measurable: $\{\lim f_n < a\}$ can be written with countably many unions and intersections of the sets $\{f_n < a - 1/k\}$. Simple random variables are more important than they look, because every non-negative random variable is an increasing limit of simple ones (chop the range into finer and finer pieces), and that's how the expectation gets defined in chapter 3.

We mentioned earlier that if we plot the frequency of outcomes for a random variable, then it would follow some distribution. The probability that X lands in a Borel set A is

$$\mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)) \tag{2.5}$$

The map $A \mapsto \mathbb{P}(X^{-1}(A))$ is itself a probability measure, but on $(\mathbb{R}, \mathcal{B})$ instead of on Ω, and it's called the **law** or **distribution** of X. This is why we needed X to be measurable, otherwise $X^{-1}(A)$ might not be an event at all. The notation makes it seem like it's more complicated than it actually is. Let's see with an example how this is actually pretty straightforward.

Problem 2.

Let X be the number of heads in a 2 coin toss experiment. Find $\mathbb{P}(X \in A)$ for every subset A of the possible values $\{0, 1, 2\}$.

Solution:

Here $\Omega = \{HH, HT, TH, TT\}$, each with probability $1/4$. It's enough to find the probability of each single value, since every other A is a disjoint union of them. All the events can be tabulated, but we will go through each of them one by one. What

is $\mathbb{P}(X = 0)$? Well, we need $X^{-1}(0)$ which in this case is just $\{TT\}$. What is the probability of $\{TT\}$? It is $1/4$. Do the same thing for $\mathbb{P}(X = 1) = \mathbb{P}(X^{-1}(1)) = \mathbb{P}(\{TH, HT\}) = 1/2$. Lastly, doing the exact same thing, $\mathbb{P}(X = 2) = \mathbb{P}(X^{-1}(2)) = \mathbb{P}(\{HH\}) = 1/4$. Then, for example, $\mathbb{P}(X \in \{1, 2\}) = 3/4$, and $\mathbb{P}(X \in \emptyset) = 0$.

2.3 Distributions

We mentioned distributions of random variables previously, but we will expand on it here. A probability measure on $\mathbb{R}$ is a complicated object (it eats sets), but because the half lines $(-\infty, x]$ generate the Borel sets, it's completely determined by what it does on half lines. We think about this as the accumulation of probability as x increases

$$F_X(x) = \mathbb{P}(X \leq x) \tag{2.6}$$

This is called the **cumulative distribution function**. There are 4 properties that such CDF has to have.

1). It must be nondecreasing. This one is really straightforward to see. If $x < y$, then $\{X \leq x\} \subseteq \{X \leq y\}$, and measures are monotone. If the CDF decreased, there would be a set with negative probability.

2). F is normalized:

$$\lim_{x \to -\infty} F_X = 0 \qquad \lim_{x \to \infty} F_X = 1 \tag{2.7}$$

This is almost immediately clear. The sets $\{X \leq n\}$ increase to all of Ω as $n \to \infty$, so by continuity from below, $F_X(n) \to \mathbb{P}(\Omega) = 1$. Similarly the sets $\{X \leq -n\}$ decrease to $\emptyset$, so by continuity from above $F_X(-n) \to 0$.

3). F_X is right continuous:

$$F_X(x) = F_X(x^+) \tag{2.8}$$

This one is not as obvious. We show this by saying that let $y_1, y_2, ...$ be a sequence of numbers such that $y_1 > y_2 > ...$ and $\lim_{i \to \infty} y_i = x$. We construct sets $A_i = (-\infty, y_i]$ and $A = (-\infty, x]$ such that $A = \lim_{i \to \infty} A_i$. The sets A_i are decreasing and their intersection is $\bigcap_i A_i = A$ (a point ω with $X(\omega) \leq y_i$ for every i has $X(\omega) \leq x$). By continuity from above, $\lim F_X(y_i) = \lim \mathbb{P}(A_i) = \mathbb{P}(A) = F_X(x)$. Why not left continuous? If y_i increases to x, then $\bigcup_i (-\infty, y_i] = (-\infty, x)$, which is missing the point x, so the limit from the left is $\mathbb{P}(X < x)$, and the size of the jump at x is exactly $\mathbb{P}(X = x)$. To see an example of this, let's look at the distribution for the 2 coin toss in problem 2, shown in figure 2.3. We can

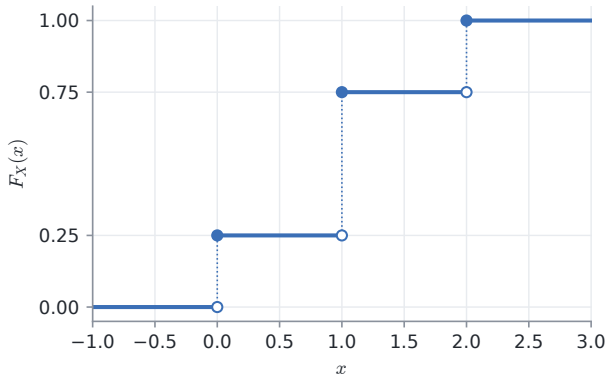


Figure 2.3: The CDF for the number of heads in 2 coin tosses (problem 2). The filled dots show which side of each jump the function takes, and the jump sizes 1/4, 1/2, 1/4 are the probabilities of 0, 1 and 2 heads.

see that this satisfies all the properties of a CDF, even the right continuity condition (at each jump, the function takes the upper value). The reason that we care about distributions is because they are going to be the thing we will use to characterize a random variable, and they will be the object that we work with when we do transformations of random variables. But before we can get to the fun stuff, I have to bore you with a bit of terminology. The reason the terminology exists is because we built up measure theory just to expand our view on what constitutes a random variable.

Let's start with the most basic. We say that a distribution is **discrete** if it takes on countably many possible outcomes so that

$$\sum_{i=1}^{\infty} P(X = x_i) = 1 \tag{2.9}$$

Oftentimes, you will hear people call $P(X = x_i)$ the distribution of X. The 2 coin toss would be an example of a discrete distribution (or sometimes we will say that it is a discrete random variable) because there are only 3 values X can take on.

Stepping up from the discrete, we say that a distribution is **continuous** if there are no jumps and if the distribution is **absolutely continuous**, then it has a density, meaning that the probability $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\,dx$ where f_X is the probability **density** of the random variable X. As a consequence of continuity, we have that $\mathbb{P}(X = x) = 0$ for every x. Perhaps we should give a bit of lip service to the difference between continuity and absolute continuity. The bare bones definition of continuity just tells us that the function is has no sudden jumps and nothing more. Absolute continuity tells us that the function

is "smoother"; and we should be careful about what we mean by that. If we change the input to the function f by some small amount dx, then the function itself should respond by a small df. Here is an an example: consider 2 functions $f_1(x) = \sin(\frac{1}{x})$ and $f_2(x) = \sqrt{x}$. Clearly, f_1 has no sudden jumps, but we consider intervals very close to $x = 0$, then a small change in x causes the function to oscillate wildly. In fact, if we take the derivative, we see that $f_1'(x) = -\frac{\cos(1/x)}{x^2}$ blows up in oscillation as $x \to 0$. What about f_2? Its derivative $1/(2\sqrt{x})$ also blows up at 0, so should we also consider it to be just continuous? Actually, no. f_2 is absolutely continuous because the total amount it can change over a collection of tiny intervals is controlled by their total length, and that's because its derivative, while unbounded, is still integrable: $\sqrt{x} = \int_0^x \frac{dt}{2\sqrt{t}}$. That's really the content of absolute continuity for a CDF: F is the integral of its derivative, and that derivative is the density. We are being a bit cavalier with these definitions and this is on purpose so that we don't open the flood gates of real analysis.

Here's the last but wackiest distribution: if F_X is continuous (no jumps), but the derivative F_X' is 0 almost everywhere, so that there is no density, then we call it **singular continuous.** The poster child is the Cantor distribution, which puts all of its probability uniformly on the Cantor set from chapter 1. Since the Cantor set has length 0, the CDF is flat everywhere except on a set of measure 0, and yet it still manages to climb continuously from 0 to 1 (it's called the devil's staircase, for good reason). Remarkably, these are the only options: every distribution is a mixture of a discrete part, an absolutely continuous part and a singular continuous part (this is the Lebesgue decomposition). The mixture of a Poisson and a uniform from chapter 1 is a mixture of the first two types.

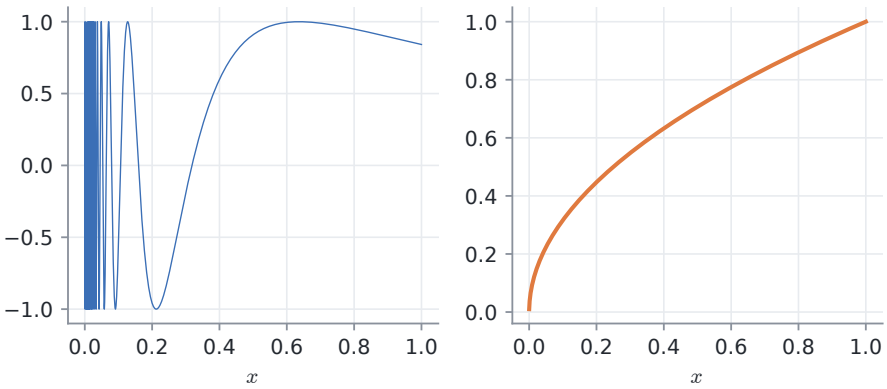


Figure 2.4: Plot of $\sin(1/x)$ (left) and $\sqrt{x}$ (right) on $(0, 1]$. Notice the oscillatory behavior of the first as $x \to 0$; the limit does not even exist. The second also has an infinite slope at 0, but it is the integral of its derivative.

2.4 The Standard Distributions

In this section, we will go through the standard distributions and how to derive them. The first one, and the most boring, is the **degenerate** distribution, which is just one outcome only. You can think of it as flipping a coin that has heads on both sides, so you always land on heads. The second is probably equally as boring and that is just the **Bernoulli**, where are only 2 possible outcomes. This is given as

$$\mathbb{P}(X = x_1) = p \quad \text{or} \quad \mathbb{P}(X = x_2) = 1 - p \tag{2.10}$$

Where p is the probability of getting one of the 2 outcomes. Since probabilities have to add up to unity, the second and only other outcome has probability $1 - p$. Pretty boring stuff. But now that we've gotten that out of the way, we can get to the interesting stuff.

2.4.1 Geometric

Suppose your best friend just got into a relationship with someone. You, being who you are, are a bitter and jealous person. You want to know the probability that they will break up after n days. If you know nothing more than the fact that each day the probability of them staying together is $1 - p$ and the probability of them breaking up is p, then we can find the probability distribution for X being the number of days until they break up (assuming they don't ever get back together) is

$$\mathbb{P}(X = x) = (1 - p)^{x-1}p \tag{2.11}$$

If we assume each day is independent from each other, then in order for them to break up on the x-th day, then they need to stay together for the first $x - 1$ days and then break up on the last day.

Problem 3.

When is the most likely day that they are going to break up?

Solution:

This is a discrete distribution, and thus we can't really just take a derivative and set it equal to 0. We can use a bit of intuition here. We know that probabilities are always less than or equal to 1, and any numbers smaller than 1 multiplied together will give a number that is even smaller. Thus, by looking at the geometric probabilities, we see that each day's probability is the previous day's multiplied by $1 - p < 1$, so it's a

decreasing function of x. Thus, the most likely day for them to break up is the next day (this is assuming that they have to be together on the first).

In the literature, you will often see a geometric random variable be described as **the number of trials until the first success** (here a "success" is the break up, which is a bit grim). Some books count the number of failures before the first success instead, which shifts everything by one, so always check which convention is being used.

2.4.2 Negative Binomial

Do not be fooled. This is not a binomial; the name comes from the binomial coefficient in front (and a negative binomial series that shows up when you sum it). If anything, this is a generalization of the geometric distribution. Instead of your friends breaking up, what's the probability that they finally separate after x break ups? Or another way to phrase this: let X be the random variable describing **the number of trials (days) until the n-th break up**. Notice that we've added an additional parameter, which is the number of break ups after which the process stops. Each day they break up with probability p and stay together with probability $1-p$. One particular sequence of x days containing exactly n break ups has probability

$$(1-p)^{x-n}p^n$$

But this is for 1 instance. How many of these sequences are there? It might be tempting to say $\binom{x}{n}$, where x is the number of days and n is the number of break ups. But this is not the case, because the last day MUST be a break up (otherwise the process would have stopped earlier or not yet). Thus, the correct form of the probability must be

$$\mathbb{P}(X=x) = \binom{x-1}{n-1}(1-p)^{x-n}p^n \tag{2.12}$$

because the last trial is fixed, you only have $n-1$ break ups to place among the first $x-1$ days. Setting $n=1$ gives back the geometric distribution, as it should.

2.4.3 Binomial

The last 2 distributions had domains that were potentially infinite. Here, we talk about a distribution that is discrete but has finite domain. Let X be the random variable that counts the number of (independent) successes with probability p in N trials. Then the probability follows as

$$\mathbb{P}(X=x) = \binom{N}{x}p^x(1-p)^{N-x} \tag{2.13}$$

If there are x successes, then there must be $N-x$ failures, and the number of ways to choose x successes from N trials is $\binom{N}{x}$. As simple as it is, it surprisingly shows up a great deal.

Problem 4.

The second law of thermodynamics is a deep statement about the nature of the universe. It tells us that all processes must either increase the entropy (number of configurations of states) of the universe or keep it the same. Entropy CANNOT decrease. Let's discuss why!
Suppose you have 3 marbles in box A and 7 marbles in box B (all marbles are distinguishable, meaning you can label them marble 1 and marble 2). They are at first separated from each other. Now, you bring both boxes into contact. If the marbles are allowed to transfer from one box to the other, what is the most probable configuration for box A and B
(that is, how many marbles in box A and how many marbles in box B) and what is the corresponding probability?
Solution:
If you can see it immediately, then you would see that it is a binomial distribution. But if you can't, let's see why. How many ways can you have 0 balls in box A and 10 balls in box B? Well, only 1 way; which is exactly the situation we just described. What about 1 ball in A and 9 balls in B? There are 10 choose 1 way that this could happen (if the balls are distinguishable). What about 4 balls? That's 10 choose 4. In general, we see that the number of possible configurations for x balls in box A is $\binom{10}{x}$. Since this distribution is symmetric and $\binom{10}{x}$ is largest in the middle, the most likely configuration is 5 marbles in both box A and B. The corresponding probability is $\binom{10}{5}/2^{10} = 252/1024 \approx 0.25$. Why is it being divided by 2^{10}? Because each of the 10 marbles can be in either box, so if we sum up all the possible configurations, there are 2^{10} of them (this is just $\sum_x \binom{10}{x} = 2^{10}$). Notice that the starting split of 3 and 7 doesn't matter at all once the marbles have had time to mix.
The connection: let's suppose you can generalize this to, instead of 10 total marbles, you have multiples Avogadro's number (approximately $6 * 10^{23}$) of marbles in either boxes. If we define entropy as the number of possible arrangements, what can we conclude about processes that are in thermal equilibrium? Well, the most likely configuration would be **thermal equilibrium**. And with 10^{23} marbles, the binomial is so sharply peaked that the relative fluctuations away from the even split are of order $1/\sqrt{N} \sim 10^{-12}$ (we'll see exactly why in chapter 8). The second law is "just"

the statement that the system goes to the overwhelmingly most likely configuration and stays there.

2.4.4 Poisson

The Poisson distribution is an interesting one, and not just because it means fish in french. Let X be the random variable describing the number of times a rare event happens given that we know that on average it occurs at some rate λ. How do we even derive the probability of such a distribution? Let's start with the binomial distribution

$$\binom{N}{x} p^x (1-p)^{N-x}$$

and let $\lambda = Np$ and send the limit that N becomes very large and p becomes very very small. Then we can say that $p = \frac{\lambda}{N}$ and substitute this into the binomial

$$\implies \lim_{N\to\infty} \binom{N}{x} \left(\frac{\lambda}{N}\right)^x (1-\frac{\lambda}{N})^{N-x}$$

We can expand the definition of the choose function

$$= \lim_{N\to\infty} \frac{N!}{x!(N-x)!} \left(\frac{\lambda}{N}\right)^x \left(1-\frac{\lambda}{N}\right)^{N-x}$$

Notice that there is a λ^x term that we can just pull out of the limit because the limit only depends of N, and same with $x!$. We can also split up the $(1-\frac{\lambda}{N})^{N-x}$ term and doing so leads us to

$$= \frac{\lambda^x}{x!} \lim_{N\to\infty} \frac{N!}{(N-x)!} \left(\frac{1}{N}\right)^x \left(1-\frac{\lambda}{N}\right)^N \left(1-\frac{\lambda}{N}\right)^{-x}$$

The last term $\left(1-\frac{\lambda}{N}\right)^{-x}$ goes to 1 in the limit. Same with the $\frac{N!}{(N-x)!}\left(\frac{1}{N}\right)^x$ term. This is true because if we expand out $N!$, it becomes $N(N-1)(N-2)...$, and how many of these terms do we see? Well, we see x terms, and this is because the remaining $N-x$ terms gets canceled out in the denominator by $(N-x)!$. Thus, we have $N(N-1)\cdots(N-x+1)/N^x = 1\cdot(1-\frac{1}{N})\cdots(1-\frac{x-1}{N})$, a fixed number of factors that each go to 1. All that we have left is

$$\frac{\lambda^x}{x!} \lim_{N\to\infty} \left(1-\frac{\lambda}{N}\right)^N$$

But we know exactly what this is from elementary calculus; this is just the definition of $e^{-\lambda}$! Thus, the probability for the Poisson random variable is

$$\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} \tag{2.14}$$

Where, once again, λ is generally interpreted as the rate in which the rare event occurs.

Problem 5.

Suppose that a radioactive isotope has a decay rate of 2 (unit/second). What is the probability that in 4 seconds, it will have decayed 5 times?
Solution:
We begin by finding our λ. It might be tempting to say that λ is 2, but the problem asks to find the probability in 4 seconds. Thus, in 4 seconds, on average, how often does the particle decay? By dimensional analysis this will be $\lambda = 8$. Now that we have our λ, we plug the numbers right into the Poisson distribution to get a probability of

$$\frac{e^{-8}8^5}{5!} \approx .092$$

2.4.5 Hyper-Geometric

Once again, don't be fooled! This is more related to the binomial than it is to the geometric. There will be a lot of parameters to keep track of so bear with me here. Let X be the random variable describing x successes (drawn randomly without replacement) from a sample size of n, given that there are K possible successes in the population with size N. An example to think of this is: If you draw 5 cards from a standard 52 deck, what is the probability that 2 of those will have a suit of hearts. In this example, $N = 52$, $K = 13$ because there are 13 possible hearts, $x = 2$ (number of hearts), and $n = 5$ because we are sampling 5. Thus the probability really just comes from basic counting

$$\mathbb{P}(X = x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}} \tag{2.15}$$

Let's give this a bit of intuition. Of the K possible successes, how many can we choose? Pretty obvious, $\binom{K}{x}$. What about failures? Well, if there are 2 successes, then there must be 3 failures as in the example that we gave. Thus, how many possible failures can we have? $(N - K)$. How many failures do we have in the sample? $(n - x)$. Thus how many possible ways can we choose a failure? $\binom{N-K}{n-x}$. What's the total number of ways we can draw n

samples? $\binom{N}{n}$. For the problem that we set up, the answer would be $\binom{13}{2}\binom{39}{3}/\binom{52}{5} \approx 0.27$. If the population is huge compared to the sample, drawing without replacement is almost the same as drawing with replacement, and the hypergeometric turns into the binomial with $p = K/N$.

2.4.6 Uniform

Up until now, we've talked about discrete random variables; but the most interesting random variables are typically (absolutely) continuous. Thus, we will encounter our most basic example here: the continuous uniform. As we mentioned before, the usual notion of $\mathbb{P}(X = x)$ doesn't work anymore, for the same reason that if I asked you to count the number of red balls in a bucket, you could do that but if I asked you to count the number of water in the bucket you would look at me funny. In a similar manner, the object that encodes the information is the density. So when we say the uniform distribution, we really just mean the uniform density on some interval $[a, b]$. This is given by

$$f_X = \frac{1}{b-a} \tag{2.16}$$

Where $\frac{1}{b-a}$ is there to ensure normalization to unity

$$\int_a^b \frac{1}{b-a} = 1$$

If we wanted to talk about probabilities on the uniform, we would use the Lebesgue measure λ (not to be confused with the Poisson λ). For example, if you knew I was in the office between 12pm and 5pm (hours 12 to 17), then the probability of seeing me between 2pm and 4pm is $\frac{\lambda([14,16])}{\lambda([12,17])} = 2/5$.

2.4.7 Exponential

The exponential distribution is, in my opinion, the most interesting distribution. This is because its interpretation is a bit surprising, and it's not something that you see all the time. The premise is this: suppose that the radioactive isotope from the Poisson problem with $\lambda = 8$ had JUST decayed, how long do we have to wait until the next decay? Notice that this is no longer a discrete random variable because time is a continuous parameter. If we want to know the number of Poisson events in a time t, then that follows the Poisson distribution

$$P(N(t) = x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}$$

Where $N(t)$ is the number of events in time t, and the λt is there for the same reason that we had 8 for λ in problem 5. Let T be the random variable that signifies the time until the first event. Since nothing has happened, we need to start with the probability of no event occurring , which is just

$$\mathbb{P}(N(t) = 0) = e^{-\lambda t}$$

Thus, $P(t \leq T) = P(N(t) = 0) = e^{-\lambda t}$ because if T is the waiting time until the first event, then $t \leq T$ means that no events occurred on the interval $[0, t]$, which has a probability distribution of $e^{-\lambda t}$. If want to find the probability density function for this random variable, we have to first find the CDF so that we can take the derivative. This is true because by definition of the CDF

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)dt$$

Then if we take the derivative of the CDF, we will get back the density by fundamental theorem of calculus. For our case above, the we need to find $P(T \leq t)$, which can be computed by using the complement rule

$$P(T \leq t) = 1 - P(T \geq t)$$

But we had already computed $P(T \geq t)$, it's just $e^{-\lambda t}$. So then the CDF is

$$F_T(t) = 1 - e^{-\lambda t}$$

and thus the density is

$$f_T(t) = \frac{d}{dt}\left(1 - e^{-\lambda t}\right) = \lambda e^{-\lambda t}, \qquad t \geq 0 \tag{2.17}$$

An interesting question is raised: why did we start with no event occurring? Why couldn't we have started with some number of events already occurring? This has to do with the memoryless property of the exponential distribution, which is just a statement that $\mathbb{P}(T > s + t|T > s) = \mathbb{P}(T > t)$. We can check it in one line: $\mathbb{P}(T > s + t|T > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. Basically all that it means is that the exponential distribution doesn't really care or remember what happened before it. Since a Poisson process has no memory either (what happens in disjoint time intervals is independent), we can start the clock whenever we want, and the waiting time to the next event is always exponential.

2.4.8 Gaussian

Now we have the big boy. There are many ways to derive the Gaussian, the most common way as being the limiting case of the binomial. You could make an entire chapter on this

distribution, but we do not have time. I will leave it as an exercise for you to derive the Gaussian distribution from the binomial distribution. Don't worry! I will work with you through it. Instead, what would be more instructive would be to look at the density of the Gaussian and do some Gaussian integrals. Although in probability and statistics, doing Gaussian integrals is not all too common, in physics they show up all the time (especially in quantum field theory). If X is a Gaussian random variable with mean μ and variance σ^2, then the Gaussian density is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \tag{2.18}$$

Problem 6.

Find the integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

Solution:

It's illustrative to work through this because later on, it will be useful when we take Fourier transforms. This integral requires us to do a trick that is not standard procedure. We are going to square the integral. That is, let I be the value of the integral. Then

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx$$

$$\implies I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2$$

The squared term can be thought of as 2 integrals that are independent from each other. Then we can introduce an new independent variable, say y

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2} dy\right)$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2} e^{-y^2} dxdy$$

Now we can do a variable substitution into polar coordinates. Let $r^2 = x^2 + y^2$. The Jacobian is $rdrd\theta$

$$\int_0^{2\pi}\int_0^{\infty} re^{-r^2} drd\theta$$

The radial integral is a u-substitution with $u = r^2$, which gives $\int_0^\infty re^{-r^2}dr = \frac{1}{2}$, and the angular integral gives 2π. Notice that this is the value of I^2, so $I^2 = \pi$, and the value for a single integral is just the square root of this, making the final answer $\boxed{\sqrt{\pi}.}$ Substituting $x \to (x-\mu)/\sqrt{2\sigma^2}$ then shows that the Gaussian density above integrates to 1, which is where the $\sqrt{2\pi\sigma^2}$ comes from.

2.5 Joint, Marginal, and Conditional Distributions

We spent a good deal of time on distributions of a single random variable. What happens now when we have multiple random variables? How do we classify them? Unsurprisingly, it's the exact same thing as the single variable case. Suppose X and Y are discrete random variables. Then the probability of getting both $X = x$ and $Y = y$ is given through their **joint probability mass function**

$$\mathbb{P}(X = x \text{ and } Y = y) \tag{2.19}$$

If X and Y were independent, then their joint PMF would split into a product of single variable PMFs. For example, consider the joint PMF in table 2.1. What is the probability

	$Y=0$	$Y=1$	
$X=0$	1/9	2/9	1/3
$X=1$	2/9	4/9	2/3
	1/3	2/3	1

Table 2.1: Example table for the joint distribution of X and Y. The last row and column are the marginals.

$\mathbb{P}(X = 1, Y = 1)$? Well, by examination of the table, it is the second column second row, which is just $\frac{4}{9}$. Notice that every entry is the product of its row and column totals, so in this example X and Y are independent. What if our random variables were absolutely continuous? Then we would have all the rules of an absolutely continuous single variable but now with 2 of everything instead of 1. This means that the density is a function of both X and Y, and we must perform multiple integrals to get the probabilities. To do a very basic example, let $f_{X,Y}$ be the joint density

$$f_{X,Y} = \begin{cases} x + y & 0 \le x, y \le 1 \\ 0 & \text{else} \end{cases}$$

We can go through the full motions to see that $f_{X,Y}$ satisfies all the properties of a distribution. Namely, the density is non-negative and the joint CDF is nondecreasing in each argument. Blah blah blah. The boring stuff. We just want to know if this integrates to 1 so we can just call it a day. We will need to do a double integral over the square domain

$$\int_0^1 \int_0^1 x + y \ dxdy$$

This one is pretty easy to do

$$= \int_0^1 \frac{x^2}{2} + xy \Big|_0^1 \ dy = \int_0^1 \frac{1}{2} + y \ dy = \frac{y}{2} + \frac{y^2}{2} \Big|_0^1 = 1$$

Voila! The integral evaluates to 1. It's a very straightforward and almost brain dead example, but it's just to demonstrate that everything we did in single variable generalizes to n variables. That is just a long winded way of saying: get used to doing multiple integrals.

Suppose now we have a joint distribution of 2 random variables $f_{X,Y}$ and we want to find just the probability of Y given by f_Y. We can do this quite easily by just summing over the probabilities of X

$$\mathbb{P}(Y = y) = \sum_x \mathbb{P}(X = x, Y = y)$$

In the case for absolutely continuous random variables, the sum turns into an integral over the densities

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \ dx \tag{2.20}$$

The resulting probability mass/density is called the **marginal PMF/PDF**. Let's do an example in the discrete case. Consider the joint PMF in table 2.1. What is the marginal distribution for $P(X = x)$? Well, all you have to do is just sum across each row of the table. More explicitly, the probability of $P(X = 0) = P(X = 0, Y = 0) + P(X = 0, Y = 1) = 1/3$. For the remainder, $P(X = 1) = P(X = 1, Y = 0) + P(X = 1, Y = 1) = 2/3$. Notice that these probabilities do indeed sum to 1! For a continuous marginal distribution, let (X, Y) have density

$$f(x, y) = \begin{cases} \frac{21}{4} x^2 y & \text{if } x^2 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

then

$$f_X(x) = \int f(x, y) \, dy = \frac{21}{4} x^2 \int_{x^2}^1 y \, dy = \frac{21}{8} x^2 (1 - x^4),$$
$$\text{for } -1 \leq x \leq 1, \quad \text{and } f_X(x) = 0 \text{ otherwise.}$$

Lastly, for conditional probabilities $\mathbb{P}(X|Y)$, we just use the definition of a conditional probability back in chapter 1 to get

$$\mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \tag{2.21}$$

As for the continuous case, you just substitute all the probability functions for densities, $f_{X|Y}(x|y) = f_{X,Y}(x, y)/f_Y(y)$, and then call it a day. (Strictly speaking this is a bit cheeky, since $\mathbb{P}(Y = y) = 0$ for a continuous Y; making it honest is one of the jobs of the conditional expectation in chapter 3.) The numerator is the joint and the denominator is the marginal. I am going through this section rather quickly because to be honest, there really isn't much to say that wouldn't be redundant. I think you get the point.

2.6 Transformations Of Random Variables

This is probably one of the most important topics that is glossed over in a lot of probability textbooks and courses. We derived a lot of distributions for various random variables, but we didn't discuss much about functions of random variables. Remember: compositions of measurable functions are still measurable functions. As such, there is an infinite number of random variables left untouched. We will talk about how to find them here.

2.6.1 Single Variable Transformations

What is our goal here? Our goal is to understand how to find the distributions of $f(X)$ where f is some measurable function of the random variable X. For example, let $X \sim \text{Uni}(0, 1)$. What is the distribution of X^2? It might be really tempting to say that it would just be the square of the uniform distribution itself; but this is NOT true. Why is this not true? Consider this thought: if you roll 2 uniformly distributed 6-sided dice and added their individual outcomes, will this also be uniformly distributed? That is, will getting a sum of 2 be as likely as a sum of 5? Well, no. There is only one way to get 2: both die must give 1. But there are more ways to get 5: 1 and 4, 2, and 3, so and so forth. So from this simple thought experiment, we see that under even just a basic transformation such as addition, random variables may not preserve the structure of their distributions. Where do we begin? The cleanest thing to transform is the CDF, because it's just a probability of an event

$$F_{f(X)} = \mathbb{P}(f(X) \leq x) \tag{2.22}$$

If f is increasing, then $f(X) \leq x$ is the same event as $X \leq f^{-1}(x)$, so

$$F_{f(X)}(x) = \mathbb{P}(X \leq f^{-1}(x)) = F_X(f^{-1}(x)) \tag{2.23}$$

(if f is not monotone, like x^2, we have to be more careful about which X values get sent below x, as we'll see in problem 8). For a discrete random variables, finding the new distribution is pretty easy; you just substitute the function of X into the probability and then solve. Here is a basic example. Let $\mathbb{P}(X = -1) = \mathbb{P}(X = 1) = 1/4$ and $\mathbb{P}(X = 0) = 1/2$. Find the distribution of X^2. Since this is a discrete random variable, we just find the probability of $\mathbb{P}(X^2 = x)$. Let's do this out: $\mathbb{P}(X^2 = 1) = \mathbb{P}(X = \pm 1)$, and the probability of each of $X = 1$ and $X = -1$ is $1/4$ by definition. Then $\mathbb{P}(X^2 = 1) = \mathbb{P}(X = 1) + \mathbb{P}(X = -1) = 1/2$. Lastly, for $\mathbb{P}(X^2 = 0) = 1/2$ which just stays the same. Thus, $\mathbb{P}(X^2 = 1) = 1/2$ and $\mathbb{P}(X^2 = 0) = 1/2$.

Problem 7.

Let X be a Poisson random variable with parameter λ. Find the distribution of X^2.
Solution:
Pretty straightforward; take the fact that the Poisson distribution follows

$$\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

Since $X \geq 0$, the event $X^2 = x$ is the same as $X = \sqrt{x}$, so

$$\mathbb{P}(X = \sqrt{x}) = \frac{e^{-\lambda}\lambda^{\sqrt{x}}}{\sqrt{x}!}$$

Note that this is only for $x \in \{0, 1, 4, 9, \dots\}$ (perfect squares), and it's 0 otherwise. The probabilities don't change at all; they just get moved to new locations. A plot of the distribution and its transformed version is in figure 2.5.

Now we turn our heads to absolutely continuous random variables. That's what we're all here to do right? For this, it's not as simple as just plugging in $f^{-1}(x)$ unfortunately. First let us find the CDF

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)dt$$

It's clear as when we did the exponential distribution that to get back the density we just take a derivative. But before we can do that, we must first find the transformation (for increasing f)

$$F_{f(X)}(x) = \mathbb{P}(f(X) \leq x) = \mathbb{P}(X \leq f^{-1}(x))$$

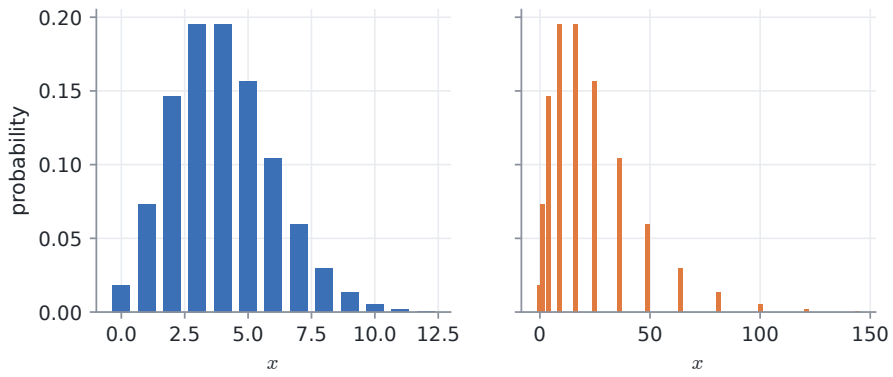


Figure 2.5: The distribution of a Poisson random variable X with $\lambda = 4$ (left) and of X^2 (right). Same heights, new locations.

Then the CDF becomes

$$F_{f(X)}(x) = \int_{-\infty}^{f^{-1}(x)} f_X(t)dt$$

Then we take the derivative. But DO NOT FORGET THE CHAIN RULE

$$f_{f(X)}(x) = f_X(f^{-1}(x)) \left| \frac{d}{dx} f^{-1}(x) \right| \tag{2.24}$$

The absolute value is there because if f is decreasing, the inequality flips, $F_{f(X)}(x) = 1 - F_X(f^{-1}(x))$, and the derivative picks up a minus sign that exactly cancels the negative slope of f^{-1}. Physicists will recognize this as conservation of probability: $f_X(x)\,|dx| = f_Y(y)\,|dy|$, the probability in a little interval doesn't care what you call the coordinate. Perhaps I should've used ρ as density as to not confuse the transformation f with the density f_X.

Problem 8.

Let X be a standard normal $N(0, 1)$. Find the distribution of X^2

Solution:

For $x > 0$,

$$F_{X^2}(x) = \mathbb{P}(X^2 \leq x)$$

Notice that when we take the square root, we get two branches, because both $\sqrt{x}$ and $-\sqrt{x}$ square to x

$$= \mathbb{P}(-\sqrt{x} \leq X \leq \sqrt{x})$$

But all that this does is split the CDF into 2

$$= F_X(\sqrt{x}) - F_X(-\sqrt{x})$$

Then to find the density, we just have to take a derivative

$$\frac{d}{dx}\left(F_X(\sqrt{x}) - F_X(-\sqrt{x})\right) = \frac{1}{2\sqrt{x}} f_X(\sqrt{x}) - \left(-\frac{1}{2\sqrt{x}} f_X(-\sqrt{x})\right) = \frac{1}{\sqrt{x}} f_X(\sqrt{x})$$

And we know what the density for the standard normal is. We can just plug it right in and get

$$f_{X^2}(x) = \frac{1}{\sqrt{2\pi x}} e^{-x/2}, \qquad x > 0 \tag{2.25}$$

Which has the famous name of the **chi squared** or χ^2 distribution (with one degree of freedom). The $1/\sqrt{x}$ blow up at 0 is real: lots of standard normals are close to 0, and squaring squashes all of them even closer.

2.6.2 Multiple Variable Transformations

Understanding how single random variables transform makes understanding how multiple random variables transform a lot easier. But even though it is a lot easier, there are still some subtleties that we must tread over carefully. How do we begin? Let X and Y be independent and identically distributed (i.i.d.) random variables. How do we find the distribution of, say, $f(X, Y) = X + Y$? The easiest thing to do is just let $Z = X + Y$ and then find the CDF $F_Z(z) = \mathbb{P}(X + Y \leq z)$. For example, let X, Y be i.i.d. uniform on $[0, 1]$. Find the distribution of $Z = X + Y$. This is now not as obvious, because the answer has a different formula on different pieces of the domain that we have to account for. Let's see why. We begin by asking ourselves: what is the range of Z? This part is easy, it's just $[0, 2]$. We have to consider 2 cases, and we will explain why there must be 2 afterwards. Case 1 is if $z \leq 1$. In which case,

$$\mathbb{P}(X + Y \leq z \big| X = x) f_X(x) = \mathbb{P}(Y \leq z - x) \cdot 1 = z - x$$

for $0 \leq x \leq z$, with the last equality being true due to Y being uniform (and independent of X, so conditioning on X doesn't change it). If $x > z$, the probability is 0, since $Y \geq 0$. Then, what we can do is integrate over all possible values of x from 0 to z (since x can take on any of these values). This statement is nothing more than just law of total probability, with the density f_X being just 1. The condition here is if x wants to take on whatever value

it wants, then Y must follow the condition that $Y \leq z - x$ to satisfy $X + Y = Z$. Therefore

$$F_Z(z) = \int_0^z (z - x)\, dx = \frac{z^2}{2}$$

We have case 1, what about case 2: $1 \leq z \leq 2$? We will do the exact same thing, but now we have to be careful, since $\mathbb{P}(Y \leq z - x)$ can't be bigger than 1

$$\mathbb{P}(Y \leq z - x) = \begin{cases} 1 & 0 \leq x \leq z - 1 \\ z - x & z - 1 \leq x \leq 1 \end{cases}$$

If x is smaller than $z - 1$, then even the largest possible Y keeps the sum below z, so it's guaranteed. Splitting the integral over x into the two pieces,

$$F_Z(z) = \int_0^{z-1} 1\, dx + \int_{z-1}^1 (z - x)\, dx = (z - 1) + \left(z - \frac{z^2}{2}\right) = 2z - \frac{z^2}{2} - 1$$

Our CDF's piecewise function is

$$F_Z(z) = \begin{cases} 0 & z \leq 0 \\ \frac{z^2}{2} & 0 \leq z \leq 1 \\ 2z - \frac{z^2}{2} - 1 & 1 \leq z \leq 2 \\ 1 & z \geq 2 \end{cases}$$

Now we can take the derivative, in which case we can get the density by taking a derivative

$$f_Z(z) = \begin{cases} z & 0 \leq z \leq 1 \\ 2 - z & 1 \leq z \leq 2 \\ 0 & \text{else} \end{cases} \tag{2.26}$$

This is the triangle distribution: getting a sum near 1 is the most likely, just like getting 7 with two dice. Notice that the density itself is continuous at $z = 1$ but it has a kink there, which is the remnant of the two cases. The point of this problem is to illustrate that while the principles are still the same as that of the single variable case, we have to take some care with the domain. Later on in the notes, we will use the convolution and the characteristic function to find sums of random variables, which provide a more "systematic" approach.

Next, we will provide a tutorial on how to find the densities of various functions on random variables; namely the minimum and maximum functions. Before we can begin, we should check whether or not $\min(X_1, X_2)$ and $\max(X_1, X_2)$ are measurable functions. Recall that

the definition of a measurable function is that $\{\omega \big| f(\omega) < a\} \in \mathcal{F}$ for all a. The maximum is less than a exactly when both are, and the minimum is less than a exactly when at least one is, so

$$\{\max(X_1, X_2) < a\} = \{X_1 < a\} \cap \{X_2 < a\},$$
$$\{\min(X_1, X_2) < a\} = \{X_1 < a\} \cup \{X_2 < a\}$$

which are both measurable. Now we must turn our heads on how to find the densities $\min(X_1, X_2)$. For sake of example, let X_1 and X_2 be i.i.d. uniform(0,1). Then $\mathbb{P}(\max(X_1, X_2) \leq z)$ means that both X_1 and X_2 must be $\leq z$ (because if one of them is less than z and that one is the larger of the other, then they both must be less than z). This just becomes $\mathbb{P}(X_1 \leq z \text{ and } X_2 \leq z)$. Using independence, we know that this is just $\mathbb{P}(X_1 \leq z)\mathbb{P}(X_2 \leq z)$. Since they are identically uniform, we can conclude that their CDF is

$$F_Z(x) = \begin{cases} 0 & z < 0 \\ z^2 & 0 \leq z \leq 1 \\ 1 & \text{else} \end{cases}$$

We can then conclude that the density is

$$\begin{cases} 2z & 0 \leq z \leq 1 \\ 0 & \text{else} \end{cases} \tag{2.27}$$

The case for $\min(X_1, X_2)$ should be fairly obvious now. If we suppose that X_1 and X_2 are the usual i.i.d. uniforms on $[0, 1]$, then $\mathbb{P}(\min(X_1, X_2) \leq z)$ is the same as $1-\mathbb{P}(\min(X_1, X_2) > z)$ by how complements work. Now, if the minimum between the X_1 and X_2 is greater than z, then they both must be larger than z (because if X_1 is larger than z and smaller X_2, then necessarily X_2 must also be larger than z. This is the same thing as "If I am older than you and my dad is older than me, then my dad must also be older than you"). Then this becomes $\mathbb{P}(X_1 > z \text{ and } X_2 > z)$. By independence, this becomes $\mathbb{P}(X_1 > z)\mathbb{P}(X_2 > z)$. With X_1 and X_2 being identical uniforms on the unit interval, the probability of $\mathbb{P}(X_i > z) = 1-\mathbb{P}(X_i \leq z)$. This makes

$$\mathbb{P}(X_1 > z)\mathbb{P}(X_2 > z) = (1 - z)^2$$

And therefore the CDF is

$$F_Z(x) = \begin{cases} 0 & z < 0 \\ 1 - (1-z)^2 & 0 \leq z \leq 1 \\ 1 & \text{else} \end{cases}$$

We can find the density pretty easily as

$$f_Z(z) = \begin{cases} 2(1-z) & 0 \leq z \leq 1 \\ 0 & \text{else} \end{cases} \tag{2.28}$$

Let us try to do a problem to get the hang of things

Problem 9.

Let X_1 and X_2 be i.i.d. exponential(λ). Find the density of $X_1 + X_2$

Solution:

It seems daunting at first but we will show that it's not too horrible if we do this systematically. We need to find the CDF

$$\mathbb{P}(X_1 + X_2 \leq z)$$

We can use our trick of

$$F_Z(z) = \int \mathbb{P}(X_1 + X_2 \leq z \big| X_2 = x) f_{X_2}(x)\, dx$$

Which again is just the law of total probability. Given $X_2 = x$, the event is $X_1 \leq z - x$, which has probability $F_{X_1}(z-x)$. Since both are non-negative, we only need $0 \leq x \leq z$

$$F_Z(z) = \int_0^z F_{X_1}(z-x) f_{X_2}(x)\, dx \tag{2.29}$$

If we differentiate under the integral, we get $f_Z(z) = \int_0^z f_{X_1}(z-x) f_{X_2}(x)\, dx$, and the astute reader might recognize this as the **convolution** of the 2 densities. But I want to spend time talking about it in a later chapter. Now, we need

$$F_{X_1}(z-x) = \int_0^{z-x} \lambda e^{-\lambda t} dt = 1 - e^{-\lambda(z-x)}$$

Then equation (2.29) becomes

$$\int_0^z \left(1 - e^{-\lambda(z-x)}\right)\left(\lambda e^{-\lambda x}\right)\, dx = \lambda \int_0^z e^{-\lambda x} - e^{-\lambda z}\, dx$$

The first term integrates to $1 - e^{-\lambda z}$ and the second is a constant, which gives $\lambda z e^{-\lambda z}$

$$F_Z(z) = 1 - e^{-\lambda z} - \lambda z e^{-\lambda z} = \mathrm{e}^{-z\lambda}\left(\mathrm{e}^{z\lambda} - z\lambda - 1\right)$$

Taking the derivative with respect to z gives us the density

$$f_Z(z) = \lambda^2 z \mathrm{e}^{-\lambda z} \tag{2.30}$$

And this distribution is called the **gamma distribution** (with shape 2 and rate λ). It makes sense: the time until the second Poisson event is the sum of two independent exponential waiting times. If you keep going, the sum of n of them has density $\lambda^n z^{n-1} e^{-\lambda z}/(n-1)!$.

2.7 Problems

1. Let X be such that $\mathbb{P}(X = 2) = \mathbb{P}(X = 3) = \frac{1}{10}$ and $\mathbb{P}(X = 5) = \frac{8}{10}$. Plot the cumulative distribution function $F(x)$. Use F_X to find $\mathbb{P}(2 < X \leq 4.8)$ and $\mathbb{P}(2 \leq X \leq 4.8)$.

2. Let $X \sim N(0, 1)$ and let $Y = e^X$.

(a) Find the PDF for Y. Plot it.

(b) **(Computational)** Generate a vector $\mathbf{x} = (x_1, \ldots, x_{10,000})$ consisting of 10,000 random standard Normal variables. Let $\mathbf{y} = (y_1, \ldots, y_{10,000})$, where $y_i = e^{x_i}$. Draw a histogram of $\mathbf{y}$ and compare it to the PDF you found in part (a).

3. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$, and assume that X and Y are independent. Show that the distribution of X given that $X + Y = n$ is $\text{Binomial}(n, \pi)$, where $\pi = \frac{\lambda}{\lambda+\mu}$. (**hint:** use the law of total probability for discrete random variables)

4. Let $X, Y \sim \text{Uniform}(0, 1)$ be independent. Find the PDF for $X - Y$ and X/Y.

5. Let $X_1, \ldots, X_n \sim \text{Exp}(\beta)$ be i.i.d. Let $Y = \max\{X_1, \ldots, X_n\}$. Find the PDF of Y.

Hint: $Y \leq y$ if and only if $X_i \leq y$ for $i = 1, \ldots, n$.

6. Show that:

(a) If X is a random variable, then so is $|X|$.

(b) The converse does not necessarily hold.

7. Suppose that X is an integer-valued random variable, and let $m \in \mathbb{N}$. Show that:

$$\sum_{n=-\infty}^{\infty} \mathbb{P}(n < X \leq n + m) = m.$$

8. Let X and Y be random variables and suppose that $A \in \mathcal{F}$. Prove that $Z = X\mathbf{1}_A + Y\mathbf{1}_{A^c}$ is a random variable.

3

Expectation

"Something Something Expectation"

–

We will now spend a good amount of time discussing expectation and variance of random variables. This is because these are the principal building blocks of statistics.

3.1 Expectation And Variance

The expected value of a random variable is the mean of its distribution. For those that are physically intuitive, we can think of the distribution as the mass of some object. The expected value, then, is the center of mass. We write the notation for the expected value as

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x) \tag{3.1}$$

This is true for discrete random variables. For the case of continuous random variables, we replace the sum with an integral and replace the probability mass function with a density f_X

$$\mathbb{E}[X] = \int x f_X(x)\, dx \tag{3.2}$$

Measure theory lets us write both of these (and the mixtures and the singular ones) as one object, the Lebesgue integral of X against $\mathbb{P}$

$$\mathbb{E}[X] = \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega) \tag{3.3}$$

The construction is the one hinted at in chapter 2. For a simple random variable $X = \sum_i x_i \mathbb{I}_{A_i}$ there's no question what the average should be: $\sum_i x_i \mathbb{P}(A_i)$. For a non-negative X we approximate it from below by simple random variables $X_k \uparrow X$ (for example, round X down to the nearest multiple of 2^{-k}, capped at k) and define $\mathbb{E}[X]$ as the limit of their expectations. For a general X we split it into positive and negative parts, $X = X^+ - X^-$, and subtract, as long as at least one of them is finite. The difference with the Riemann integral is that Lebesgue chops up the *range* of X into thin horizontal slices instead of chopping up the domain, and that's what makes it work on any probability space, not just intervals. We should note that the expectation operator is a linear operator. That is, it satisfies

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y] \tag{3.4}$$

For some constants a and b with X and Y being random variables. We will leave this statement as a basic proof (it's really obvious, you can just see it from the definition). Notice that we don't need X and Y to be independent for this to hold. Let's do an example for the case of the binomial distribution

$$\mathbb{P}(X = x) = \binom{N}{x} p^x (1-p)^{N-x}$$

Then we can find the average value of the distribution from (3.1) since it is discrete

$$\mathbb{E}[X] = \sum_{x=0}^{N} \binom{N}{x} x p^x (1-p)^{N-x}$$

This is a bit daunting because there is no systematic way to do sums. We will have to rely on this trick. Let's pretend for a moment that p and $q = 1 - p$ are two independent variables, and only set $q = 1 - p$ at the very end. Then

$$p\frac{\partial}{\partial p} p^x = x p^x$$

Which allows us to substitute the xp^x term in the previous equation for the expectation

$$\mathbb{E}[X] = \sum_{x=0}^{N} \binom{N}{x} \left(p\frac{\partial}{\partial p} p^x \right) q^{N-x}$$

Since q is being treated as independent of p, the derivative only sees p^x, and the sum is over x, so we can pull the derivative out front as well as the lonely p term

$$= p\frac{\partial}{\partial p}\sum_{x=0}^{N}\binom{N}{x}p^x q^{N-x}$$

We can see why this distribution has the name binomial, because this is exactly the binomial expansion

$$= p\frac{\partial}{\partial p}\left(p+q\right)^N = Np\left(p+q\right)^{N-1} = Np$$

where only at the very end we used $p+q=1$. The mathematically inquisitive reader might be queasy at the stunt we just pulled, but it's completely legal: the identity $\sum_x \binom{N}{x}xp^x q^{N-x} = Np(p+q)^{N-1}$ is true for all p and q, so it's certainly true on the line $q = 1-p$. What we must NOT do is set $q = 1-p$ before differentiating. Thus we have come to the conclusion that $\mathbb{E}[X] = Np$ for the binomial random variable.

What about an example with a continuous random variable? In fact, it seems as though continuous random variables are easier to work with, since integrals are usually easier than sums (and we have the nifty tool of Mathematica). A very simple example would be the uniform distribution on $[a,b]$, which has the density $f_X = \frac{1}{b-a}$

$$\mathbb{E}[X] = \int_a^b \frac{x}{b-a}dx = \frac{1}{2}\frac{b^2-a^2}{b-a} = \frac{1}{2}\left(b+a\right) \tag{3.5}$$

Notice that we integrated over the domain of the distribution.

Now that we have the expectation, what can we say about the variance? The variance is a measure of the spread of the distribution. A more technical exposition would say that the variance is the average squared distance of every point from the average

$$\mathrm{Var}[X] = \mathbb{E}\left[\left(X-\mathbb{E}[X]\right)^2\right] \tag{3.6}$$

We can simplify the calculation a bit by expanding the square

$$= \mathbb{E}\left[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2\right]$$

remember that $\mathbb{E}[X]$ is just a number, and so $\mathbb{E}[X]^2$ must also be a number. This means we can use the linearity of the expectation

$$= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]^2$$

$$= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \tag{3.7}$$

The object $\mathbb{E}[X^2]$ is called the **second moment**, where the n-th moment is $\mathbb{E}[X^n]$. The question becomes: how do we evaluate the second moment? We could find the distribution of X^2 using chapter 2 and then take its mean, but there's a shortcut (sometimes called the law of the unconscious statistician, because people use it without thinking): for a function f of the random variable X,

$$\mathbb{E}[f(X)] = \sum_x f(x)\mathbb{P}(X = x) \tag{3.8}$$

The sum is replaced by an integral for continuous random variables. It's clear, then, that the second moment can be calculated by $\sum x^2\mathbb{P}(X = x)$. Let's see how to do this for the binomial. Begin with

$$\mathbb{E}[X^2] = \sum_{x=0}^{N} \binom{N}{x} x^2 p^x q^{N-x}$$

We can again use another trick

$$p^2 \frac{\partial^2}{\partial p^2} p^x = x(x-1)p^x$$

We can isolate the quadratic term by introducing

$$x^2 p^x = x(x-1)p^x + xp^x$$

We will see how this becomes useful in one second. For now, let us just substitute in the second derivative to second moment calculation

$$\implies \sum_{x=0}^{N} \binom{N}{x} p^2 \frac{\partial^2}{\partial p^2} p^x q^{N-x}$$

But notice that we didn't actually calculate x^2p^x, we calculated $x(x-1)p^x$. We need to add back xp^x

$$\mathbb{E}[X^2] = \sum_{x=0}^{N} \binom{N}{x} \left(p^2 \frac{\partial^2}{\partial p^2} p^x q^{N-x} + xp^x q^{N-x} \right)$$

We can break this sum into 2 with the second part of the sum being the definition of the average so we can just substitute that in

$$= \left(p^2 \frac{\partial^2}{\partial p^2} \sum_{x=0}^{N} \binom{N}{x} p^x q^{N-x} \right) + Np$$

The term on the left can be evaluated using the same trick by setting the binomial expansion as $(p+q)^N$

$$= p^2 \frac{\partial^2}{\partial p^2}(p+q)^N + Np = N(N-1)p^2 + Np$$

Now that we have the second moment, we can calculate the variance using $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$$\text{Var}[X] = N(N-1)p^2 + Np - N^2p^2 = Np - Np^2$$

We can simplify this a bit further to get

$$\text{Var}[X] = Npq \tag{3.9}$$

Can we do the same for the uniform distribution? Yes we can! In fact, it's a bit easier. We start with finding the second moment

$$\mathbb{E}[X^2] = \int_a^b \frac{x^2}{b-a}\,dx = \frac{1}{3}\frac{b^3-a^3}{b-a} = \frac{1}{3}\left(b^2+ab+a^2\right)$$

We can then find the variance by subtracting the square of the average, $\frac{1}{4}(a+b)^2$. After all the algebra (it's $\frac{1}{12}(4b^2+4ab+4a^2-3a^2-6ab-3b^2)$) we find that the variance for the uniform is

$$\text{Var}[X] = \frac{1}{12}(b-a)^2 \tag{3.10}$$

Problem 1.

Suppose you are gambling with a fair coin. You start with 10 dollars. If the coin lands on heads, you multiply your current balance by 1.8 (e.g. now you have 18 dollars). If the coin lands on tails, you divide your current balance by 2 (e.g. now you have 5 dollars). Are you expected to win?

Solution:

Let M be the multiplier on your balance in one round, so $M = 1.8$ or $M = 0.5$, each with probability $1/2$. Then

$$\mathbb{E}[M] = \frac{1}{2}\cdot 1.8 + \frac{1}{2}\cdot 0.5 = 1.15$$

Since the rounds are independent, your balance after n rounds is $10M_1M_2\cdots M_n$, and the expectation of a product of independent random variables is the product of the expectations (that's problem 1 at the end of this chapter), so $\mathbb{E}[\text{balance}] = 10(1.15)^n$.

On average, you win, and you win exponentially fast!
Note: Now play it and see what happens. A win followed by a loss leaves you with $1.8 \times 0.5 = 0.9$ of what you had. So the typical player, who wins about half the time and loses about half the time, gets multiplied by 0.9 every two rounds and goes broke. The average is being propped up by a tiny number of absurdly lucky players. We'll make this precise in chapter 7, where the law of large numbers tells us that almost every player's balance goes to zero, even though the expectation goes to infinity. Hold on to this one; it's a good warning about what an expectation does and doesn't tell you.

3.2 Convergence Theorems and Fatou's Lemma

The problem that we are going to try to tackle now is the question: if a sequence of random variables converges, does that mean that the expectations converge to the expectation of the limit? That is, if

$$\lim_{n\to\infty} X_n(\omega) = X(\omega) \implies \lim_{n\to\infty} \mathbb{E}[X_n(\omega)] = \mathbb{E}[X(\omega)]$$

The answer to this, whether surprising or not, is **no**. The example that my probability professor loves to use is the skinny tall rectangles. That is, consider a random variable of the form

$$X_n(\omega) = \begin{cases} n & \omega \in [0, \frac{1}{n}] \\ 0 & \text{else} \end{cases} \tag{3.11}$$

where ω is a uniform random point of $[0, 1]$ (so $\mathbb{P}$ is the Lebesgue measure). Each X_n is a tall skinny rectangle with area exactly 1. You can imagine that as n becomes increasingly large, we just have something that looks like a delta function at 0. What is the expectation of each term in the sequence? It is

$$\mathbb{E}[X_n] = n\frac{1}{n} + 0(1 - \frac{1}{n}) = 1$$

So the limit of the expectations is 1. What if we find the expectation of the limit? Pick any $\omega > 0$. As soon as $n > 1/\omega$, the rectangle has moved past ω, and $X_n(\omega) = 0$ from then on. So $X_n \to 0$ for every ω except the single point $\omega = 0$, which has probability 0 (we say $X_n \to 0$ almost surely). The expectation of the limit is the expectation of the 0 function, which is 0. But that means that the expectation of the limit does NOT equal the limit of the expectations. Where did the probability mass go? It all piled up on a set of measure 0 and then disappeared. We have to be a bit more careful, and we see that there will need to

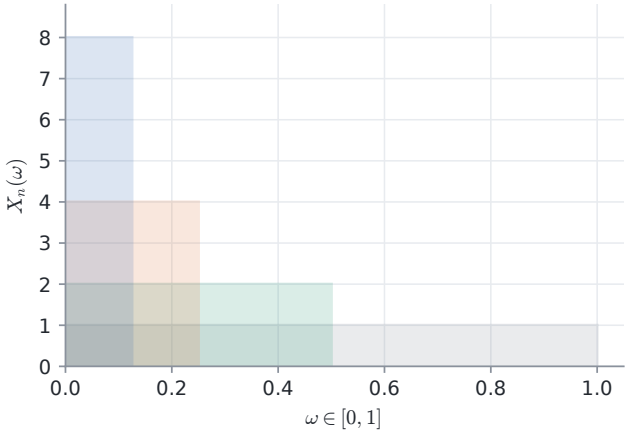


Figure 3.1: The skinny tall rectangles $X_n = n\mathbb{I}_{[0,1/n]}$ for $n = 1, 2, 4, 8$ (gray, green, orange, blue). Each rectangle has area 1, but $X_n(\omega) \to 0$ for every $\omega > 0$.

be some condition that is satisfied in order for this to work.

3.2.1 Bounded Convergence Theorem

We will have to borrow a bit from real analysis when dealing with interchange of limits of integration (because that's ultimately what an expectation is: an integral). The bounded convergence theorem gives us one such set of conditions. Suppose we have a sequence of functions $f_n(x)$ that converge to $f(x)$ in the limit that $n \to \infty$. The bounded convergence theorem tells us if the sequence is uniformly bounded (that is, if there exist a non-negative number M such that the absolutely value of every function in the sequence is less than M $|f_n(x)| \leq M \ \ \forall n$ and $\forall x$), then we may switch the limit and the integral

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n\to\infty} X_n] \tag{3.12}$$

Let's see how this is true in general. If we can work out how it is true in general, then seeing the case for expectations is pretty straightforward. Suppose that we have a sequence of functions $f_n(x)$ that converges pointwise to $f(x)$ and that every function in the sequence is (absolute valued) bounded by some number M. Define

$$E_n = \{x \in X : |f_n(x) - f(x)| \geq \epsilon\}$$

to be the set of points where f_n is still more than ϵ away from f. We would expect that the measure of this set goes to 0 because $f_n \to f$ pointwise. To see it, notice that a point that's in infinitely many of the E_n can't have $f_n(x) \to f(x)$, so $\mu(\limsup E_n) = 0$, and then by the

result of problem 8 in chapter 1 (which needs $\mu(X) < \infty$), $\limsup \mu(E_n) \leq \mu(\limsup E_n) = 0$. Let us now set up the integral

$$\int_X |f_n(x) - f(x)|\, d\mu = \int_{E_n} |f_n(x) - f(x)|\, d\mu + \int_{X \setminus E_n} |f_n(x) - f(x)|\, d\mu$$

Where we have split up the domain the integral into E_n and its complement. Let us tackle the E_n integral first. We know that

$$|f_n(x) - f(x)| \leq |f_n(x)| + |f(x)|$$

by the triangle inequality. Since each term in the sequence is bounded by M (and therefore so is the limit f), this allows us to rewrite

$$|f_n(x) - f(x)| \leq M + M = 2M$$

And so the integral becomes

$$\int_{E_n} |f_n(x) - f(x)|\, d\mu \leq 2M\mu(E_n)$$

The integral evaluates to just the measure of E_n since M is a constant. We've already discussed the fact that $\mu(E_n)$ is 0 in the limit that $n \to \infty$, so this integral is 0. This leaves the $X \setminus E_n$ integral remaining. This one is also pretty straightforward. Since $|f_n(x) - f(x)| < \epsilon$, can just take the value of the integral to just be the ϵ times the set measure

$$\int_{X \setminus E_n} |f_n(x) - f(x)|\, d\mu \leq \epsilon \cdot \mu(X)$$

Putting the two pieces together,

$$\limsup_{n\to\infty} \int_X |f_n(x) - f(x)|\, d\mu \leq 2M \limsup_{n\to\infty} \mu(E_n) + \epsilon\, \mu(X) = \epsilon\, \mu(X)$$

We have to be careful; in order for this to die as $\epsilon \to 0$, we have to make the assumption that the measure $\mu(X)$ is finite. But for the case of probabilities, $\mathbb{P}(\Omega) = 1$, so we don't need to worry! (On the whole real line with Lebesgue measure the theorem is false: $f_n = \mathbb{I}_{[n,n+1]}$ is bounded by 1 and goes to 0 pointwise, but always has integral 1.) Since ϵ was arbitrary, Which leads us to conclude that

$$\lim_{n\to\infty} \int_X f_n(x)\, d\mu = \int_X f(x)\, d\mu$$

To specify this for expectations, if we allow

$$\mathbb{E}[X] = \int_\Omega X(\omega)\, d\mathbb{P}(\omega)$$

(which is the definition of the expectation) then we get back the interchange of limit and expectation for uniformly bounded random variables.

3.2.2 Monotone Convergence

Before we move forward to the generalization of the bounded convergence theorem, we need the one convergence theorem that is basically built into the definition of the integral. Suppose $0 \le f_1 \le f_2 \le \cdots$ and $f_n \uparrow f$ pointwise. Then

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu \tag{3.13}$$

even if both sides are infinite. This is the **monotone convergence theorem**. Why should it be true? Monotonicity of the integral gives $\int f_n \le \int f$ for every n, so the limit (which exists because the integrals are increasing) is at most $\int f$. For the other direction, remember that $\int f$ was *defined* as the supremum of integrals of simple functions sitting below f. Take any such simple function $s \le f$ and shrink it a tiny bit, $(1-\delta)s$. Since $f_n \uparrow f$, the sets $A_n = \{f_n \ge (1-\delta)s\}$ increase to the whole space, so by continuity from below, $\int f_n \ge \int_{A_n} (1-\delta)s \to (1-\delta)\int s$. Letting $\delta \to 0$ and taking the supremum over s gives $\lim \int f_n \ge \int f$. The thing to notice is that the only way the skinny rectangles could cheat was by having mass that moves around and then disappears. Monotone sequences can't do that, since they only ever add mass.

3.2.3 Fatou's Lemma

Next, we will need a mystery tool from our tool kit (anyone watch Mickey Mouse Clubhouse?) called **Fatou's lemma**. The statement says that if we have a sequence of nonnegative functions $f_n(x)$, then

$$\int_X \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int_X f_n \, d\mu. \tag{3.14}$$

We can show this by beginning with the definition of a lim inf

$$\liminf_{n\to\infty} f_n(x) = \lim_{n\to\infty} g_n(x), \qquad g_n(x) = \inf_{k\ge n} f_k(x).$$

Which is nothing more than our notion of lim inf of sets but for functions. The point of the g_n is that they are increasing in n (we take the infimum over fewer and fewer functions), so

they are exactly the kind of sequence that the monotone convergence theorem is made for. Since g_n is the infimum of f_k over $k \geq n$, it sits below every one of them

$$\int g_n \, d\mu \leq \int f_k \, d\mu \quad \text{for all } k \geq n \quad \Longrightarrow \quad \int g_n \, d\mu \leq \inf_{k \geq n} \int f_k \, d\mu$$

Now we send $n \to \infty$. On the left, monotone convergence turns $\lim \int g_n$ into $\int \lim g_n = \int \liminf f_n$. On the right, the infimum over $k \geq n$ becomes the lim inf by definition. That's it! The inequality can be strict, and the skinny tall rectangles are exactly the example: $\int \liminf X_n = 0 < 1 = \liminf \int X_n$. The way to remember which direction the inequality goes is that mass can be lost in the limit (it escapes to a set of measure zero, or off to infinity), but it can never be created. Remembering that we are doing probability and not analysis, we can see then that the expectations relation must be

$$\mathbb{E}[\liminf X_n] \leq \liminf \mathbb{E}[X_n] \tag{3.15}$$

3.2.4 Dominated Convergence Theorem

The bounded convergence theorem makes a statement about sequences of functions bounded by a constant. We want to generalize this to boundedness by other functions. Let us consider a sequence of functions bounded above by another function $g(x)$

$$|f_n(x)| \leq g(x) \quad \forall n \text{ and almost all x}$$

If $g(x)$ is integrable (meaning the integral of the absolute value of $g(x)$ is finite), then we can say that

$$\lim_{n \to \infty} \int f_n \, d\mu = \int f \, d\mu \tag{3.16}$$

This is known as the **dominated convergence theorem**. We can see that the bounded convergence theorem is a special case of the dominated convergence theorem when $g(x)$ is a constant M. We will use Fatou's lemma as the essential tool. Suppose that $|f_n(x)| \leq g(x)$, then

$$|f - f_n| \leq |f| + |f_n| \leq 2g$$

by triangle inequality. Since $f_n \to f$, the lim sup of the difference is zero

$$\limsup_{n \to \infty} |f - f_n| = 0$$

Using linearity we get

$$\left|\int_S f d\mu - \int_S f_n d\mu\right| = \left|\int_S (f - f_n)\, d\mu\right| \leq \int_S |f - f_n|\, d\mu$$

Now here's the trick. The functions $2g - |f - f_n|$ are non-negative, so we can apply Fatou's lemma to them

$$\begin{aligned}\int 2g\, d\mu = \int \liminf_{n\to\infty} (2g - |f - f_n|)\, d\mu &\leq \liminf_{n\to\infty} \int (2g - |f - f_n|)\, d\mu \\ &= \int 2g\, d\mu - \limsup_{n\to\infty} \int |f - f_n|\, d\mu\end{aligned}$$

Since g is integrable, $\int 2g\, d\mu$ is a finite number and we can cancel it from both sides. This is sometimes called the **Reverse Fatou's Lemma** (where we use lim sups instead of lim infs), and it gives

$$\limsup_{n\to\infty} \int_S |f - f_n|\, d\mu \leq \int_S \limsup_{n\to\infty} |f - f_n|\, d\mu = 0$$

This is where we needed the integrable dominating function: it's what stops mass from escaping, because there's nowhere for f_n to put mass that g isn't already paying for. All of this leads us to conclude

$$\lim_{n\to\infty} \left|\int_S f d\mu - \int_S f_n d\mu\right| \leq \lim_{n\to\infty} \int_S |f - f_n|\, d\mu = 0$$

When the absolute value of a difference between two quantities goes to zero, this implies that the two quantities themselves converge to the same limit. Therefore

$$\lim_{n\to\infty} \int_S f_n d\mu = \int_S f d\mu \tag{3.17}$$

In the language of probability, this just means that we can swap the order of limits for expectations

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n\to\infty} X_n] \tag{3.18}$$

So far, this has been one giant yap session. But that's what all analysis is like, so we shouldn't be surprised. Let's do a problem.

Problem 2.

Let X be a non-negative random variable. Show that

$$\lim_{n\to\infty} n\mathbb{E}\left[\frac{1}{X}I\{X > n\}\right] = 0,$$

Solution:
Notice that since n is just a constant, we can put it inside the expectation

$$\lim_{n\to\infty} \mathbb{E}\left[\frac{n}{X}I\{X > n\}\right]$$

On the event $\{X > n\}$ we have $\frac{n}{X} < 1$, and off of it the indicator is zero, so the random variables $\frac{n}{X}I\{X > n\}$ are bounded by 1. We can use the bounded convergence theorem to now send the limit inside

$$\mathbb{E}\left[\lim_{n\to\infty} \frac{n}{X}I\{X > n\}\right]$$

For each fixed ω, $X(\omega)$ is a finite number, so eventually $n > X(\omega)$. From then on the indicator kills the whole thing, so the limit inside is 0 for every ω, and we are left with just the expectation of 0, which is equal to...0.

3.3 Conditional Expectation

We reach conditional expectation. There was a quote that I read some time ago that says that all of probability theory is just measure theory + conditional expectation. It seems a bit reductive, but to be honest I don't know what else to make of it other than perhaps being true. In chapter 2, we touched a bit on conditional distributions. That is,

$$\mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

And for the continuous case, we have

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

The question that we are going to answer now is: how do we calculate the mean value of X when Y is some number? Or vice versa. Well, it's pretty straightforward from the above. We define the conditional expectation to be

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y \mid x)\, dy \tag{3.19}$$

There is a subtle point to be made here. For regular expectations $\mathbb{E}[Y]$, this is a number. For conditional expectations, $\mathbb{E}[Y|X = x]$ is a function of x (the y has been integrated out). If we plug the random variable X into that function, we get $\mathbb{E}[Y|X]$, which is itself a random variable: our best guess for Y given whatever we learn about X. In the measure theory language, it's the average of Y over the information in the σ-algebra $\sigma(X)$, and that point of view is what makes conditioning on events of probability 0 (like $X = x$ for a continuous X) honest. Here is a really simple example

Problem 3.

Suppose Y represents the number of heads in 10 coin flips, and X represents the number of heads in the first 5 flips. Find $\mathbb{E}[Y \mid X = x]$

Solution:

Using the definition of a conditional expectation

$$\mathbb{E}[Y|X = x] = \sum_{y=0}^{10} y \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)}$$

The question now is: what is the probability distribution? Well, it must be a binomial because we have a fixed number of trials and want to find how many successes there are. We need to find the joint distribution, and we can do this by deduction. Suppose we have x heads on the first 5 tries

$$\implies \binom{5}{x} p^x (1-p)^{5-x}$$

If there are y heads in 10 tries, and x of them are in the first 5, then the remaining $y - x$ must be in the second half

$$\implies \binom{5}{y-x} p^{y-x} (1-p)^{5-y+x}$$

The two halves are independent, so the joint probability is the product

$$\binom{5}{x}\binom{5}{y-x} p^x (1-p)^{5-x} p^{y-x} (1-p)^{5-y+x}$$

When we divide through by probability of $\mathbb{P}(X = x)$, we just retrieve

$$\mathbb{P}(Y = y|X = x) = \binom{5}{y-x} p^{y-x} (1-p)^{5-y+x}$$

Now, we have to perform the expectation

$$\mathbb{E}[Y|X=x] = \sum_{y=x}^{x+5} y \binom{5}{y-x} p^{y-x} (1-p)^{5-y+x}$$

This is a bit cluttered, so let's do a change of variables. Let $z = y - x$. Then the expectation becomes

$$= \sum_{z=0}^{5} (x+z) \binom{5}{z} p^z (1-p)^{5-z}$$

What is the interpretation of z here? It is the number of heads after the first 5, since x head in the first 5 already happened. This is where we can be a little slick, because it is a lot more obvious to see now. The above equation is nothing more than just

$$\mathbb{E}[x+Z] = x + \mathbb{E}[Z]$$

The expectation is with respect to the distribution of Z. Here $X = x$ is just a constant, and Z is binomially distributed. We know the expected value of a binomial random variable is Np, where $N = 5$ this case. Thus, the conditional expectation reduces to

$$\mathbb{E}[Y|X=x] = x + 5p$$

If the coin is fair, then we have

$$= x + 2.5$$

Can we do the same for a continuous distribution? Sure we can! It's just the same formula but with densities instead of probability masses.

Problem 4.

Let X be uniformly distributed on [0,1]. What is the expected value of Y, where Y is uniform $[x, 1]$. What is expected value of $\mathbb{E}[Y|X=x]$?

Solution:

Pretty straightforward. Here we are actually handed the conditional density directly: given $X = x$, Y is uniform on $[x, 1]$, so

$$f_{Y|X}(y|x) = \frac{1}{1-x}, \qquad x \le y \le 1$$

The joint density is $f_{X,Y}(x,y) = f_X(x)f_{Y|X}(y|x) = \frac{1}{1-x}$ on $0 \leq x \leq y \leq 1$. (Notice that this is NOT $f_X f_Y$, since X and Y are very much not independent.) The conditional expectation is

$$\mathbb{E}[Y|X=x] = \int_x^1 \frac{y}{1-x} dy$$

$$= \frac{1}{2}\frac{1-x^2}{1-x} = \frac{1}{2}(x+1)$$

We noted earlier that the conditional expectation was not a number but rather a function of X, which makes $\mathbb{E}[Y|X]$ a random variable. Because of this, we can take its expectation, and we should get back just the regular expectation. Let's see what we mean by this. Consider

$$\mathbb{E}[Y|X=x] = \int_{-\infty}^{\infty} y\ f_{Y|X}\ dy$$

If we take the expectation again, we get

$$\mathbb{E}[\mathbb{E}[Y|X]] = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y\ f_{Y|X}(y|x)\, dy \right) f_X(x)\ dx$$

From the definition of a conditional probability, we know that $f_{Y|X}f_X$ is the joint distribution between X and Y. Fubini's theorem tells us that the order doesn't matter for as long as the functions are measurable (which densities are), then we can just integrate the joint with respect to X. Marginalizing the joint over X just gives us the distribution for Y

$$= \int_{-\infty}^{\infty} y f_Y\ dy$$

But we recognize that this is just the expectation of Y

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y] \tag{3.20}$$

This result is known as the **Law Of Total Expectation**. A little thought can reveal that other way must be true too

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X] \tag{3.21}$$

For example, consider problem 4. If we find the expectation of $\frac{1}{2}(X+1)$, we should see that

$$\mathbb{E}[\frac{1}{2}(X+1)] = \frac{1}{2}\left(\mathbb{E}[X]+1\right)$$

Knowing that X is a uniform $[0,1]$ random variable, we've previously calculated the expectation to be $1/2$. Thus, the expectation is $3/4$. The great thing about this rule is that we don't need to calculate the joint and then marginalize. If we can just take the expectation again, then we can retrieve $\mathbb{E}[Y]$ for free. This is true for every pair of random variables (as long as $\mathbb{E}|Y| < \infty$), which is what makes it such a powerful tool for computations, in mathematical finance and everywhere else. The physicist's way to remember it is "average over the fast variable, then average over the slow one."

Problem 5.

Here is an example of how this is powerful in mathematical finance. Suppose that we know for one reason or another that the price of a stock fluctuates like a binomial random variable (that is, it goes up a unit price with probability p and goes down with probability $1-p$. This is an example of a stochastic process called **random walks**). If we know nothing about the value of p, then we can just assume that it is uniformly distributed on $[0,1]$. A reasonable first guess. What is the expected change in value for the stock?

Solution:

Over N days, let Y be the number of price increases, so that the change in price is $Y-(N-Y)=2Y-N$. Given p, Y is Binomial(N,p). We don't know what p is, so we have to ask: what is the expected value for Y given that p is some number? Here p is a uniform random variable on $[0,1]$. The conditional probability is

$$\mathbb{P}(Y=y|p)=\binom{N}{y}p^y(1-p)^{N-y}$$

We don't see the probability density for p because it is just 1 due to it being a uniform(0,1). Finding the average of y just gives us Np, which is the conditional expectation. We can now find the total expectation by finding the average of Np, which is just $N/2$. Thus average number of price increases for uniform probability of success is $N/2$.

So the expected change in price is $\mathbb{E}[2Y-N]=0$. This makes sense because if we know nothing about p other than it is uniformly probable, then we should expect no biases towards a particular direction.

3.3.1 An Illustrative Example: Random Sums

We are going to play around with a toy model. Suppose that $\{X_k\}$ are i.i.d. real valued random variables. We want to compute

$$S_N = \sum_{k=1}^{N} X_k$$

where N is itself a random variable. My probability professor once used the example of N being the number of car accidents and X_k being the cost to repair that specific car. The random sum is the total cost of fixing the cars. If X_k and N are independent, what is the distribution of S_N? Whenever we see 2 random variables intertwining like so, we resort to the law of total probability

$$\mathbb{P}(S_N \leq x) = \sum_{n=0}^{\infty} \mathbb{P}(S_N \leq x \big| N = n)\mathbb{P}(N = n)$$

This can be simplified down to

$$= \sum_{n=0}^{\infty} \mathbb{P}(S_n \leq x \big| N = n)\mathbb{P}(N = n)$$

where S_n is the ordinary sum of n terms, and we were allowed to replace S_N by S_n inside because we're conditioning on $N = n$ and N is independent of the X_k. Without knowing how N and X_k are distributed, there is nothing more we can do. But at the very least, we can deduce some basic structure. Let us find the expectation of S_N

$$\mathbb{E}[S_N] = \mathbb{E}[\mathbb{E}[S_N|N]]$$

We can rewrite the law of total probability as

$$= \sum_{n=0}^{\infty} \mathbb{E}[S_N|N = n]\mathbb{P}(N = n)$$

Which is just the definition of the expectation of a function of N. Since X_k and N are independent random variables, the sum of the X_k must also be independent of N, giving us

$$\implies \sum_{n=0}^{\infty} \mathbb{E}[S_n]\mathbb{P}(N = n)$$

$$= \sum_{n=0}^{\infty} \mathbb{E}\left[\sum_{k=1}^{n} X_k\right] \mathbb{P}(N = n)$$

Using the fact that the expectation is linear and all the X_k are i.i.d., this simplifies our calculation down to

$$= \sum_{n=0}^{\infty} n\mathbb{E}[X_1]\mathbb{P}(N = n)$$

Since X_k and N are independent, the expectation of X_1 is a constant with respect to the sum. All that is left of the sum is

$$= \mathbb{E}[X_1] \sum_{n=0}^{\infty} n\mathbb{P}(N = n)$$

Which is just the expectation of the random variable N. This tells us that the average value of the sum is

$$\mathbb{E}[S_N] = \mathbb{E}[X_1]\mathbb{E}[N] \tag{3.22}$$

This is exactly what you'd guess: the average number of accidents times the average cost of each one. The calculation for the variance gives

$$\mathrm{Var}[S_N] = \mathrm{Var}[X_1]\,\mathbb{E}[N] + \mathbb{E}[X_1]^2\,\mathrm{Var}[N] \tag{3.23}$$

which also has a nice reading: the first term is the randomness in the costs, and the second is the randomness in how many costs there are. A proof will be left as an exercise. Trivially.

3.4 Problems

1. Let X and Y be independent random variables. Show that

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

2. Look back to the random sums example. Show that

$$\mathrm{Var}[S_N] = \mathrm{Var}[X_1]\,\mathbb{E}[N] + \mathbb{E}[X_1]^2\,\mathrm{Var}[N]$$

by first finding the second moment and then subtracting the square of the expectation.

3. (Computational: Simulating the Stock Market.) Let $Y_1, Y_2, \ldots$ be independent random variables such that $\mathbb{P}(Y_i = 1) = \mathbb{P}(Y_i = -1) = \frac{1}{2}$. Let $X_n = \sum_{i=1}^{n} Y_i$. Think of $Y_i = 1$ as "the stock price increased by one dollar," $Y_i = -1$ as "the stock price decreased

by one dollar," and X_n as the value of the stock on day n.

(a) Find $\mathbb{E}(X_n)$ and $\text{Var}(X_n)$.

(b) Simulate X_n and plot X_n versus n for $n = 1, 2, \ldots, 10,000$. Repeat the whole simulation several times. Notice two things. First, it's easy to "see" patterns in the sequence even though it is random. Second, you will find that the four runs look very different even though they were generated the same way. How do the calculations in (a) explain the second observation?

4. Let X be a continuous random variable with cdf F. Suppose that $\mathbb{P}(X > 0) = 1$ and that $\mathbb{E}(X)$ exists. Show that:

$$\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X > x)\, dx.$$

Hint: Consider integrating by parts. The following fact is helpful: if $\mathbb{E}(X)$ exists, then $\lim_{x\to\infty} x[1 - F(x)] = 0$.

5. Let X be a non negative random variable. Show that

$$\lim_{n\to\infty} \frac{1}{n}\mathbb{E}\left(\frac{1}{X}\mathbf{1}\{X > \frac{1}{n}\}\right) = 0.$$

6. Let $\{A_n, n \geq 1\}$ be independent events, and suppose that $P(A_n) < 1$ for all n. Prove that

$$P(A_n \text{ i.o. }) = 1 \Longleftrightarrow P\left(\bigcup_{n=1}^{\infty} A_n\right) = 1.$$

Why is $P(A_n) = 1$ forbidden?

7. Let $X \sim N(0, 1)$ and let $Y = e^X$. Find $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

4

Inequalities

"Who is equalities and why are we inside them?"

– Unknown

In almost every probability textbook I've read, the section on inequalities is measure 0 (ba dum tss). There isn't much of a story to be told as much as it's just a dictionary. Nevertheless, we move forward, and I'll try to convince you that there actually is a story: almost every inequality in this chapter is Markov's inequality wearing a different hat.

4.1 Probability Inequalities

Why do we care about inequalities? Sometimes probabilities are hard to compute exactly but bounds are not. They put bounds on our probabilities and that could potentially give us some insight. The most important inequality to discuss is **Markov's inequality**. Let X be a non-negative random variable with finite expected value $\mathbb{E}[X]$. For any $t > 0$, Markov's inequality states:

$$P(X \geq t) \leq \frac{\mathbb{E}(X)}{t} \tag{4.1}$$

Where do we begin? The probability on the left can be calculated via (for a continuous X, though the argument works for anything)

$$P(X \geq t) = \int_t^\infty f_X(x)\, dx$$

Here's the trick: notice that for $x \geq t$, the function x is always at least as large as the constant t (figure 4.1). If that's the case, then

$$t f_X(x) \leq x f_X(x) \qquad \text{for } x \geq t$$

since f_X is non-negative. When we integrate both sides from t to ∞, we get

$$t\,\mathbb{P}(X \geq t) = \int_t^\infty t f_X(x)\,dx \;\leq\; \int_t^\infty x f_X(x)\,dx \;\leq\; \int_0^\infty x f_X(x)\,dx = \mathbb{E}[X]$$

where the last step just adds back the non-negative piece from 0 to t. Dividing by t gets us where we want to go. The slicker way to say all of this is the one-line inequality between random variables $t\,\mathbb{I}_{\{X \geq t\}} \leq X$ (check it for $X < t$ and $X \geq t$ separately), and then you take expectations of both sides.

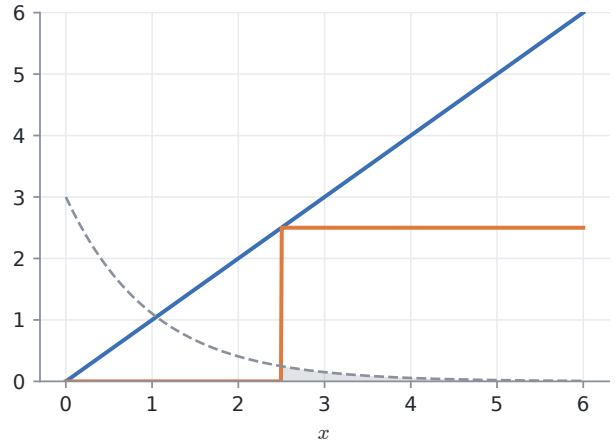


Figure 4.1: Markov in a picture. For a non-negative variable, the step $t\,\mathbb{I}_{\{x \geq t\}}$ (orange) lies below the line x (blue) everywhere. Averaging both against the density (dashed gray, rescaled) gives $t\,\mathbb{P}(X \geq t) \leq \mathbb{E}[X]$, where $\mathbb{P}(X \geq t)$ is the shaded tail.

Problem 1.

Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}[X]$. Show that

$$\mathbb{P}\left(|X - \mu| \geq t\right) \leq \frac{\sigma^2}{t^2} \quad \text{and} \quad \mathbb{P}(|Z| \geq k) \leq \frac{1}{k^2}$$

Where $Z = (X - \mu)/\sigma$. This is known as **Chebyshev's Inequality**

Solution:
The event $|X-\mu| \geq t$ is the same event as $|X-\mu|^2 \geq t^2$. So apply Markov's inequality to the non-negative random variable $|X-\mu|^2$

$$\mathbb{P}\left(|X-\mu|^2 \geq t^2\right) \leq \frac{\mathbb{E}[|X-\mu|^2]}{t^2}$$

But $\mathbb{E}[|X-\mu|^2]$ is the definition of variance. Thus term becomes σ^2 and we get

$$\mathbb{P}\left(|X-\mu| \geq t\right) \leq \frac{\sigma^2}{t^2} \tag{4.2}$$

Now if we let $Z = (X-\mu)/\sigma$, then we would have

$$\mathbb{P}\left(\frac{|X-\mu|}{\sigma} \geq \frac{t}{\sigma}\right) \leq \frac{\sigma^2}{t^2}$$

Now if we let $t = k\sigma$, then we get

$$\mathbb{P}(|Z| \geq k) \leq \frac{1}{k^2} \tag{4.3}$$

To show a quick calculation, suppose we test a prediction method, a neural net for example, on a set of n new test cases. Let $X_i = 1$ if the predictor is wrong and $X_i = 0$ if the predictor is right. Then $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$ is the observed error rate. Each X_i may be regarded as a Bernoulli with unknown mean p. We would like to know the true - but unknown - error rate p. Intuitively, we expect that $\bar{X}_n$ should be close to p. How likely is $\bar{X}_n$ to not be within ϵ of p ? We have that $\mathrm{Var}\left[\bar{X}_n\right] = \mathrm{Var}\left[X_1\right]/n = p(1-p)/n$ and

$$\mathbb{P}\left(\left|\bar{X}_n - p\right| > \epsilon\right) \leq \frac{\mathrm{Var}\left[\bar{X}_n\right]}{\epsilon^2} = \frac{p(1-p)}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2}$$

since $p(1-p) \leq \frac{1}{4}$ for all p. For $\epsilon = .2$ and $n = 100$ the bound is $1/16 = .0625$. Notice the shape of this: the probability of being off by more than ϵ dies like $1/n$. That is already the weak law of large numbers, which we'll get to in chapter 7.

My claim is that Markov's inequality is all that you will ever really see and all that you will ever need. Chebyshev was Markov applied to $(X-\mu)^2$. What if we apply it to e^{sX} instead? Since e^{sx} is increasing for $s > 0$,

$$\mathbb{P}(X \geq t) = \mathbb{P}\left(e^{sX} \geq e^{st}\right) \leq e^{-st}\,\mathbb{E}\left[e^{sX}\right] \tag{4.4}$$

for every $s > 0$, and then we get to pick the best s. This is called the **Chernoff bound**, and the expectation on the right is the moment generating function that we'll meet in chapter 5. The reason it's so much stronger than Chebyshev is that it uses *all* of the moments at once instead of just the second one, and for sums of independent variables the moment generating function factorizes. The most famous consequence is **Hoeffding's inequality**. Let $X_1, X_2, \ldots, X_N$ be independent random variables such that each X_i is bounded in the interval $[a_i, b_i]$. Define the sum $S_N = \sum_{i=1}^{N} X_i$. Hoeffding's inequality states that for any $t > 0$,

$$\mathbb{P}\left(S_N - \mathbb{E}(S_N) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{N}(b_i - a_i)^2}\right)$$

In the special case that the expectation of S_N is 0, and we let $S_N = \sum Y_i$, then sometimes we might see

$$\mathbb{P}\left(\sum_{i=1}^{n} Y_i \geq \epsilon\right) \leq e^{-t\epsilon} \prod_{i=1}^{n} e^{t^2(b_i - a_i)^2/8} \tag{4.5}$$

for every $t > 0$, which honestly is such a nicer look on the eyes. This second form is really the derivation. It's (4.4) applied to the sum, with the moment generating function factorized by independence, $\mathbb{E}[e^{t\sum Y_i}] = \prod_i \mathbb{E}[e^{tY_i}]$, and then each factor bounded using **Hoeffding's lemma**: a mean zero variable stuck in $[a, b]$ has $\mathbb{E}[e^{tY}] \leq e^{t^2(b-a)^2/8}$ (the worst case is a coin flip between the two endpoints). Then we optimize: the exponent $-t\epsilon + t^2 \sum (b_i - a_i)^2/8$ is a parabola in t, minimized at $t = 4\epsilon / \sum (b_i - a_i)^2$, and plugging that in gives exactly the first form. So Hoeffding's inequality is just Markov's inequality, applied cleverly, with a much tighter bound: the probability of a deviation of size $n\epsilon$ in a sum of n terms dies *exponentially* in n, instead of like $1/n$ (figure 4.2). To show how one might calculate this, suppose you have a fair six-sided die, and you roll it $n = 100$ times. Define X_i as the result of the i-th roll, where each X_i is an independent random variable taking values in $\{1, 2, 3, 4, 5, 6\}$. We want to use Hoeffding's inequality to find an upper bound on the probability that the average roll differs from the expected value by at least 1. Well, we know that each X_i is bounded in the interval $[1, 6]$, so $a_i = 1$ and $b_i = 6$. The expected value is

$$\mathbb{E}(X_i) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Making the expected sum to be 350 since there are 100 rolls. Applying Hoeffding's inequality, we want to find $P\left(\left|\frac{S_{100}}{100} - 3.5\right| \geq 1\right)$ which leads to

$$\mathbb{P}\left(S_{100} - 350 \geq 100\right) \leq \exp\left(-\frac{2 \times 100^2}{\sum_{i=1}^{100}(6-1)^2}\right) = e^{-8}$$

That's for the average being too high. Being too low by 1 has the same bound by symmetry,

so

$$\mathbb{P}\left(\left|\frac{S_{100}}{100} - 3.5\right| \geq 1\right) \leq 2e^{-8} \approx 6.7 \times 10^{-4}$$

A fairly small chance! For comparison, Chebyshev with $\text{Var}[X_i] = 35/12$ only gives $\frac{35/12}{100 \cdot 1^2} \approx 0.029$.

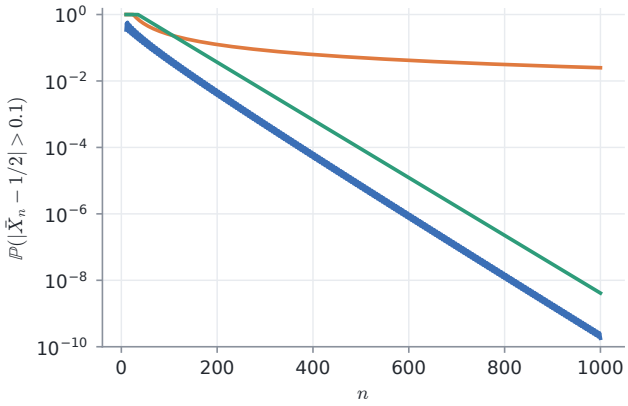


Figure 4.2: The exact probability that the fraction of heads in n fair coin flips is off by more than 0.1 (blue), with the Chebyshev bound (orange) and the Hoeffding bound (green), on a log scale. Chebyshev only decays like $1/n$, while Hoeffding captures the exponential decay of the true probability (with the wrong constant).

Next on our bucket list is Kolmogorov's inequality. Let $X_1, X_2, \ldots, X_n$ be independent random variables with zero mean ($\mathbb{E}(X_i) = 0$) and finite variances $\text{Var}(X_i) = \sigma_i^2$. Define the partial sum $S_k = \sum_{i=1}^{k} X_i$. Kolmogorov's inequality states that for any $t > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq t\right) \leq \frac{\sum_{i=1}^{n} \sigma_i^2}{t^2} \tag{4.6}$$

Look at what this says. Chebyshev applied to S_n alone would give exactly the same right hand side for $\mathbb{P}(|S_n| \geq t)$. Kolmogorov says that you get control over the whole path $S_1, S_2, \ldots, S_n$ for free! The reason is that if the walk ever wanders above t at some time k, the remaining steps are just as likely to push it further out as to bring it back, so the end point S_n "remembers" the excursion often enough. The one and only time I've ever seen Kolmogorov's inequality being used was in lecture, and it was used to prove the convergence of a random series (we'll use it again for exactly this in chapter 7). The argument goes as follows: let X_k be independent with

$$X_k = \begin{cases} 1/k & p = .5 \\ -1/k & p = .5 \end{cases}$$

so $S_n = \sum_{k \le n} \pm 1/k$ is the harmonic series with random signs. Then for the sum $S_n = \sum_{k=1}^n X_k$,

$$\mathbb{P}\left(\max_{1 \le k \le n} |S_k| \ge x\right) \le \frac{\sum_{i=1}^n \sigma_i^2}{x^2}$$

Since each X_k has 0 mean, the variance is just equal to the second moment, $\sigma_k^2 = 1/k^2$

$$\implies \mathrm{Var}[S_n] = \sum_{k=1}^n \frac{1}{k^2}$$

Subbing this back into Kolmogorov's inequality gives us

$$\mathbb{P}\left(\max_{1 \le k \le n} |S_k| \ge x\right) \le \frac{\sum_{k=1}^n \frac{1}{k^2}}{x^2}$$

To show that the sum converges, we use the same inequality on the tail of the series, starting at some j. The increments $S_k - S_j$ for $k > j$ are again a sum of independent mean zero terms, so

$$\mathbb{P}\left(\max_{j < k \le n} |S_k - S_j| \ge x\right) \le \frac{1}{x^2} \sum_{k=j+1}^n \frac{1}{k^2}$$

The right hand side doesn't care about n if we just sum to infinity, so we can let $n \to \infty$ (the events on the left increase with n, so continuity from below lets us do this)

$$\mathbb{P}\left(\sup_{k > j} |S_k - S_j| \ge x\right) \le \frac{1}{x^2} \sum_{k=j+1}^\infty \frac{1}{k^2} \underset{j \to \infty}{\longrightarrow} 0$$

Now, how far apart can two partial sums beyond j be? Let $D_j = \sup_{k,l > j} |S_k - S_l|$. By the triangle inequality, $D_j \le 2 \sup_{k > j} |S_k - S_j|$. The D_j are decreasing in j (we take the sup over fewer pairs), so they have a limit D, and $D \le D_j$ for every j. Therefore

$$\mathbb{P}(D \ge 2x) \le \mathbb{P}(D_j \ge 2x) \le \frac{1}{x^2} \sum_{k > j} \frac{1}{k^2} \quad \text{for every } j$$

and letting $j \to \infty$ gives $\mathbb{P}(D \ge 2x) = 0$ for every $x > 0$. So $D = 0$ almost surely, which is exactly the statement that the partial sums are a Cauchy sequence, which means S_n converges with probability 1. This is pretty remarkable: the harmonic series $\sum 1/k$ diverges, but if you flip a coin for each sign, it converges with probability 1 (figure 4.3). The cancellations win.

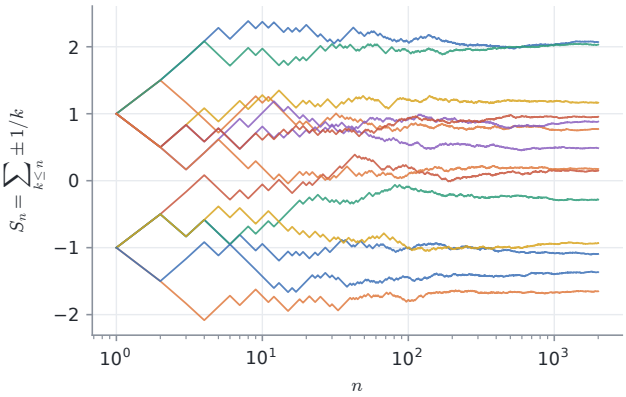


Figure 4.3: Partial sums $S_n = \sum_{k \le n} \pm 1/k$ of the harmonic series with random signs, for 14 independent sequences of signs. Every path converges, but to a different (random) limit.

4.2 Inequalities For Expectation

This is where things get a bit interesting, but not so much so that it's really worth doing any crazy proofs. The most important inequalities are Jensen's, Cauchy-Schwarz, and Hölder's inequalities. I'll throw in Minkowski just because I love relativity.

First up is probably the most important and widespread of all these. Before we begin, we should discuss convexity of functions. We say that a function $f(x)$ is convex if every chord lies above the graph, $f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y)$; for twice differentiable functions this is the same as $f''(x) \ge 0$. We say that a function is concave if $-f(x)$ is convex. I know that there is a lot of formalism in these definitions, but we need not go into them. But with that being said, we can now give **Jensen's inequality**. If g is convex, then

$$\mathbb{E}[g(X)] \ge g(\mathbb{E}[X]) \tag{4.7}$$

If g is concave, then

$$\mathbb{E}[g(X)] \le g(\mathbb{E}[X]) \tag{4.8}$$

The proof of this one is pretty straightforward, so I'll give it to you. Let $L(x) = a + bx$ be a line, tangent to $g(x)$ at the point $\mathbb{E}(X)$. Since g is convex, it lies above the line $L(x)$. So,

$$\mathbb{E}[g(X)] \ge \mathbb{E}[L(X)] = \mathbb{E}[a + bX] = a + b\mathbb{E}[X] = L(\mathbb{E}[X]) = g(\mathbb{E}[X])$$

Nice and easy right! The chord definition of convexity is Jensen's inequality for a random variable taking two values, and the general statement just says it keeps working for any average. Two consequences you have already used: $\mathbb{E}[X^2] \ge \mathbb{E}[X]^2$ (that's $\mathrm{Var}[X] \ge 0$), and

$\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$, which is exactly why the gambler in chapter 3 goes broke while their expected balance grows.

Problem 2.

Show that, for any positive random variable X,

$$\mathbb{E}[X] + \mathbb{E}\left[\frac{1}{X}\right] \geq 2$$

Solution:
Let's use Jensen's inequality (because what choice do we have?). The function $1/x$ is convex on $(0, \infty)$ (its second derivative is $2/x^3 > 0$), so $\mathbb{E}\left[\frac{1}{X}\right] \geq 1/\mathbb{E}[X]$, which just means

$$\mathbb{E}[X] + \mathbb{E}\left[\frac{1}{X}\right] \geq \frac{\mathbb{E}[X]^2 + 1}{\mathbb{E}[X]}$$

The smallest possible value that the right hand side can take on is 2, since $m + \frac{1}{m} - 2 = \frac{(m-1)^2}{m} \geq 0$ for $m > 0$. Thus, by Jensen's inequality,

$$\mathbb{E}[X] + \mathbb{E}\left[\frac{1}{X}\right] \geq 2$$

The next big boy is the **Hölder inequality**. Let X and Y be random variables, and let p and q be positive real numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. Then Hölder's inequality states

$$\mathbb{E}(|XY|) \leq (\mathbb{E}(|X|^p))^{1/p} \cdot (\mathbb{E}(|Y|^q))^{1/q} \tag{4.9}$$

We will not prove Hölder's inequality, because then we would have to unpack Young's inequality and blah blah blah. I don't believe it is really that instructive and honestly I just want to get to characteristic functions. But let's do a problem anyways!

Problem 3.

Let X be a random variable. Show that, if $\mathbb{E}X^2 = 1$ and $\mathbb{E}X^4 < \infty$, then

$$\mathbb{E}[|X|] \geq \frac{1}{\sqrt{\mathbb{E}X^4}}$$

Solution:

We will use a trick, and split $X^2 = |X|^{2/3}|X|^{4/3}$. In order for us to use Hölder's inequality, we need to choose p and q such that $1/p + 1/q = 1$, and we want $(|X|^{2/3})^p$ to be $|X|$ and $(|X|^{4/3})^q$ to be X^4. That forces $p = 3/2$ and $q = 3$, and indeed $2/3 + 1/3 = 1$. Applying the inequality,

$$1 = \mathbb{E}\left[X^2\right] = \mathbb{E}\left[|X|^{2/3}|X|^{4/3}\right] \leq (\mathbb{E}|X|)^{2/3}\left(\mathbb{E}X^4\right)^{1/3}$$

with the first equality using the fact that the second moment equals 1. We can cube both sides, dividing the forth moment term to the left hand side then taking the square root to find

$$\mathbb{E}[|X|] \geq \frac{1}{\sqrt{\mathbb{E}X^4}}$$

Now, onto the Cauchy-Schwarz; probably one the most famous inequalities of all time. If X and Y have finite variances then

$$\mathbb{E}|XY| \leq \sqrt{\mathbb{E}\left[X^2\right]\mathbb{E}\left[Y^2\right]} \tag{4.10}$$

The proof of this is super simple. It is just the Hölder inequality but with $p = q = 2$. A nice consequence: applying it to $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$ gives $|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}[X]\operatorname{Var}[Y]}$, which is why correlation coefficients live in $[-1, 1]$. One really famous use of the Cauchy-Schwarz inequality is in quantum mechanics (where it's applied to vectors in a Hilbert space instead of random variables, but it's the same inequality), which states that the standard deviation of the measurement in momentum Δp multiplied by the standard deviation in the position Δx must always be greater than or equal to $\frac{\hbar}{2}$

$$\Delta p \Delta x \geq \frac{\hbar}{2} \tag{4.11}$$

where the reduced Planck's constant $\hbar$ sets the scale for quantum mechanics. The uncertainty principle is a deep statement about the universe; it says that the universe forbids us from knowing measurements of momentum and position with arbitrary precision. That is, the more you know about a particle's position, the less you know about where it is going. This is in stark contrast to every day classical mechanics, which forbids no such thing!

The last, and probably not too talked about inequality is the **Minkowski inequality**. Suppose X and Y are random variables, and $p \geq 1$. The Minkowski inequality in probability theory states:

$$(\mathbb{E}[|X + Y|^p])^{1/p} \leq (\mathbb{E}[|X|^p])^{1/p} + (\mathbb{E}[|Y|^p])^{1/p} \tag{4.12}$$

You can kinda think about it as the triangle inequality for the "length" $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$, which is what turns the random variables with finite p-th moment into a normed space, L^p. We'll see these norms again when we talk about convergence in chapter 6.

4.3 Problems

1. Show that, for any positive random variable X,

$$\mathbb{E}\left[\max\left\{X, \frac{1}{X}\right\}\right] \geq 1$$

2. Let $X \sim \text{Poisson}(\lambda)$. Use Chebyshev's inequality to show that $\mathbb{P}(X \geq 2\lambda) \leq 1/\lambda$.

3. Prove Mill's inequality Let $Z \sim N(0,1)$. Then,

$$\mathbb{P}(|Z| > t) \leq \sqrt{\frac{2}{\pi}} \frac{e^{-t^2/2}}{t}$$

4. Prove that, for any random variable X, the minimum of $E(X-a)^2$ is attained for $a = EX$.

5. Prove that, for any random variable X, the minimum of $E|X-a|$ is attained for $a = \text{med}(X)$.

6. Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Bound $\mathbb{P}\left(|\ \bar{X}_n - \ p\ | > \epsilon\right)$ using Chebyshev's inequality and using Hoeffding's inequality. Show that, when n is large, the bound from Hoeffding's inequality is smaller than the bound from Chebyshev's inequality.

5

Integral Transforms and Convolutions

"Fourier analysis is just a vibe check"

– Fourier, probably

In my humble opinion, the characteristic function is probably the most interesting topic in all of probability. But what exactly are the problems we're trying to solve with them? Why do we care? As it turns out, sometimes it's a lot easier to work with the characteristic function than it is to work with the density itself. We will begin by motivating a problem, finding a standard way to solve it, and see how the characteristic function expedites the process.

5.1 Convolutions

Let us look back to chapter 2, specifically on transformations of multiple random variables. We wanted to find the sum of 2 uniform random variables, and saw that it was not as straightforward as we thought. We had to look at the domain, saw that the answer had different formulas on different pieces, exploit the law of total probability, it was kinda a mess. Let us now try to generalize sums of random variables. Suppose that X and Y are independent random variables. They need not be identical. How do we find the sum $Z = X + Y$? Let us start with the law of total probability

$$\mathbb{P}(Z = z) = \sum_x \mathbb{P}(X + Y = z | X = x)\mathbb{P}(X = x) = \sum_x \mathbb{P}(Y = z - x)\mathbb{P}(X = x)$$

where in the last step we used independence to drop the conditioning. To generalize this to probability densities, we remember to start with finding the CDF

$$F_Z(z) = \int_{-\infty}^{\infty} \mathbb{P}(X + Y \leq z | X = x) f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} \mathbb{P}(Y \leq z - x) f_X(x)\, dx$$

But the $\mathbb{P}(Y \leq z - x)$ is just the CDF of Y up to $z - x$, so we can substitute this in as

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_Y(y) f_X(x)\, dy dx$$

Now if we wanted to find the density of Z, we need to take derivatives of the CDF as before. But what does taking the derivative do in this case? Only the inner integral depends on z, and by the fundamental theorem of calculus, the derivative kills it and evaluates the integrand at the upper limit

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) dx \tag{5.1}$$

This result is known as the **convolution** of 2 independent random variables. The picture to have in mind is: to land at z, X has to land somewhere, at x, and then Y has to make up the difference $z - x$; add up over all the ways of splitting z. Although it's technically not anything new, it allows us to somewhat systematically expedite the process. In the case of the sum of uniforms, because the densities have edges, we still have to be mindful of our domain. If you keep convolving uniforms with each other, something interesting happens (figure 5.1): the edges get smoothed out, and the shape starts looking suspiciously like a bell curve. Hold that thought until chapter 8.

Problem 1.

We saw at the end of chapter 2 that sums of exponential random variables are not exponential; they are **gamma** distributed. Same with uniforms, sums of uniform random variables are not uniform. What about normal distributions? Are sums of standard normals also normal?

Solution:

So given X_1 and X_2 are i.i.d. standard normals. We can find their sum using the convolution integral. We know what the densities of standard normals are, so we can

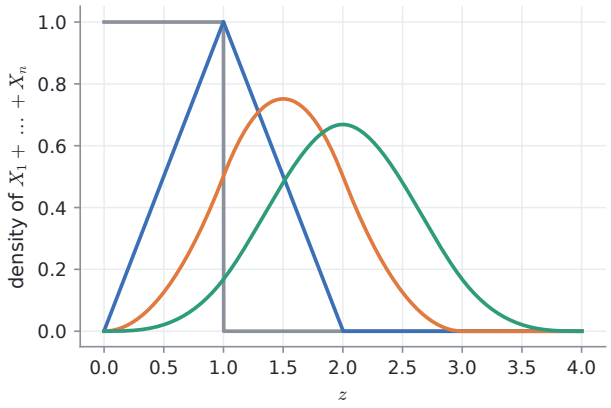


Figure 5.1: Densities of sums of n independent uniforms on $[0, 1]$ for $n = 1, 2, 3, 4$ (gray, blue, orange, green), computed by repeated convolution. $n = 2$ is the triangle from chapter 2.

just plug it right in

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{2\pi} \exp\left[-\frac{(z-x)^2}{2}\right] \exp\left[-\frac{x^2}{2}\right] dx$$

We can simplify this down by expanding the square

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left[-\frac{(z^2 - 2xz + 2x^2)}{2}\right] dx$$

This is a standard Gaussian integral that can be computed by completing the square in the exponent (or by a secret trick called throwing it into Mathematica). Completing the square, $2x^2 - 2xz + z^2 = 2\left(x - \frac{z}{2}\right)^2 + \frac{z^2}{2}$, so

$$f_Z(z) = \frac{e^{-z^2/4}}{2\pi} \int_{-\infty}^{\infty} e^{-(x-z/2)^2} dx = \frac{e^{-z^2/4}}{2\pi} \sqrt{\pi} = \frac{e^{-\frac{z^2}{4}}}{2\sqrt{\pi}}$$

using the integral from problem 6 of chapter 2. Comparing with the Gaussian density, $e^{-z^2/4}/\sqrt{4\pi}$ is a normal with $2\sigma^2 = 4$. Therefore the sum of two standard normals is still a normal, with mean 0 but a variance of 2 instead of 1 (and a standard deviation of $\sqrt{2}$). Variances of independent things add.

The logical question that one will want to ask is: if we have a sum of 2 random variables, how do we compute more than a sum of 2 terms? Suppose we have a string of random variables, how do we compute their sum? Or an even better question: Is an infinite sum of normal random variables also normal? Well, with what we know now, that would be a

nightmare! We would have to compute an infinite number of convolutions! What can we do?

5.2 The Moment Generating Function

We're going to motivate a seemingly completely unrelated problem but it will all connect soon enough. Suppose we want to compute the expectation and the variance and even higher moments. This would be dreadful! That's so many integrals to do! But we can use a trick, because physicists are just a walking bag of tricks. We know that the exponential function $e^{\lambda x}$ is an eigenfunction of the derivative operator with eigenvalue λ (if you don't believe me, check for yourself that the derivative gives back the same function but scaled by λ). So what? Let us consider an object

$$\mathbb{E}\left[e^{tX}\right] \tag{5.2}$$

where X is a random variable. Let us take derivatives of this object with respect to t. Since we are motivated by the fact that the exponential function is an eigenfunction of the derivative, we should expect to get back the same function

$$\frac{d}{dt}\mathbb{E}\left[e^{tX}\right] = \mathbb{E}\left[Xe^{tX}\right]$$

by linearity of the expectation and the derivative. Now if we send $t \to 0$, what do we get?

$$\implies \mathbb{E}[X]$$

Which is the first moment! Now what if we take second derivatives? Well, we should expect to get the second moment. With n derivatives, we retrieve the n-th moment! Another way to see it is to Taylor expand the exponential inside the expectation

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{n=0}^{\infty} \frac{t^n}{n!}\mathbb{E}[X^n]$$

so the moments are just the Taylor coefficients. This object is called the **moment generating function**. For those that are savvy in physics, this is the source term when computing path integrals in quantum field theory. How does this help us? Well, for computing mean and variance, all that we have to do now is calculate the moment generating function and then take its derivatives to get the moments. No longer do we have to compute integrals. Let's see an example

Problem 2.

Find the moment generating function for the exponential(λ) random variable. Verify that this indeed is the MGF by taking derivatives and showing that we do get the correct moments.

Solution:

First start with

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Then we compute $\mathbb{E}\left[e^{tX}\right]$ by

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} f_X(x)\, dx$$

$$= \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{(t-\lambda)x}\, dx$$

This integral evaluates to

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

To verify that this indeed is the MGF, we take the first derivative with respect to t and evaluate it at $t = 0$

$$\frac{d}{dt} M_X(t) = \frac{0 \cdot (\lambda - t) - \lambda \cdot (-1)}{(\lambda - t)^2} = \frac{\lambda}{(\lambda - t)^2}$$

$$\implies \frac{\lambda}{(\lambda - 0)^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

Voila! We get the correct first moment. Now for the second moment

$$\frac{d^2}{dt^2} M_X(t) = \lambda \cdot \frac{2}{(\lambda - t)^3} = \frac{2\lambda}{(\lambda - t)^3}$$

At 0 this gives

$$\frac{2\lambda}{(\lambda - 0)^3} = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}$$

Which is indeed also the correct second moment! We can find the variance by subtracting the square of the first moment and we get

$$\text{Var}[X] = \frac{1}{\lambda^2}$$

The moment generating function is a powerful tool for calculating what we need. But there is one major problem: it doesn't always exist. This is actually quite obvious when you look at what the moment generating function is: it's (up to the sign of t) the two-sided Laplace transform of the PMF/PDF, and the Laplace transform may not always exist. If the tails of X are heavier than exponential, e^{tx} wins against them and the integral blows up for every $t \neq 0$. The Cauchy distribution and the log-normal e^Z are examples. I have a name for these types of functions: Laplaceable. Some functions are not Laplaceable, and therefore the MGF is not defined for some random variables. But there does exist an integral transform that is guaranteed to exist and that is the **Fourier transform**

5.3 The Characteristic Function

Instead of taking the expectation of e^{tX}, what if we took the expectation of e^{itX}

$$\varphi_X(t) = \mathbb{E}\left[e^{itX}\right] \tag{5.3}$$

This is called the **characteristic function**, and it's the Fourier transform of the distribution (physicists would put a minus sign in the exponent, but it doesn't matter). We can see that due to the exponential property, the characteristic function still gives us the moments when we take derivatives, but they will be off by a phase factor

$$\frac{d}{dt}\varphi_X(t) = \frac{d}{dt}\mathbb{E}\left[e^{itX}\right]\Bigg|_{t=0} = i\mathbb{E}[X]$$

No matter! We can still retrieve the moment generating property by dividing by powers of i

$$\mathbb{E}[X^n] = \frac{1}{i^n}\frac{d^n}{dt^n}\varphi_X(t)\Bigg|_{t=0}$$

(as long as those moments exist). But that's not why we are interested in the characteristic function. We said before that the characteristic function always exists. If we start with the definition

$$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx$$

we can use the fact that e^{itx} has unit magnitude for all values of x and all values of t. So

$$|\varphi_X(t)| \leq \int_{-\infty}^{\infty} |e^{itx}| f_X(x)\, dx = 1$$

The integral converges absolutely, for every random variable and every real t, and that's existence. Unlike e^{tX}, the oscillating e^{itX} never grows, so heavy tails can't hurt it. The same bound gives us continuity. Since $|e^{itx} - e^{isx}| \leq 2$ and $e^{itx} \to e^{isx}$ as $t \to s$, the dominated convergence theorem (with the integrable dominating function $2f_X$) lets us take the limit inside

$$\lim_{t\to s} \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx = \int_{-\infty}^{\infty} \lim_{t\to s} e^{itx} f_X(x)\, dx = \varphi_X(s)$$

so the characteristic function is a continuous function of t. Now that we have guaranteed existence, do we also have guaranteed uniqueness? That is, if two random variables have the same characteristic function, must they have the same distribution? This is where our knowledge of Fourier transforms comes in. The Fourier transform is invertible (it's a unitary operator; learn quantum mechanics), and when φ_X is integrable the inverse is given by

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \varphi_X(t)\, dt \tag{5.4}$$

So if $\varphi_X(t) = \varphi_Y(t)$ for all t, then the right hand side is the same for both, and they must have the same density. When there's no density (or φ isn't integrable) there's a version of the inversion formula for $\mathbb{P}(a < X < b)$ instead, which you can find in any of the references, and it gives the same conclusion. Thus, the characteristic function uniquely determines the distribution of the random variable. (It does NOT determine the random variable itself; remember that X and $1-X$ have the same distribution when X is uniform on $[0, 1]$.) There's one more theorem about characteristic functions that we will lean on very heavily later, the **Lévy continuity theorem**, which says that convergence of characteristic functions is the same thing as convergence of distributions. We'll see it in chapter 6, and it's the engine behind the central limit theorem.

5.3.1 Properties Of The Characteristic Function

Let us collect everything we know about the characteristic function in one place

(1) **Existence:** We have shown that previously
(2) **Uniqueness:** We have also shown that
(3) **Normalization:** The characteristic function must equal 1 at $t = 0$

$$\varphi_X(0) = \mathbb{E}[e^{i\cdot 0\cdot X}] = 1$$

(4) **Boundedness:** $|\varphi_X(t)| \leq 1$ for all t.
(5) **Uniform Continuity:** The characteristic function is uniformly continuous, because $|\varphi_X(t+h) - \varphi_X(t)| \leq \mathbb{E}\left|e^{ihX} - 1\right|$, and the right hand side doesn't depend on t and

goes to 0 as $h \to 0$ by dominated convergence.

(6) **Symmetry:**

$$\varphi_X(-t) = \overline{\varphi_X(t)}$$

This one is really easy to show. Given the definition $\varphi_X(t) = \mathbb{E}[e^{itX}]$, if we send $t \to -t$, then this is $\mathbb{E}[e^{-itX}]$ which is the same thing as $\overline{\varphi_X(t)}$ by definition of complex conjugate.

(7) **Positive Definiteness:** The characteristic function is positive definite. For any finite set of real numbers $t_1, t_2, \ldots, t_n$ and complex numbers $c_1, c_2, \ldots, c_n$, the following holds:

$$\sum_{j=1}^{n}\sum_{k=1}^{n} c_j \overline{c_k} \varphi_X(t_j - t_k) \geq 0$$

(8) **Scaling and Shifting:** If $Y = aX + b$ where a and b are constants, then

$$\varphi_Y(t) = e^{itb}\varphi_X(at)$$

(9) **Independence:** Probably most notable of all, if 2 random variables X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t)$$

Let us show that the last property holds. Suppose we define $Z = X + Y$, then $\varphi_Z(t) = \mathbb{E}[e^{itZ}] = \mathbb{E}[e^{it(X+Y)}] = \mathbb{E}[e^{itX} \cdot e^{itY}] = \mathbb{E}[e^{itX}] \cdot \mathbb{E}[e^{itY}] = \varphi_X(t) \cdot \varphi_Y(t)$ Where we are allowed to break up the expectation of the product into $\mathbb{E}[e^{itX}] \cdot \mathbb{E}[e^{itY}]$ because X and Y are independent (functions of independent random variables are independent, and expectations of products of independent things factor). Why is this great? Because now, if we wanted to find the distribution of a sum of n i.i.d. random variables, we can just find the characteristic function of one, raise it to the n-th power, then take the inverse transform, and that will uniquely determine the distribution of the sum! This is the convolution theorem from Fourier analysis: convolution in x becomes multiplication in t, and n convolutions become one power. I should note that the definition $\mathbb{E}[e^{itX}]$ never cared whether X had a density, so all of this analysis works perfectly well for discrete random variables (where it's a sum instead of an integral) and everything in between.

Problem 3.

In chapter 2, we found the distribution of X^2 where X was a standard normal random variable. Let us now take the sum of n of these, where the X_i are i.i.d. standard normals. What is the distribution of $\sum_{i=1}^{n} X_i^2$?

Solution:
We know what the distribution of 1 of these X^2 terms is. It is just the χ^2 distribution. Here the moment generating function actually exists (for $t < 1/2$), and it's a Gaussian integral

$$M_{X^2}(t) = \mathbb{E}[e^{tX^2}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx^2} e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1-2t)x^2/2}\, dx$$

Substituting $u = \sqrt{1-2t}\, x$ turns this back into the standard normal integral, which is 1, leaving

$$M_{X^2}(t) = (1-2t)^{-1/2} \quad \text{for } t < \frac{1}{2}$$

(and the characteristic function is the same with $t \to it$, $\varphi_{X^2}(t) = (1-2it)^{-1/2}$). This is for one term. Since all the X_i^2 are i.i.d., all that we need to do is use the product property and raise it to the n-th power

$$\implies (1-2t)^{-n/2}$$

There are 2 things we can do now: either we go and look up what this characteristic function corresponds to (which feels kind of cheaty) or we take the inverse Fourier transform. Using our trusty tool Mathematica, we find that the inverse transform gives us

$$f_Z(z) = \frac{1}{2^{n/2}\Gamma(n/2)} z^{n/2-1} e^{-z/2}$$

for $z > 0$, where $\Gamma(n/2)$ is the **gamma function**, which is just the generalized factorial ($\Gamma(k) = (k-1)!$ and $\Gamma(1/2) = \sqrt{\pi}$). The above distribution is called the χ^2 distribution with **n degrees of freedom**. As a check, setting $n = 1$ gives $\frac{1}{\sqrt{2}\sqrt{\pi}} z^{-1/2} e^{-z/2}$, which is exactly what we found in chapter 2. A few of these densities are shown in figure 5.2.

We also notice that in calculating the χ^2 distribution that we saw another way of calculating transformations of single random variables. We just need to find the characteristic function $\mathbb{E}\left[e^{itf(X)}\right]$ where $f(X)$ is the function on the random variable. This method, while theoretically is correct, is not very reliable simply due to the fact that taking integrals is very difficult. Nevertheless, if you wanted to extract information like the mean and variance from the characteristic function, that is still doable. Let us try another example to get our feet wet

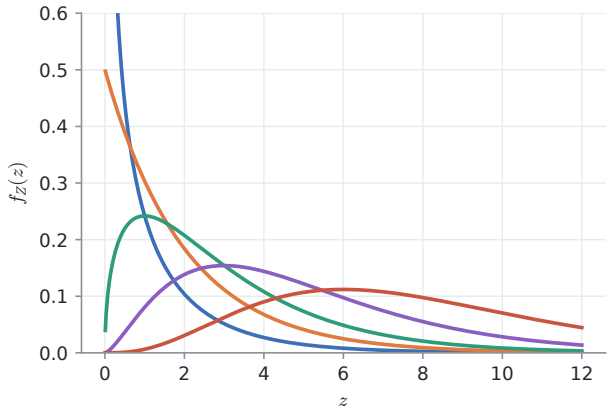


Figure 5.2: The χ^2 density with $n = 1, 2, 3, 5, 8$ degrees of freedom (blue, orange, green, purple, red). As n grows, the sum of n squared normals has mean n and variance $2n$, and its shape becomes more and more symmetric.

Problem 4.

Show, by using characteristic, or moment generating functions, that if

$$f_X(x) = \frac{1}{2}\mathrm{e}^{-|x|}, \quad -\infty < x < \infty,$$

then $X \stackrel{d}{=} Y_1 - Y_2$, where Y_1 and Y_2 are independent, exponentially distributed random variables.

Solution:

Remember, if X and Y share the same characteristic function, then they must have the same distribution. We begin by finding the characteristic function of the double exponential

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx$$

$$\implies \int_{-\infty}^{\infty} e^{itx} \frac{1}{2} e^{-|x|}\, dx$$

Now, remember that the absolute value function is a piecewise function, and therefore must be broken up as 2 integrals

$$= \frac{1}{2}\left(\int_{-\infty}^{0} e^{itx} e^{x}\, dx + \int_{0}^{\infty} e^{itx} e^{-x}\, dx\right)$$

Using our magic helper mathematica to evaluate these integrals, we find that the characteristic function is

$$\varphi_X(t) = \frac{1}{2}\left(\frac{1}{it+1} + \frac{1}{1-it}\right)$$

But notice that this can be combined into a single function of t

$$\varphi_X(t) = \frac{1}{2}\left(\frac{2}{(it+1)(1-it)}\right)$$

which is just $\varphi_X(t) = \frac{1}{1+t^2}$. But hold on, now, the factors in the denominator look like characteristic functions. For an exponential with $\lambda = 1$, $\varphi_Y(t) = \int_0^\infty e^{itx}e^{-x}dx = \frac{1}{1-it}$, and by the symmetry property $\varphi_{-Y}(t) = \varphi_Y(-t) = \frac{1}{1+it}$. So

$$\varphi_X(t) = \frac{1}{1-it}\cdot\frac{1}{1+it} = \varphi_{Y_1}(t)\,\varphi_{-Y_2}(t) = \varphi_{Y_1-Y_2}(t)$$

When we see a product, it must be the characteristic function of a sum of independent random variables. Thus, by uniqueness, $X \stackrel{d}{=} Y_1 - Y_2$ with Y_1 and Y_2 independent exponentials.

5.4 Parseval's Theorem

We will consider one consequence of the Fourier transform, and it is to make calculations a bit more tenable. Let X and Y be independent random variables with characteristic functions φ_X and γ_Y, respectively. Consider the single number

$$\mathbb{E}\left[e^{iXY}\right]$$

There are two ways to compute it, using the law of total expectation from chapter 3. If we condition on Y first, then for fixed $Y = y$, the inner expectation over X is $\mathbb{E}[e^{iyX}] = \varphi_X(y)$. If we condition on X first, then for fixed $X = x$, the inner expectation over Y is $\mathbb{E}[e^{ixY}] = \gamma_Y(x)$. So

$$\mathbb{E}\left[\varphi_X(Y)\right] = \mathbb{E}\left[e^{iXY}\right] = \mathbb{E}\left[\gamma_Y(X)\right] \tag{5.5}$$

If both have densities f_X and g_Y, this reads

$$\int_{-\infty}^{\infty} \varphi_X(y)g_Y(y)\,dy = \int_{-\infty}^{\infty} \gamma_Y(x)f_X(x)\,dx \tag{5.6}$$

which is called **Parseval's relation** (it's a probabilistic cousin of the Parseval/Plancherel theorem from Fourier analysis, and swapping the order of the two expectations is just Fubini's theorem, which is fine because $|e^{iXY}| = 1$). The beauty of this is that we can use it backwards to compute integrals that look like they have nothing to do with probability. As an example, we want to show

$$\int_{-\infty}^{\infty} \frac{\cos y}{1+y^2}\, \mathrm{d}y = \frac{\pi}{\mathrm{e}}$$

Whenever I see a trig integral, my mind goes to 2 places: residues and Parseval's relations. We need to recognize the two pieces. Let X be a coin-tossing random variable, $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$. Then $\varphi_X(y) = \frac{1}{2}e^{iy} + \frac{1}{2}e^{-iy} = \cos y$. And let Y be a standard Cauchy random variable, with density $g_Y(y) = \frac{1}{\pi(1+y^2)}$. By inversion (since $\frac{1}{1+t^2}$ is the characteristic function of $\frac{1}{2}e^{-|x|}$ from problem 4, the Cauchy density is the Fourier transform of that), the Cauchy characteristic function is $\gamma_Y(x) = e^{-|x|}$. Then

$$\int_{-\infty}^{\infty} \frac{\cos y}{1+y^2}\, \mathrm{d}y = \pi \int_{-\infty}^{\infty} \varphi_X(y) g_Y(y)\, dy = \pi\, \mathbb{E}\left[\gamma_Y(X)\right]$$

and X only takes two values, so the expectation on the right is a two term sum

$$\pi\left(\frac{1}{2}\mathrm{e}^{-|-1|} + \frac{1}{2}\mathrm{e}^{-|1|}\right) = \frac{\pi}{\mathrm{e}}$$

No contour integration needed. (If you like delta functions, the coin flip "density" is $\frac{1}{2}\delta(x-1) + \frac{1}{2}\delta(x+1)$, and this is the same calculation.) Voila! Let's do another problem to get the hang of it

Problem 5.

Prove that

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{(\cos y)^2}{1+y^2}\, \mathrm{d}y = \frac{1+\mathrm{e}^{-2}}{2}$$

Solution:

When I first did this problem, my mind went straight to contour integration; and I was able to get the right answer. But as it turns out, not many people know how to do contour integration, so we will resort to using Parseval's relation

$$\int_{-\infty}^{\infty} \varphi_X(y) g_Y(y)\, dy = \int_{-\infty}^{\infty} \gamma_Y(x) f_X(x)\, dx$$

The Cauchy density will take care of the $\frac{1}{\pi(1+y^2)}$ again, so we need a random variable whose characteristic function is $\cos^2 y$. Using the double angle formula,

$$\cos^2 y = \frac{1+\cos 2y}{2} = \frac{1}{4}e^{-2iy} + \frac{1}{2} + \frac{1}{4}e^{2iy}$$

which is the characteristic function of a random variable X with $\mathbb{P}(X=-2) = \frac{1}{4}$, $\mathbb{P}(X=0) = \frac{1}{2}$, $\mathbb{P}(X=2) = \frac{1}{4}$ (it's the sum of two independent coin flips!). Now using the Parseval relation with $\gamma_Y(x) = e^{-|x|}$

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{(\cos y)^2}{1+y^2}\,\mathrm{d}y = \mathbb{E}\left[\varphi_X(Y)\right] = \mathbb{E}\left[e^{-|X|}\right] = \frac{1}{4}e^{-2} + \frac{1}{2} + \frac{1}{4}e^{-2}$$

$$= \frac{1}{4}\left(2e^{-2}+2\right) = \frac{1+e^{-2}}{2}$$

5.5 Probability Generating Functions

We end this chapter with a discussion on probability generating functions, as well as other integral transforms. We spent a great deal of time with the characteristic function and the moment generating function, both of which generate moments. We now turn to an object that is used to generate probabilities, and behaves effectively the same way as the moment generating function. Because the object generates probabilities, this object only works for non-negative integer valued random variables (basically discrete). We define it here

$$G_X(t) = \mathbb{E}[t^X] = \sum_{k=0}^{\infty} P(X=k)t^k \tag{5.7}$$

It functions in basically all the same way that the moment generating function works (in fact $G_X(e^t) = M_X(t)$). The biggest difference is that $G_X(t)$ extracts the probabilities instead of moments

$$\mathbb{P}(X=k) = \frac{1}{k!}\frac{d^k}{dt^k}G_X(t)\bigg|_{t=0}$$

The argument for why this is true is the same. If we just find $G_X(0)$, then all of the terms in the power series are 0 except for the $k=0$ term, where $t^0 = 1$. What we would be left with is

$$G_X(0) = \mathbb{P}(X=0)$$

retrieving the first probability. What if we take the first derivative? Taking

$$\frac{d}{dt}G_X(t) = \frac{d}{dt}\sum_{k=0}^{\infty}\mathbb{P}(X=k)t^k = \sum_{k=1}^{\infty}k\mathbb{P}(X=k)t^{k-1}$$

Notice that as $t \to 0$, the only term that is independent of t is the $k = 1$ term. If we divide out by k, then we are left with just $\mathbb{P}(X = 1)$. As we can see, if we take subsequent derivatives, this will just give us the k-th probability. You can kind of think of this as the Taylor series expansion: when we take the k-th derivative and set t to 0, all that we are left with is $k!$ times the coefficient multiplying t^k. Derivatives at $t = 1$ give moments instead: $G'_X(1) = \mathbb{E}[X]$ and $G''_X(1) = \mathbb{E}[X(X-1)]$. Let us do a problem

Problem 6.

Find the probability generating function for the Poisson(λ).
Solution:
Start with the Poisson distribution

$$\mathbb{P}(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

The probability generating function can be found by

$$G_X(t) = \mathbb{E}[t^X] = \sum_{k=0}^{\infty}P(X=k)t^k$$

$$G_X(t) = \sum_{k=0}^{\infty}\frac{\lambda^k e^{-\lambda}}{k!}t^k$$

We can factor out the $e^{-\lambda}$ term and group the t^k with λ^k. Doing so gives

$$G_X(t) = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda t)^k}{k!}$$

But we know that the sum is just the Taylor series expansion for e^x, so we can find the probability generating function to be

$$G_X(t) = e^{\lambda(t-1)}$$

As we can see, taking $G_X(0) = e^{-\lambda}$ and $G'_X(0) = \lambda e^{-\lambda}$, which indeed are the probabilities for $k = 0$ and $k = 1$ respectively.

The question that is on everyone's mind is: who cares? Why not just plug it in. And honestly, that's so valid. But there are a couple of uses for the probability generating the function. The first is that sometimes we might not know what the distribution is, but we know the generating function (similar to the characteristic). This is...very uncommon. The second reason and the most important is that it allows us to take sums of discrete random variables pretty easily. This is due to the fact that sums of independent random variables are just products of the generating function. The proof of this is identical to that of the characteristic function. As a proof of concept, let us do a problem

Problem 7.

Let X_1, X_2 i.i.d. Poisson(λ) random variables. Find the distribution of their sums $Z = X_1 + X_2$

Solution:

We can solve this via probability generating function. We already have the probability generating function for a single Poisson random variable. We can just multiply them together to get

$$G_{X_1+X_2}(t) = e^{2\lambda(t-1)}$$

What we see is that we get back another Poisson distribution but with 2λ. Both the Poisson and the normal have the property that sums of independent copies stay in the same family, just with new parameters.

In general, it is very difficult if not impossible to get back the probability distribution once we have the generating function...but who cares? If you have the probability generating function, you can just plot the distribution by calculating all the probabilities. If you have the characteristic function and the probability generating function, there's no need to have a close form analytic solution because all of the information is stored in both objects.

5.5.1 More Generating Functions

There are still a couple we didn't talk about, but the fact that we didn't talk about them means that they weren't too important. Well, except this first one, which will be the star of chapter 8. It's the **cumulant generating function**, which generates all the cumulants κ_n.

We can find it by taking the log of the moment generating function

$$K_X(t) = \log M_X(t) = \sum_{n=1}^{\infty} \kappa_n \frac{t^n}{n!} \tag{5.8}$$

(or the log of the characteristic function, if the moment generating function doesn't exist, in which case the derivatives pick up powers of i). The first cumulant is the mean, the second is the variance, the third is the third central moment $\mathbb{E}[(X-\mu)^3]$ (which measures skewness), and the fourth is $\mathbb{E}[(X-\mu)^4] - 3\sigma^4$ (which measures the excess kurtosis), so on and so forth. Why bother with the log? Because the characteristic function of a sum of independent things is a product, and the log turns products into sums: the cumulants of independent random variables simply add, $\kappa_n(X+Y) = \kappa_n(X) + \kappa_n(Y)$. Physicists know these as the connected correlation functions, and K is the free energy.

The other one is the characteristic function for random vectors. Up until now, we didn't really discuss random vectors, which is funny because they show up a lot if you are to talk about machine learning and physics. I want to save an entire chapter on random vectors and random matrices. For a random vector $\boldsymbol{X} \in \mathbb{R}^d$, the characteristic function is $\varphi_{\boldsymbol{X}}(\boldsymbol{t}) = \mathbb{E}[e^{i\boldsymbol{t}\cdot\boldsymbol{X}}]$, and the analysis is identical to that of single random variables, except now you'll have to perform matrix calculus and vector calculus and all that. However, in terms of substance, there really isn't anything new.

5.6 Problems

1. Let $X_1, X_2, \ldots, X_n$ be independent random variables with expectation 0 and finite third moments. Show, with the aid of characteristic functions, that

$$\mathbb{E}\left(X_1 + X_2 + \cdots + X_n\right)^3 = \mathbb{E}X_1^3 + \mathbb{E}X_2^3 + \cdots + \mathbb{E}X_n^3$$

2. Which of these cannot be characteristic functions? Explain.

1). $\cos(t)$

2). e^{t^2}

3). 1

3. Let $X \sim N(0,1)$ and let $Y = e^X$. Find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$ (A repeated problem, but now with the characteristic function)

4. Prove that

$$\int_{-\infty}^{\infty} \frac{1}{(1+y^2)^2}\, \mathrm{d}y = \frac{\pi}{2},$$
$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{1-\cos y}{y^2} \mathrm{e}^{-|y|} \mathrm{d}y = \frac{1}{2} - \frac{\log 2}{\pi}.$$

5. Let X be a random variable, such that

$$EX^n = \frac{2n!}{3\lambda^n}, \quad n = 1, 2, \ldots$$

Find the distribution of X. Is it unique?

6. (Computational) Let $X_1, X_2, ..., X_n$ be i.i.d. geometric random variables. Using the probability generating function, plot the distribution of $\sum_{i=1}^{n} X_1 + X_2 + ... + X_n$ by calculating the probabilities.

7. Let X be an exponential random variable with $\lambda = 1$ and Y be a Poisson random variable with $\lambda = 1$. Using the convolution, find the distribution of $X + Y$

8. Find the characteristic function of the normal distribution with mean μ and variance σ^2. Once you have the characteristic function, take the log to find the cumulant generating function. Find the first 4-5 cumulants. What do you notice? Generalize your results.

9. Given the function

$$e^{-c|t|^\alpha}$$

Where $c > 0$ is some constant and α takes on positive integer values. For what values of α can the above be a characteristic function?

6

Convergence

"Harmonic convergence is about to begin!"

– Vaatu

This chapter marks the end of a first-semester course on probability theory. Oftentimes in probability, we are given a sequence of random variables or a family of random variables. It would be really good if we could classify their convergence properties; or better yet, determine whether or not they converge at all. The thing that makes this subtle is that a random variable is a function, and there are a lot of different ways for functions to be "close". We will discuss the different types of convergence and how they relate, then uniform integrability (the condition that lets us swap limits and expectations), and finally the Lévy continuity theorem, which connects convergence in distribution to the characteristic functions of chapter 5. This is all set up for the next two chapters, where we will bring all that we've learned together and formulate the crowning achievements of all of probability and statistics: the law of large numbers and the central limit theorem.

6.1 Types Of Convergences

I will try to keep this section pretty short. I find that there is a lot of fluff in probability textbooks, but there are only 4 main types of convergence that we really care about. We will discuss them here and how they connect to each other.

6.1.1 Weak Convergence

Weak convergence is sometimes called **convergence in distribution**. It is called weak because...well, it is the weakest of the types of convergence that we will be talking about. We will discuss what we mean by weak later when we discuss how all these types of convergences relate to each other. Nevertheless, we say that a sequence of random variables $\{X_1, X_2, X_3, ..., X_n\}$ converges in distribution to X if

$$\lim_{n\to\infty} F_n(t) = F(t) \qquad \text{at every } t \text{ where } F \text{ is continuous} \tag{6.1}$$

where $F_n(t)$ are the CDFs of random variables in the sequence and $F(t)$ is the CDF of X. Why only at continuity points? Take $X_n = 1/n$ (a constant). Surely this should converge to $X = 0$. But $F_n(0) = \mathbb{P}(1/n \leq 0) = 0$ for every n, while $F(0) = 1$. The jump of the limit is exactly where things can approach from the wrong side, so we forgive that point. Sometimes we will see an alternative definition

$$\lim_{n\to\infty} \int f \, d\mu_n = \int f \, d\mu \qquad \text{for every bounded continuous } f \tag{6.2}$$

where μ_n and μ are the distributions (laws) of X_n and X, so in random variable language this is $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. It's a statement that averages of nice functions converge. The two definitions are equivalent (this is part of the Portmanteau theorem that we'll meet at the end of the chapter): taking f to be a smoothed out version of the indicator $\mathbb{I}_{(-\infty,t]}$ gets you from the second to the first, and the continuity requirement is what you pay for the smoothing. Perhaps we should give some lip service as to what this means. If we have a sequence of random variables, and the distribution of the sequence converges to the distribution of another random variable, then we say that the sequence converges in distribution. If 2 random variables have the same distribution, then we say that they are equal in distribution. This does NOT mean that the limiting random variable X_∞ in the sequence is the same as the random variable X, it just means that they share the same distribution. It is possible for 2 random variables to have the same CDF but be different random variables.

Problem 1.

Give an example of 2 random variables that are not the same but have the same distribution.
Solution:

Let X be a uniform random variable on $[0,1]$. Let Y be a random variable such that $Y = 1 - X$. To find the CDF for Y, we can transform the random variable as

$$\mathbb{P}(Y \leq t) = \mathbb{P}(1 - X \leq t).$$

Solving the inequality we find that the CDF for Y is

$$F_Y(t) = \mathbb{P}(X \geq 1 - t).$$

For $0 \leq t \leq 1$ this is $1 - (1 - t) = t$, so

$$F_Y(t) = \begin{cases} 0 & t < 0 \\ t & 0 \leq t \leq 1 \\ 1 & t > 1 \end{cases}$$

Which is exactly the same CDF as X. We then conclude that these random variables are equal in distribution, but X clearly is not the same thing as Y. In fact, they are almost never equal: $X = Y$ only when $X = 1/2$, which has probability 0.

This idea can also extend to sequences of random variables. Let us now show an example of a sequence of random variables that converges in distribution.

Problem 2.

Let $Z_1, Z_2, \ldots$ be i.i.d. random variables with density f. Suppose that $\mathbb{P}(Z_i > 0) = 1$ and that $\lambda = \lim_{x \downarrow 0} f(x) > 0$. Let

$$X_n = n \min\{Z_1, \ldots, Z_n\}.$$

Show that $X_n \rightsquigarrow Z$ where Z has an exponential distribution with mean $1/\lambda$. ($\rightsquigarrow$ indicates convergence in distribution)

Solution:

We've known how to compute the minimum of a sequence of random variables all the way back in chapter 2. The minimum is bigger than x exactly when all of them are, so since the Z_i are i.i.d.,

$$\mathbb{P}(\min\{Z_1, ..., Z_n\} > x) = (1 - F_Z(x))^n$$

For X_n, scaling the minimum by n just rescales the argument

$$\mathbb{P}(X_n > x) = \mathbb{P}\left(\min\{Z_1, ..., Z_n\} > \frac{x}{n}\right) = (1 - F_Z(x/n))^n$$

For a fixed $x > 0$ and n sufficiently large, we would be evaluating F_Z near 0. But we know that near 0, $F_Z(\delta) = \int_0^\delta f \approx \lambda\delta$ by a first order Taylor series expansion (the density at 0 is λ, and $F_Z(0) = 0$ since $Z > 0$). Thus

$$\mathbb{P}(X_n > x) = \left(1 - \frac{\lambda x}{n} + o\left(\frac{1}{n}\right)\right)^n \underset{n\to\infty}{\longrightarrow} e^{-\lambda x}$$

so $F_{X_n}(x) \to 1 - e^{-\lambda x}$, which is the CDF of an exponential distribution with mean $1/\lambda$. The intuition: the n samples are crowded near 0 with density $\approx \lambda$ each, so near the origin they look like a Poisson process with rate $n\lambda$, whose first arrival is exponential with mean $1/(n\lambda)$. Multiplying by n undoes the crowding.

6.1.2 Convergence In Probability

Now that we have convergence in distribution, we take it one step higher. In the last section, we showed that if 2 random variables have the same distribution, it does not necessarily mean that the random variables are the same. Now we will define a notion of describing when 2 random variables are identical. Let $\{X_1, X_2, X_3, ..., X_n\}$ be a sequence of random variables. If

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \epsilon) = 0 \quad \forall \epsilon > 0 \tag{6.3}$$

Then we say that X_n **converges to X in probability**. We said that this was a stronger notion of convergence than weak convergence. To see this, fix any x that is a continuity point of F_X. We want to show

$$\lim_{n\to\infty} \mathbb{P}\left(X_n \leq x\right) = \mathbb{P}(X \leq x).$$

Fix a small $\delta > 0$. If $X_n \leq x$, then either $X \leq x + \delta$, or X and X_n are more than δ apart. That's just a union bound

$$\mathbb{P}\left(X_n \leq x\right) \leq \mathbb{P}\left(X \leq x + \delta\right) + \mathbb{P}\left(|X_n - X| > \delta\right)$$

The same logic the other way around (if $X \leq x - \delta$ then either $X_n \leq x$ or they are more than δ apart) gives

$$\mathbb{P}\left(X \leq x - \delta\right) \leq \mathbb{P}\left(X_n \leq x\right) + \mathbb{P}\left(|X_n - X| > \delta\right)$$

Now let $n \to \infty$. By convergence in probability, the $\mathbb{P}(|X_n - X| > \delta)$ terms die, so we've squeezed

$$F_X(x-\delta) \leq \liminf_{n\to\infty} \mathbb{P}(X_n \leq x) \leq \limsup_{n\to\infty} \mathbb{P}(X_n \leq x) \leq F_X(x+\delta)$$

Since δ was arbitrary and F_X is continuous at x, both ends go to $F_X(x)$ as $\delta \to 0$, ensuring

$$\lim_{n\to\infty} \mathbb{P}\left(X_n \leq x\right) = \mathbb{P}(X \leq x)$$

The above are the distributions of X_n and X by definition. Thus what we have shown is that if X_n converges in probability, then will also converge in distribution. The opposite is not necessarily true. Convergence in distribution only looks at the distributions one at a time, and never compares X_n and X on the same ω. Let us see an example.

Problem 3.

Let X_n be a sequence of independent random variables such that

$$\mathbb{P}(X_n = 0) = \mathbb{P}(X_n = 1) = \frac{1}{2}$$

Show that this does not converge in probability to any random variable but does converge in distribution.

Solution:

First let us show non-convergence in probability. Suppose $X_n \to Y$ in probability for some random variable Y. Then X_n and X_m would both be close to Y, and therefore close to each other, for large n and m. More precisely, by the triangle inequality, $|X_n - X_m| > \epsilon$ forces either $|X_n - Y| > \epsilon/2$ or $|X_m - Y| > \epsilon/2$, so

$$\mathbb{P}(|X_n - X_m| > \epsilon) \leq \mathbb{P}(|X_n - Y| > \epsilon/2) + \mathbb{P}(|X_m - Y| > \epsilon/2) \to 0$$

But for $n \neq m$, X_n and X_m are independent fair coin flips, so they disagree with probability $1/2$, and when they disagree $|X_n - X_m| = 1$. So $\mathbb{P}(|X_n - X_m| > 1/2) = 1/2$ for all $n \neq m$, which never goes to 0. Contradiction! Thus this sequence of random variables can never converge in probability. However, by construction, every X_n has the exact same CDF as the Bernoulli random variable with $p = 1/2$. Thus, the sequence X_n never converges in probability but trivially converges in distribution. (If we had dropped the word independent, then $X_n = X_1$ for all n would be allowed, and that certainly converges. Convergence in probability cares about how the X_n are built

on the same Ω, convergence in distribution does not.)

This was a non example. But what about the case in which it does converge in probability? How would we go about doing such a problem? Let us do an illustrative problem.

Problem 4.

Let $\{X_n, n \geq 1\}$ be a sequence of independent random variables with common density

$$f(x) = \begin{cases} \mathrm{e}^{-(x-a)}, & \text{for } x \geq a, \\ 0, & \text{for } x < a \end{cases}$$

Set $Z_n = \min\{X_1, X_2, \ldots, X_n\}$. Show that

$$Z_n \xrightarrow{p} a \quad \text{as } n \to \infty.$$

Solution:
This problem is a bit more involved, and will require us to remember back to transformations of random variables again. But no matter! We're used to it at this point. First, let us write the convergence in probability

$$\mathbb{P}(|Z_n - a| > \epsilon) \to 0$$

Since every $X_k \geq a$, the minimum is also at least a, so $|Z_n - a| > \epsilon$ just means $Z_n > a + \epsilon$. As in problem 2, the minimum is bigger than $a + \epsilon$ exactly when all of the X_k are, and $\mathbb{P}(X_k > a + \epsilon) = \int_{a+\epsilon}^{\infty} e^{-(x-a)} dx = e^{-\epsilon}$. By independence,

$$\mathbb{P}(|Z_n - a| > \epsilon) = \mathbb{P}(X_1 > a + \epsilon)^n = e^{-n\epsilon}$$

In the limit that $n \to \infty$, this goes to 0. Thus, we conclude that $Z_n \xrightarrow{p} a$

6.1.3 Convergence In L_p Mean

The next type of convergence is called **convergence in** L_p. Suppose that $\{X_1, ..., X_n\}$ is a sequence of random variables. If

$$\lim_{n\to\infty} \mathbb{E}\left[|X_n - X|^p\right] = 0 \tag{6.4}$$

then we say that X_n converges to X in L_p mean. This is the convergence in the norm $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$ that the Minkowski inequality of chapter 4 told us about. Most famously, and the result that we will be using a lot, is quadratic mean ($p = 2$). In fact, for any value of $p > 0$, if the sequence converges in L_p, then it converges in both probability and distribution. To see this, use the Markov inequality

$$\mathbb{P}(|X_n - X| > \epsilon) = \mathbb{P}(|X_n - X|^p > \epsilon^p) \leq \frac{\mathbb{E}\left[|X_n - X|^p\right]}{\epsilon^p}$$

If the term on the right hand side (the L_p mean) converges for all ϵ, then the probability of $|X_n - X|^p$ must tend to 0. This ensures convergence in probability. We've shown that convergence in probability implies convergence in distribution, therefore convergence in L_p mean also implies convergence in distribution. Obviously, this does not generalize going backwards. The skinny tall rectangles of chapter 3 converge to 0 in probability ($\mathbb{P}(X_n \neq 0) = 1/n$), but $\mathbb{E}|X_n - 0| = 1$ for every n, so they don't converge in L_1. A rare but enormous value is invisible to convergence in probability and fatal to convergence in mean. Let us do a problem.

Problem 5.

Let $X_1, \ldots, X_n$ be i.i.d. and let $\mu = \mathbb{E}(X_1)$. Suppose that the variance is finite. Show that $\bar{X}_n \xrightarrow{\text{qm}} \mu$.

Solution:

We want to show that

$$\lim_{n\to\infty} \mathbb{E}[(\bar{X}_n - \mu)^2] = 0$$

We can expand the quadratic

$$\implies \mathbb{E}\left[\bar{X}_n^2 - 2\mu\bar{X}_n + \mu^2\right]$$

Using the fact that μ is a constant allows us to use the linearity of the expectation

$$\implies \mathbb{E}[\bar{X}_n^2] - 2\mu\mathbb{E}[\bar{X}_n] + \mu^2$$

We know that the expectation of $\bar{X}$ is μ since each term in the sum contributes a μ and it is divided by n so the final result is just μ.

$$\implies \mathbb{E}[\bar{X}_n^2] - \mu^2$$

Now we need to find a way to deal with the second moment. We can rewrite this as

$$= \text{Var}[\bar{X}_n] + \mathbb{E}[\bar{X}_n]^2 - \mu^2$$

But we just showed that the expectation of the sample mean is μ. So the squared mean cancels out with the $-\mu^2$ term. What about the variance of the sample mean? Well, it has

$$\text{Var}[\bar{X}_n] = \text{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

Since the $1/n$ term is a constant (it comes out squared), the variances are finite (let us call them σ^2), and the variance of a sum of independent random variables is the sum of the variances, this simplifies to

$$\implies \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}$$

Now if we send the limit to ∞, we see that the variance goes to 0. Thus, the sample average converges to the true average in quadratic mean, and therefore in probability. What we have shown is a version of the **law of large numbers**, the subject of chapter 7. Notice how cheap this was, but also notice that we needed a finite variance; most of the work in chapter 7 is about getting rid of that assumption.

For a lot of problems, showing convergence in quadratic mean is really straightforward because you end up using expectation properties that we learned a while back.

6.1.4 Almost Sure Convergence

There is one more, and in some ways it's the most natural one, because it's the one you'd write down if you'd only ever taken real analysis. A random variable is a function of ω, so why not just ask that $X_n(\omega) \to X(\omega)$ as a sequence of numbers, for each ω? We don't need it for literally every ω, only for all of them except a set of probability zero. If

$$\mathbb{P}\left(\left\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1 \tag{6.5}$$

then we say that X_n converges to X **almost surely**, and write $X_n \xrightarrow{\text{a.s.}} X$. This is a statement about each sample path: if you run the experiment once and watch the whole sequence $X_1(\omega), X_2(\omega), \ldots$ unfold, it converges (with probability 1). Convergence in probability is a statement about each time n: at a fixed large n, most ω's have $X_n(\omega)$ close

to $X(\omega)$, but which ω's are the bad ones is allowed to change with n.

How do the two compare? We can rewrite almost sure convergence using the limits of sets from chapter 1. $X_n(\omega) \not\to X(\omega)$ means that for some $\epsilon > 0$, $|X_n(\omega) - X(\omega)| > \epsilon$ happens infinitely often. So (taking ϵ along $1/k$ to keep things countable)

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}\left(|X_n - X| > \epsilon \text{ i.o.}\right) = 0 \text{ for every } \epsilon > 0 \tag{6.6}$$

and by problem 8 of chapter 1, $\limsup \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n)$, so if the right hand side holds then $\mathbb{P}(|X_n - X| > \epsilon) \to 0$. **Almost sure convergence implies convergence in probability.** This also hands us the most useful tool for proving almost sure convergence. By the first Borel-Cantelli lemma,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(|X_n - X| > \epsilon\right) < \infty \text{ for every } \epsilon > 0 \implies X_n \xrightarrow{\text{a.s.}} X \tag{6.7}$$

So convergence in probability *fast enough* (summably fast) upgrades to almost sure convergence. The classic example of what goes wrong without the speed is the **typewriter sequence** in figure 6.1. Start with all of $[0, 1]$, then sweep an interval of length $1/2$ across it, then an interval of length $1/4$, then $1/8$, and so on, and let X_n be the indicator of the n-th interval. Then $\mathbb{P}(X_n \neq 0)$ is the length of the interval, which goes to 0, so $X_n \to 0$ in probability (and in every L_p). But every single ω gets swept over once in every pass, so $X_n(\omega) = 1$ infinitely often and $X_n(\omega)$ converges for no ω at all. The interval used at step n has length between $1/n$ and $2/n$, so the probabilities $\mathbb{P}(X_n \neq 0)$ go to zero like $1/n$, which is too slowly to be summable. Going the other way, the skinny tall rectangles converge almost surely but not in L_1, so neither of almost sure and L_p convergence implies the other. What saves the day there is uniform integrability, which is the subject of the next section.

6.1.5 Connection Of All Convergences

We've hinted at the relationship between all of these convergences in the past couple of sections. We can formally state the relation as follows: almost sure $\implies$ probability $\implies$ distribution, and $L_p \implies$ probability. Convergence in distribution implies convergence in probability only if the limit is a point mass (a constant c), because then $\mathbb{P}(|X_n - c| > \epsilon) = F_n(c - \epsilon) + 1 - F_n(c + \epsilon)$ (roughly) only depends on the distribution of X_n. The other arrows need extra conditions: convergence in probability plus uniform integrability gives convergence in L_1, and convergence in probability always has a subsequence that converges almost surely (pick the subsequence fast enough to make the probabilities summable). I will also give but not prove the following theorems (they're all variations on "limits respect arithmetic", and the proofs are ϵ-chasing). Let X_n, X, Y_n, Y be random variables. Let g be

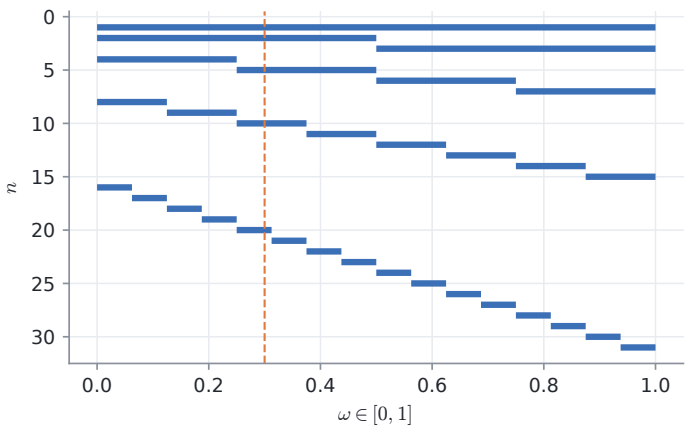


Figure 6.1: The typewriter sequence $X_n = \mathbb{I}_{I_n}$. Row n (counting downward) shows the interval I_n where $X_n = 1$. The intervals shrink, so $X_n \to 0$ in probability, but every vertical line, like the dashed one at a fixed ω, crosses infinitely many of them, so $X_n(\omega)$ never converges.

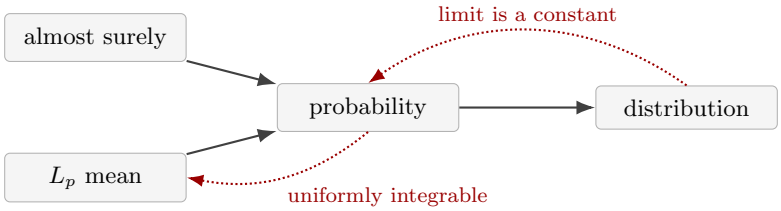


Figure 6.2: Relationship between all the convergences. Solid arrows always hold, dotted arrows need the extra condition written on them.

a continuous function.

(a) If $X_n \xrightarrow{\text{P}} X$ and $Y_n \xrightarrow{\text{P}} Y$, then $X_n + Y_n \xrightarrow{\text{P}} X + Y$.

(b) If $X_n \xrightarrow{\text{qm}} X$ and $Y_n \xrightarrow{\text{qm}} Y$, then $X_n + Y_n \xrightarrow{\text{qm}} X + Y$.

(c) If $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow c$, then $X_n + Y_n \rightsquigarrow X + c$.

(d) If $X_n \xrightarrow{\text{P}} X$ and $Y_n \xrightarrow{\text{P}} Y$, then $X_n Y_n \xrightarrow{\text{P}} XY$.

(e) If $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow c$, then $X_n Y_n \rightsquigarrow cX$.

(f) If $X_n \xrightarrow{\text{P}} X$, then $g(X_n) \xrightarrow{\text{P}} g(X)$.

(g) If $X_n \rightsquigarrow X$, then $g(X_n) \rightsquigarrow g(X)$.

For those that are familiar, parts (c) and (e) are known as **Slutsky's theorem**, and (f)

and (g) are the **continuous mapping theorem**. Notice that (c) needs one of the limits to be a constant. If $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow Y$ separately, we know nothing about $X_n + Y_n$: take $X_n = Z$ and $Y_n = -Z$ for a standard normal Z, so that $Y_n \rightsquigarrow Z$ as well, but $X_n + Y_n = 0$, which is not $Z + Z$ in distribution. Distributions alone don't know how the variables are glued together.

6.2 Uniform Integrability

In this section, we are going to build the tool that fixes the skinny tall rectangles: the extra condition under which convergence in probability (or almost surely) lets us swap limits and expectations. Suppose we have a family of random variables $\{X_\alpha\}_{\alpha \in A}$. We want a notion of describing uniformity across all members of the family. That is, we do not want random variables that have disproportionately heavy tails (or rare events with extremely large values influencing the average). If it is the case that

$$\lim_{c \to \infty} \sup_{\alpha \in A} \mathbb{E}\left[|X_\alpha| \, \mathbb{I}_{\{|X_\alpha| > c\}}\right] = 0 \tag{6.8}$$

then we say that the family of random variables is **uniformly integrable.** Let us try to break this down piece by piece, starting with $|X_\alpha|$: this is the absolute value of the random variable X_α. It measures how large X_α is, regardless of its sign. $\mathbb{I}_{\{|X_\alpha| > c\}}$ is an indicator function. For a given constant $c > 0$,

$$\mathbb{I}_{\{|X_\alpha| > c\}} = \begin{cases} 1 & \text{if } |X_\alpha| > c \\ 0 & \text{otherwise} \end{cases}$$

The indicator picks out the part of the probability space where $|X_\alpha|$ exceeds the threshold c. Multiplying $|X_\alpha|$ by the indicator restricts our attention to the event $\{|X_\alpha| > c\}$. This product equals $|X_\alpha|$ on the set where $|X_\alpha| > c$ and is 0 elsewhere. That is, we're looking at the magnitude of X_α only in the tails beyond c. Taking the average $\mathbb{E}\left[|X_\alpha| \, \mathbb{I}_{\{|X_\alpha| > c\}}\right]$ tells us how much of the expectation of $|X_\alpha|$ comes from values beyond c (the tails). Now we take the supremum of these tail expectations over all random variables in the family $\{X_\alpha\}$. This gives us a "worst-case" tail expectation at the cutoff c. Now, when we take the limit that $c \to \infty$, we send the threshold further along the tails. If the worst case tail expectation goes to 0, then every member in the family has small tails, and the tail contribution can be made arbitrarily small uniformly across the family. A single integrable random variable is always uniformly integrable (that's dominated convergence), and the point is that here one c has to work for everybody. The skinny tall rectangles fail exactly this test: for $n > c$, $\mathbb{E}[X_n \mathbb{I}_{\{X_n > c\}}] = \mathbb{E}[X_n] = 1$, so the sup is always 1. All of their expectation lives in the tail.

And the payoff is the **Vitali convergence theorem**: if $X_n \to X$ in probability and $\{X_n\}$ is uniformly integrable, then $\mathbb{E}|X_n - X| \to 0$, and in particular $\mathbb{E}[X_n] \to \mathbb{E}[X]$. It's the sharp version of dominated convergence. Let us do an obvious example

Problem 6.

Consider a family of random variables $\{X_\alpha\}$ such that $|X_\alpha| \leq M$ for all values of α, where M is some nonnegative constant. Show that this is uniformly integrable.
Solution:
Obviously this is uniformly integrable. First start with

$$\sup_{\alpha \in A} \mathbb{E}\left[|X_\alpha| \, \mathbb{I}_{\{|X_\alpha| > c\}}\right]$$

The worst case scenario. Now, if we send c to be very large, there will be a value for c such that $c > M$. But every random variable is bounded by M, so the expectation at these tails will be 0. Thus, this family is uniformly integrable.

The above problem points at something rather interesting. We don't need the random variables to be bounded, we just need their tails to be under control, and a uniform bound on a higher moment does that. Suppose that for some $p > 1$,

$$\sup_{\alpha \in A} \mathbb{E}\left[|X_\alpha|^p\right] \leq M < \infty$$

On the event $\{|X_\alpha| > c\}$ we have $|X_\alpha|/c > 1$, so $|X_\alpha| \leq |X_\alpha| \left(|X_\alpha|/c\right)^{p-1} = |X_\alpha|^p / c^{p-1}$. Thus

$$\mathbb{E}\left[|X_\alpha| \, \mathbb{I}_{\{|X_\alpha| > c\}}\right] \leq \frac{1}{c^{p-1}} \mathbb{E}\left[|X_\alpha|^p\right] \leq \frac{M}{c^{p-1}}$$

which goes to 0 as $c \to \infty$ uniformly in α, since $p - 1 > 0$. So bounded in L_p for some $p > 1$ implies uniformly integrable. (Bounded in L_1 is not enough, and the skinny tall rectangles are again the counterexample.) Let us try another non obvious example

Problem 7.

Suppose that $X_n \in \text{Exp}\left(\lambda_n\right), n \geq 1$. When is $\{X_n, n \geq 1\}$ uniformly integrable?
Solution:

First, let us apply the definition of uniform integrability

$$\lim_{c\to\infty}\sup_{\alpha\in A}\mathbb{E}\left[|X_\alpha|\,\mathbb{I}_{\{|X_\alpha|>c\}}\right]=0$$

Now, we know that the tail expectation of an exponential random variable is

$$\mathbb{E}\left[X_n\mathbb{I}_{\{X_n>c\}}\right]=\lambda_n\int_c^\infty xe^{-\lambda_n x}dx=\left(c+\frac{1}{\lambda_n}\right)\mathrm{e}^{-c\lambda_n}$$

For a single n, this goes to 0 as $c\to\infty$. But we need the supremum over n. The right hand side is a decreasing function of λ_n (both factors shrink as λ_n grows), so the worst case is the smallest rate. If $\inf_n\lambda_n=\lambda_*>0$, then the sup is at most $(c+1/\lambda_*)e^{-c\lambda_*}\to 0$, and the family is uniformly integrable. If $\inf_n\lambda_n=0$, we can pick λ_n small enough that $\frac{1}{\lambda_n}e^{-c\lambda_n}$ is as large as we want for any fixed c, so the family is not uniformly integrable. The answer: exactly when $\inf_n\lambda_n>0$, which is the same as saying the means $1/\lambda_n$ are bounded.

6.3 Vague Convergence

We've covered the meat and bone of this chapter in the previous sections; the following will just be some seasoning on top. Is vague convergence and Helly's theorem necessary? To be honest, no. But if we are to consider the complete theory of probability, it might be good to see. Suppose (μ_n) is a sequence of (finite or σ-finite) measures on a locally compact Hausdorff space X, such as $\mathbb{R}^d$. We say μ_n converges vaguely to a measure μ if for every continuous function $f:X\to\mathbb{R}$ with compact support, we have:

$$\lim_{n\to\infty}\int_X f d\mu_n=\int_X f d\mu.$$

This form looks similar to that of weak convergence. However, vague convergence is a weaker form of weak convergence. Weak convergence tests against all bounded continuous functions, while vague convergence only tests against continuous functions with compact support. This is weaker, because a compactly supported function can't see what happens far away, so it allows the total mass to leak out past any fixed compact set. The condition that stops the leaking is called **tightness**: for every $\epsilon>0$ there is one compact set $[-K,K]$ that holds all but ϵ of the mass of every μ_n. Vague convergence plus tightness is weak convergence. Here is a basic example of a mass escaping out to infinity.

Consider a sequence of measures μ_n on the real line $\mathbb{R}$, defined by placing a unit mass at

the point n :

$$\mu_n = \delta_n$$

where δ_n is the Dirac measure at n. This means $\mu_n(\{n\}) = 1$ and $\mu_n(B) = 0$ for any Borel set B that does not contain n. As $n \to \infty$, the single "spike" of mass moves further and further out to the right. If we test these measures against any continuous function with compact support $f : \mathbb{R} \to \mathbb{R}$, what happens? Since f has compact support, there exists some large number $M > 0$ such that $f(x) = 0$ for all $|x| > M$. That means for large enough n, the point n (where μ_n has its mass) is outside the support of f. Hence $f(n) = 0$. To restate formally

$$\int_{\mathbb{R}} f(x) d\mu_n(x) = f(n).$$

For large n, because n goes beyond the finite support of f, we have $f(n) = 0$. Thus

$$\lim_{n\to\infty} \int_{\mathbb{R}} f(x) d\mu_n(x) = \lim_{n\to\infty} f(n) = 0.$$

This is exactly the integral of f against the zero measure μ (which assigns zero mass to every set)

$$\int_{\mathbb{R}} f(x) d\mu(x) = 0.$$

Therefore, μ_n converges vaguely to the zero measure. In symbols

$$\mu_n \xrightarrow{\text{vague}} 0 \quad \text{(the zero measure)}.$$

6.3.1 Helly's Theorem

The reason we care about this is because we want to get to Helly's theorem. Consider a sequence (F_n) of distribution functions on the real line. Each F_n is non-decreasing, right-continuous, and satisfies $\lim_{x\to-\infty} F_n(x) = 0$ and $\lim_{x\to\infty} F_n(x) = 1$. Helly's theorem states that

1. There exists a subsequence (F_{n_k}) and a non-decreasing, right-continuous function F such that:

$$F_{n_k}(x) \to F(x) \quad \text{for all } x \text{ at which } F \text{ is continuous.}$$

2. The limit F satisfies $0 \le F \le 1$, but it is NOT always a distribution function. It might have $F(\infty) < 1$ or $F(-\infty) > 0$, because some mass could have escaped to infinity, exactly like the δ_n example above (there $F_n(x) \to 0$ for every x). It is a genuine distribution function if and only if the sequence is tight.

In the language of the previous paragraphs, Helly's theorem says every sequence of distributions has a vaguely convergent subsequence. It's the probabilistic version of Bolzano-Weierstrass (every bounded sequence has a convergent subsequence), and tightness is what upgrades it to weak convergence.

To be honest I don't really care too much about vague convergence because...it's so weak. But Helly plus tightness is exactly what's needed to prove the Lévy continuity theorem, which is the one we really care about.

6.4 Continuity Theorems And The Characteristic Function

In chapter 5 we promised the Lévy continuity theorem. Here, we will state it and see how it makes sense in the bigger picture. Consider a sequence of random variables (X_n) and another random variable X. Let φ_n and φ be their characteristic functions, defined by:

$$\varphi_n(t) = \mathbb{E}\left[e^{itX_n}\right], \quad \varphi(t) = \mathbb{E}\left[e^{itX}\right]$$

Lévy's continuity theorem has two halves.

(1) If $X_n \rightsquigarrow X$, then $\varphi_n(t) \to \varphi(t)$ for every $t \in \mathbb{R}$.
(2) Conversely, if $\varphi_n(t)$ converges for every t to some function $\varphi(t)$, and that limit is continuous at $t = 0$, then φ is the characteristic function of some random variable X, and $X_n \rightsquigarrow X$.

The first half is easy, once you have the second definition of weak convergence, $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for every bounded continuous f (this equivalence is part of a theorem with the excellent name of the **Portmanteau theorem**; I'm looking at its proof and it's long and not very instructive, so we will just take it as is). Then $\cos(tx)$ and $\sin(tx)$ are bounded and continuous, so $\varphi_n(t) = \mathbb{E}[\cos tX_n] + i\mathbb{E}[\sin tX_n] \to \varphi(t)$. Done.

The second half is the useful one, because it's how we prove things converge: compute the characteristic functions, which are often easy, and take their limit. The proof goes through Helly's theorem. Some subsequence of the F_n converges to some F, and the only question is whether F is a real distribution or whether mass has leaked off to infinity. This is where continuity at 0 comes in. Mass escaping to infinity means X_n takes huge values, and e^{itX_n} then spins around the unit circle extremely fast for any fixed $t \neq 0$, so its average is small. That makes $\varphi_n(t)$ drop away from $\varphi_n(0) = 1$ as soon as t moves off zero, and in the limit φ would be discontinuous at 0. Making that quantitative gives tightness, and

then every subsequence converges to a distribution with characteristic function φ, which by uniqueness is the same distribution every time. You can see the failure mode in the δ_n example: $\varphi_n(t) = e^{itn}$ doesn't converge at all for $t \neq 0$. And for $X_n \sim N(0, n)$, whose mass spreads out to infinity, $\varphi_n(t) = e^{-nt^2/2} \to \mathbb{I}_{\{t=0\}}$, which is not continuous at 0.

Although this section is not very technical, it's the engine behind the central limit theorem: in chapter 8 we'll compute the characteristic function of a standardized sum, watch it converge to $e^{-t^2/2}$, and Lévy will do the rest. Let's do an example.

Problem 8.

Suppose that $X_n \in \text{Geometric}\left(\frac{\lambda}{n+\lambda}\right), n = 1, 2, \ldots$, where λ is a positive constant. Show that X_n/n converges in distribution to an exponential distribution as $n \to \infty$, and determine the parameter of the limit distribution.

Solution:

We can solve this problem by finding the characteristic function of the geometric random variable (counting trials until the first success, $\mathbb{P}(X = k) = p(1-p)^{k-1}$ for $k \geq 1$). It's a geometric series

$$\varphi_X(t) = \sum_{k=1}^{\infty} e^{itk} p(1-p)^{k-1} = pe^{it} \sum_{j=0}^{\infty} \left((1-p)e^{it}\right)^j$$

which sums to

$$\varphi_X(t) = \frac{pe^{it}}{1-(1-p)e^{it}}$$

We can substitute $p = \frac{\lambda}{n+\lambda}$, and the scaling property says that dividing by n just replaces t with t/n

$$\varphi_{X_n}(t) = \frac{\frac{\lambda}{n+\lambda} e^{it/n}}{1-(1-\frac{\lambda}{n+\lambda})e^{it/n}}$$

Multiplying the top and bottom by $n + \lambda$ and then simplifying gives

$$\frac{\lambda e^{it/n}}{\lambda + n\left(1 - e^{it/n}\right)}$$

In the limit that $n \to \infty$, the argument it/n of the exponential becomes very small and we can do a Taylor series expansion, $e^{it/n} = 1 + \frac{it}{n} + O(n^{-2})$, so that $n(1 - e^{it/n}) \to -it$

$$\implies \frac{\lambda(1+\frac{it}{n})}{\lambda + n\left(1-(1+\frac{it}{n})\right)} = \frac{\lambda(1+\frac{it}{n})}{\lambda - it}$$

Now in the limit that $n \to \infty$, we are left with

$$\varphi_{X_n/n}(t) = \frac{\lambda}{\lambda - it}$$

Which is the characteristic function of the exponential distribution with parameter λ (rate λ, mean $1/\lambda$). It's continuous at 0, so by Lévy, X_n/n converges in distribution to it. This makes sense: a geometric is the discrete-time version of waiting for an event, the exponential is the continuous-time version, and measuring time in units of n steps with success probability $\approx \lambda/n$ per step turns one into the other.

6.5 Problems

1. Let $X_1, X_2, \ldots$ be a sequence of random variables. Show that $X_n \xrightarrow{\text{qm}} b$ if and only if

$$\lim_{n\to\infty} \mathbb{E}(X_n) = b \quad \text{and} \quad \lim_{n\to\infty} \mathbb{V}(X_n) = 0$$

2. Let $X_1, X_2, \ldots$ be a sequence of random variables such that

$$\mathbb{P}\left(X_n = \frac{1}{n}\right) = 1 - \frac{1}{n^2} \quad \text{and} \quad \mathbb{P}(X_n = n) = \frac{1}{n^2}.$$

Does X_n converge in probability? Does X_n converge in quadratic mean?

3. Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. Prove that

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{\text{P}} p \quad \text{and} \quad \frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{\text{qm}} p.$$

4. Suppose that $X_n \sim N(0, 1/n)$ and let X be a random variable with distribution $F(x) = 0$ if $x < 0$ and $F(x) = 1$ if $x \geq 0$. Does X_n converge to X in probability? (Prove or disprove). Does X_n converge to X in distribution? (Prove or disprove).

5. Suppose that $X_n \in \text{Ge}(p_n), n \geq 1$. Show that

(a) if $p_n \to p > 0$, then $X_n \xrightarrow{d} \text{Ge}(p)$ as $n \to \infty$.

(b) if $p_n \to 0$ and $np_n \to \lambda > 0$, then X_n/n converges in distribution to an exponential with mean $1/\lambda$ as $n \to \infty$.

6 Let $X_1, X_2, \ldots$ be independent random variables with common characteristic function

$$\varphi(t) = \begin{cases} 1 - \sqrt{|t|(2-|t|)}, & \text{for } |t| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

and let $\{S_n, n \geq 1\}$ denote the partial sums. Show that $\frac{S_n}{n^2}$ converges in distribution as $n \to \infty$ and determine the limit.

7. Consider a sequence of random variables $\{X_n\}$ defined by:

$$X_n = \begin{cases} n & \text{with probability } \frac{1}{2n} \\ 0 & \text{with probability } 1 - \frac{1}{2n} \end{cases}$$

Is it uniformly integrable?

7

The Law Of Large Numbers

"Just give it time."

– Hieu

The law of large numbers is one of the most important results of probability theory (second only to the central limit theorem). It is the central ingredient in what makes statistics possible, and its statement is, while intuitive, also deeply profound. In this chapter, we will discuss both the **weak law of large numbers** and the **strong law of large numbers**. My goal is less to prove them in their most general form and more to understand where they come from, so we'll derive each one a few different ways, starting with the cheapest derivation and slowly paying for more generality. We will also provide commentary on what consequences these laws hold for statistics, gambling, and physics.

Throughout this chapter, $X_1, X_2, \ldots$ will be i.i.d. random variables, and we write

$$S_n = \sum_{k=1}^{n} X_k, \qquad \bar{X}_n = \frac{S_n}{n} \tag{7.1}$$

for the partial sum and the sample average.

7.1 The Weak Law Of Large Numbers

The weak law of large numbers is the statement that if the X_k are i.i.d. with a finite mean $\mu = \mathbb{E}[X_1]$, then the sample average converges in probability to the true population mean

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu, \qquad \text{that is,} \qquad \lim_{n\to\infty} \mathbb{P}\left(|\bar{X}_n - \mu| > \epsilon\right) = 0 \text{ for every } \epsilon > 0 \tag{7.2}$$

Why is this such a profound statement? Note that we didn't make any assumptions on the underlying distribution of the random variables X_k, other than that they have a mean. In real life, we oftentimes have no idea what the distribution of the X_k is, and we need to make conclusions on population parameters like μ. The law of large numbers tells us that if our sample size is large enough, then the sample average is a reasonable approximation of the population mean. A very intuitive example of this is the coin toss. Suppose we flip a coin n times, and we denote the outcome of heads to be 1 and the outcome of tails to be 0. If one were to estimate the probability of heads p, then the obvious answer would be the number of heads divided by the number of coin tosses. But the number of heads is just the sum of the X_k, so our estimate is

$$\hat{p}_n = \frac{1}{n}\sum_{k=1}^{n} X_k = \bar{X}_n$$

and since $\mathbb{E}[X_k] = p$, the law of large numbers says that $\hat{p}_n \to p$. This is the bridge between the axiomatic notion of probability from chapter 1 (a number assigned by a measure) and the frequentist one (the long run fraction of times something happens). Without the law of large numbers, those would just be two different things that happen to share a name.

I've mentioned before that there exists a strong law of large numbers, and the only difference is that the convergence is almost sure as opposed to in probability. Well, if we could show almost sure convergence, why even bother with the weak law? There are a couple things to consider. The first and most obvious answer is that the strong law requires a great deal of stuff from analysis, while the weak law can be done in two lines (if you're willing to assume a little more). The second reason is that sometimes the weak law is all that we need, and in statistics it usually is: we care about the probability that *this* estimate, with *this* sample size, is off.

7.1.1 The Two Line Derivation

Let's start with the cheapest possible derivation, which needs one extra assumption: a finite variance $\sigma^2 = \text{Var}[X_1]$. Since the X_k are independent, the variances add, so

$$\mathbb{E}\left[\bar{X}_n\right] = \mu, \qquad \text{Var}\left[\bar{X}_n\right] = \frac{1}{n^2}\,\text{Var}\left[S_n\right] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

and Chebyshev's inequality from chapter 4 finishes the job

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2} \underset{n\to\infty}{\longrightarrow} 0 \tag{7.3}$$

That's it. (We did the same calculation in chapter 6, phrased as convergence in quadratic mean.) This derivation is worth staring at, because it tells you *why* averaging works. The sum S_n has a spread that grows like $\sqrt{n}$ (its standard deviation is $\sigma\sqrt{n}$), because independent fluctuations partially cancel each other. Its mean grows like n. Dividing by n squeezes the spread down to $\sigma/\sqrt{n}$, which goes to zero. Figure 7.1 shows this happening: every path of running averages funnels into the band around μ. Notice also what we didn't use. We only needed the variances to add, which is true as long as the X_k are uncorrelated (pairwise independence is plenty), and they don't even need to be identically distributed as long as the variances stay bounded.

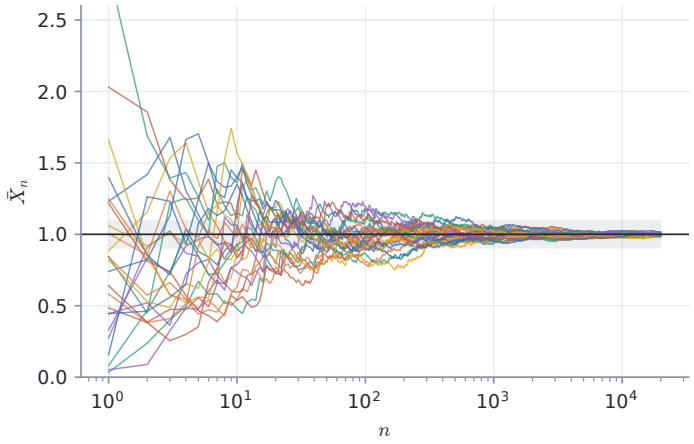


Figure 7.1: Running averages $\bar{X}_n$ for 25 independent sequences of Exp(1) samples (so $\mu = 1$, the black line). The gray band is $\mu \pm 0.1$. The weak law says that at any fixed large n, most paths are inside the band. The strong law says that each path eventually enters the band and never leaves.

7.1.2 A Derivation With Characteristic Functions

Can we get rid of the finite variance? Here is a derivation that only needs the mean, and it's the one I find the prettiest, because it uses everything we built in chapters 5 and 6. Let $\varphi(t) = \mathbb{E}[e^{itX_1}]$. By independence and the scaling property, the characteristic function of the sample average is

$$\varphi_{\bar{X}_n}(t) = \mathbb{E}\left[e^{i\frac{t}{n}(X_1+\cdots+X_n)}\right] = \left[\varphi\left(\frac{t}{n}\right)\right]^n$$

Now we only need to know φ near $t = 0$, because t/n is getting small. If $\mathbb{E}|X_1| < \infty$, then φ is differentiable at 0 with $\varphi'(0) = i\mu$ (differentiate under the expectation; the difference quotient $(e^{ihX} - 1)/h$ is bounded by $|X|$, which is integrable, so dominated convergence lets us take $h \to 0$ inside). So the first order Taylor expansion is

$$\varphi(s) = 1 + i\mu s + o(s) \qquad \text{as } s \to 0$$

and therefore

$$\varphi_{\bar{X}_n}(t) = \left[1 + \frac{i\mu t}{n} + o\left(\frac{1}{n}\right)\right]^n \underset{n\to\infty}{\longrightarrow} e^{i\mu t} \tag{7.4}$$

by the same limit that defines the exponential. But $e^{i\mu t}$ is the characteristic function of the constant μ (a random variable that is always μ has $\mathbb{E}[e^{it\mu}] = e^{it\mu}$). It's continuous at 0, so by the Lévy continuity theorem, $\bar{X}_n \rightsquigarrow \mu$ in distribution. And we showed in chapter 6 that convergence in distribution to a *constant* implies convergence in probability. That's the weak law, with nothing but a finite mean!

There's a nice way to see what happened. Taking logs, $\log \varphi_{\bar{X}_n}(t) = n \log \varphi(t/n)$. The log of the characteristic function is the cumulant generating function, so this says that the m-th cumulant of $\bar{X}_n$ is $n \cdot \kappa_m/n^m = \kappa_m/n^{m-1}$. The mean ($m = 1$) survives untouched, and every other cumulant (the variance, the skewness, and so on) dies. A distribution with only a first cumulant is a point mass. Keep this picture in your head, because in chapter 8 we'll zoom in by a factor of $\sqrt{n}$ instead of n, and then the variance will survive too.

7.1.3 Truncation And Convergence Equivalence

The characteristic function derivation is slick, but it hides the mechanism, and it doesn't generalize well to almost sure convergence. The "by hand" way to deal with random variables that might not have a variance is **truncation**: chop off the large values, which are rare, and use Chebyshev on what's left, which is bounded and so has a variance. The large values are rare enough that chopping them off doesn't change anything in the limit. To make "doesn't change anything" precise, we need some sort of definition for when two sequences are interchangeable.

Suppose that we have 2 sequences of random variables $\{X_n\}$ and $\{Y_n\}$. We say that X_n and Y_n are **convergence equivalent** if

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) < \infty \tag{7.5}$$

What this means is, by the first Borel-Cantelli lemma, $X_n \neq Y_n$ happens only finitely many

times, almost surely. So for almost every ω, the two sequences are *identical* from some point on. That has two consequences that we will use constantly. First, $\sum_n (X_n - Y_n)$ is a sum with only finitely many nonzero terms, so $\sum X_n$ converges exactly when $\sum Y_n$ does. Second, $\frac{1}{a_n}\sum_{k\le n}(X_k - Y_k) \to 0$ almost surely for any $a_n \to \infty$, since the numerator stops changing after a while. So for the purposes of averages, we can swap one sequence for the other.

When is a truncated sequence convergence equivalent to the original? Here's the key estimate. For any non-negative random variable Y,

$$\sum_{n=1}^{\infty} \mathbb{P}(Y > n) \le \mathbb{E}[Y] \le 1 + \sum_{n=1}^{\infty} \mathbb{P}(Y > n) \tag{7.6}$$

The reason is in figure 7.2. Problem 4 at the end of chapter 3 had you show that $\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y > x)\,dx$, the area under the survival function. Since $\mathbb{P}(Y > x)$ is decreasing, on the interval $[n-1, n]$ it's at least $\mathbb{P}(Y > n)$ and at most $\mathbb{P}(Y > n-1)$. So the sum is a staircase of unit-width rectangles that sits just under the curve (giving the left inequality) or, shifted by one, just over it (giving the right one). Summing over a lattice and integrating are the same up to the first step.

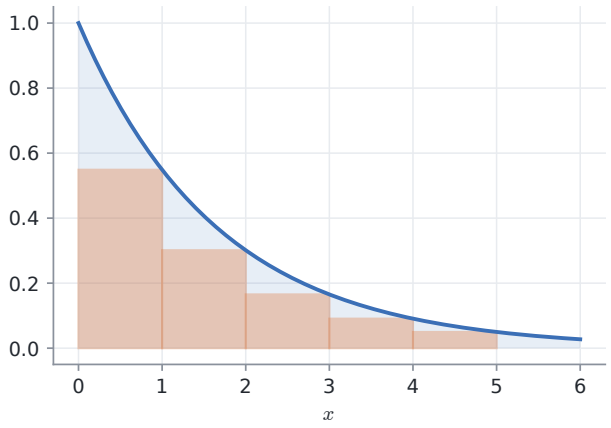


Figure 7.2: The area under the survival function $\mathbb{P}(Y > x)$ (blue curve, shaded) is $\mathbb{E}[Y]$. The orange rectangles of height $\mathbb{P}(Y > n)$ on $[n-1, n]$ fit underneath, so $\sum_n \mathbb{P}(Y > n) \le \mathbb{E}[Y]$.

Let us explore an example. Suppose that $\{X_n\}$ are i.i.d. with $\mathbb{E}[|X_1|^r] < \infty$ for some $r > 0$. Let us define $Y_n = X_n \mathbb{I}_{\{|X_n| \le n^{1/r}\}}$, so we chop X_n off at a level that grows with n. Are Y_n and X_n convergence equivalent? The two differ exactly when the truncation does something,

which is when $|X_n| > n^{1/r}$, so

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_n| > n^{1/r}) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1|^r > n) \leq \mathbb{E}\left[|X_1|^r\right] < \infty$$

where we used that the X_n are identically distributed, and then (7.6) with $Y = |X_1|^r$. So yes. In words: if $|X|^r$ has a finite mean, then the n-th sample essentially never exceeds $n^{1/r}$. With $r = 1$, this is the statement we'll use for the strong law: if $\mathbb{E}|X_1| < \infty$, then $|X_n| \leq n$ eventually, almost surely, and we lose nothing by truncating X_n at level n.

7.1.4 The Weak Law By Truncation

Now let's prove the weak law using only $\mathbb{E}|X_1| < \infty$, without characteristic functions. Without loss of generality, we can assume that $\mu = 0$ (otherwise replace X_k by $X_k - \mu$). We want to show that $\mathbb{P}(|S_n| > n\epsilon) \to 0$. First, a tool. If we truncate at a level c, with $Y_k = X_k \mathbb{I}_{\{|X_k| \leq c\}}$ and $S'_n = \sum_{k=1}^{n} Y_k$, then

$$\mathbb{P}\left(|S_n - \mathbb{E}\left[S'_n\right]| > x\right) \leq \frac{n \operatorname{Var}\left[Y_1\right]}{x^2} + n\mathbb{P}(|X_1| > c) \tag{7.7}$$

which is sometimes called the **truncated Chebyshev inequality**. It's two lines. Either $S_n = S'_n$, in which case Chebyshev applies to S'_n (a sum of n independent bounded variables, with variance $n \operatorname{Var}[Y_1]$), or some X_k got truncated, which by the union bound has probability at most $n\mathbb{P}(|X_1| > c)$.

The art is in choosing c. It has to be big enough that the truncation rarely does anything, but small enough that the truncated variables have a small variance. Let's pick $c = n\epsilon^3$ (the weird power will make the arithmetic work), so $Y_{k,n} = X_k \mathbb{I}_{\{|X_k| \leq n\epsilon^3\}}$, and set $x = n\epsilon$ in (7.7)

$$\mathbb{P}\left(|S_n - \mathbb{E}\left[S'_n\right]| > n\varepsilon\right) \leq \frac{\operatorname{Var}\left[Y_{1,n}\right]}{n\varepsilon^2} + n\mathbb{P}\left(|X_1| > n\varepsilon^3\right) \tag{7.8}$$

We can get rid of the variance by bounding it with the second moment, $\operatorname{Var}[Y] \leq \mathbb{E}[Y^2]$. Then we use a bit of a trick: $Y_{1,n}^2 = |Y_{1,n}| \cdot |Y_{1,n}|$, and one of the two factors is at most the truncation level $n\epsilon^3$, so $Y_{1,n}^2 \leq n\epsilon^3 |X_1|$. Thus

$$\frac{\operatorname{Var}\left[Y_{1,n}\right]}{n\varepsilon^2} \leq \frac{n\epsilon^3 \mathbb{E}|X_1|}{n\epsilon^2} = \epsilon \mathbb{E}|X_1|$$

This is great, because $\mathbb{E}|X_1|$ is just a constant, so this term can be made arbitrarily small with ϵ. We focus our attention on the right most term, regarding the probability. For any

$x > 0$, on the event $\{|X_1| > x\}$ we have $x < |X_1|$, so

$$x\,\mathbb{P}\left(|X_1| > x\right) \leq \mathbb{E}\left[|X_1|\mathbb{I}_{\{|X_1|>x\}}\right] \underset{x\to\infty}{\longrightarrow} 0 \tag{7.9}$$

where the limit is dominated convergence (or just: the tail of a convergent integral goes to 0). Setting $x = n\epsilon^3$, we get $n\epsilon^3\,\mathbb{P}(|X_1| > n\epsilon^3) \to 0$, and ϵ^3 is a fixed constant, so $n\,\mathbb{P}(|X_1| > n\epsilon^3) \to 0$ too. Putting it together,

$$\limsup_{n\to\infty} \mathbb{P}\left(|S_n - \mathbb{E}\left[S_n'\right]| > n\varepsilon\right) \leq \epsilon\,\mathbb{E}|X_1|$$

Fantastic work so far! All that is left to show is that the centering $\frac{1}{n}\mathbb{E}[S_n']$ goes to 0. Let us take a look at that term

$$|\mathbb{E}\left[S_n'\right]| = \left|\sum_{k=1}^{n} \mathbb{E}\left[X_k\mathbb{I}_{\{|X_k|\leq n\varepsilon^3\}}\right]\right| = n\left|\mathbb{E}\left[X_1\mathbb{I}_{\{|X_1|\leq n\varepsilon^3\}}\right]\right|$$

Since the expectation $\mathbb{E}[X_1] = 0$, splitting it into the truncated part and the tail gives

$$0 = \mathbb{E}\left[X_1\mathbb{I}_{\{|X_1|\leq n\varepsilon^3\}}\right] + \mathbb{E}\left[X_1\mathbb{I}_{\{|X_1|>n\varepsilon^3\}}\right]$$

so the truncated mean is minus the tail, and by the triangle inequality

$$\left|\frac{\mathbb{E}\left[S_n'\right]}{n}\right| \leq \mathbb{E}\left[|X_1|\,\mathbb{I}_{\{|X_1|>n\varepsilon^3\}}\right] \underset{n\to\infty}{\longrightarrow} 0$$

by the same tail argument as (7.9). So for n large, $|\mathbb{E}[S_n']| < n\epsilon$, and then $|S_n| > 2n\epsilon$ forces $|S_n - \mathbb{E}[S_n']| > n\epsilon$. Therefore

$$\limsup_{n\to\infty} \mathbb{P}\left(\left|\bar{X}_n\right| > 2\epsilon\right) \leq \epsilon\,\mathbb{E}|X_1| \tag{7.10}$$

Since the probability on the left only gets bigger when we shrink ϵ, and the right hand side goes to 0 as $\epsilon \to 0$, the limit must be 0 for every ϵ. This completes the proof of the weak law of large numbers.

Let us take a deep breath, because there's a lot happening. The proof has the same skeleton as the two line derivation (Chebyshev), with two extra pieces of bookkeeping: the probability that truncation did anything, and the bias that truncation introduced. Both are controlled by the tail $\mathbb{E}[|X_1|\mathbb{I}_{\{|X_1|>x\}}]$, and that's the only place the finite mean was used. In fact, looking back, all we needed was $x\,\mathbb{P}(|X_1| > x) \to 0$ (plus a little care with the centering), which is *weaker* than a finite mean. This will come back in the problems.

7.1.5 When The Mean Doesn't Exist

What happens when our assumptions are wrong? Do we still get stabilization? Consider the following situation. Let $\{X_n\}$ be a sequence of i.i.d. Cauchy random variables. If we define S_n to be the partial sum, then what does $\bar{X}_n$ converge to? We can answer easily by taking a look at the characteristic function of the Cauchy distribution, which we found in chapter 5

$$f_X(x) = \frac{1}{\pi\left(x^2+1\right)} \quad \Longrightarrow \quad \varphi_X(t) = \mathbb{E}\left[e^{itX}\right] = e^{-|t|}$$

Since the $\{X_n\}$ are independent, the characteristic function of the average is

$$\varphi_{\bar{X}_n}(t) = \left[\varphi\left(\frac{t}{n}\right)\right]^n = e^{-\left|\frac{t}{n}\right|n} = e^{-|t|}$$

which just means that the average of n independent Cauchy random variables is again a standard Cauchy random variable. It doesn't matter if you average 1 or a million of them, the average is exactly as spread out as a single sample! What does this tell us? Clearly, the weak law of large numbers failed to stabilize the average, but this makes sense. Recall that the Cauchy distribution does not have a mean ($\mathbb{E}|X| = \int \frac{|x|}{\pi(1+x^2)}dx$ diverges logarithmically), and so we violated the key assumption of the weak law. Where exactly did the characteristic function derivation break? The Cauchy $\varphi(t) = e^{-|t|} = 1 - |t| + \cdots$ has a kink at $t = 0$, so there's no linear term $i\mu t$ to exponentiate. Figure 7.3 shows what this looks like: every so often, a single enormous sample comes along and knocks the average somewhere new.

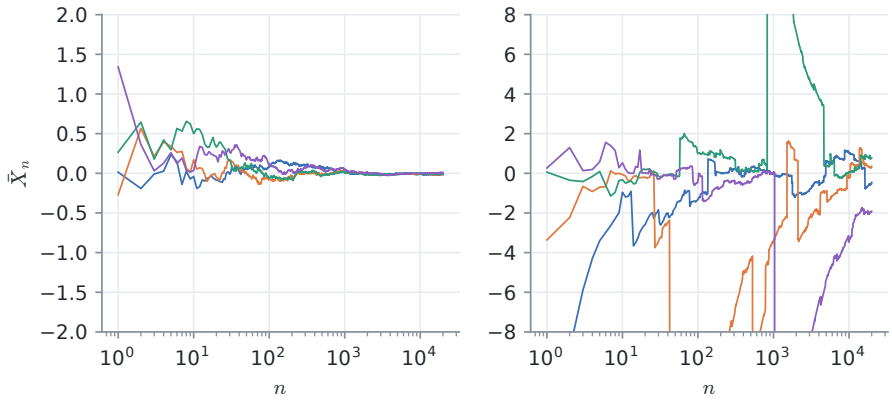


Figure 7.3: Running averages of standard normal (left) and standard Cauchy (right) samples, 4 sequences each. The Cauchy averages keep getting kicked by single huge samples and never settle down.

7.1.6 The Marcinkiewicz-Zygmund Weak Law

In the last section, we saw that if the mean does not exist, then averaging over n is not guaranteed to stabilize the partial sum. Is it still possible to discuss convergence? The answer is yes, if we divide by something bigger than n. Here's the intuition. If $\mathbb{E}|X_1|^r < \infty$, then the convergence equivalence calculation above says that $|X_n| \leq \epsilon\, n^{1/r}$ eventually, for any $\epsilon > 0$ (apply it to X_n/ϵ). So the largest of the first n samples is $o(n^{1/r})$. When $r < 1$, the tails are so heavy that the sum is dominated by its largest few terms (there's no time for cancellations to matter), so S_n should also be $o(n^{1/r})$. This is the **Marcinkiewicz-Zygmund weak law**: if the X_n are i.i.d. with $\mathbb{E}[|X_1|^r] < \infty$ for some $r \in (0, 1)$, then

$$\frac{S_n}{n^{1/r}} \xrightarrow{\mathbb{P}} 0 \tag{7.11}$$

(The same is true for $r \in [1, 2)$ if we also assume $\mathbb{E}[X_1] = 0$, and the proof follows an almost exact procedure as the weak law above, so we will consider the $r \in (0, 1)$ case. For $r = 1$ it *is* the weak law.) We will use convergence in L_r to imply convergence in probability. Let $\epsilon > 0$ and choose $M > 0$ such that

$$\mathbb{E}\left[|X_1|^r\, \mathbb{I}_{\{|X_1|>M\}}\right] < \epsilon \tag{7.12}$$

which is always possible because it's the tail of the convergent expectation $\mathbb{E}[|X_1|^r]$. We will also define 2 new random variables, the small part and the big part

$$Y_k = X_k\, \mathbb{I}_{\{|X_k|\leq M\}}, \qquad Z_k = X_k\, \mathbb{I}_{\{|X_k|>M\}}$$

and this has the consequence that $X_k = Y_k + Z_k$. To continue, we need the c_r **inequality**, which for $r \leq 1$ says $|a + b|^r \leq |a|^r + |b|^r$ (the function x^r is concave, so it's subadditive), and therefore $\mathbb{E}|X + Y|^r \leq \mathbb{E}|X|^r + \mathbb{E}|Y|^r$. (For $r > 1$ there's a factor of 2^{r-1} in front.) Using this on the partial sum, we get

$$\mathbb{E}\left[|S_n|^r\right] = \mathbb{E}\left[\left|\sum_{k=1}^n Y_k + \sum_{k=1}^n Z_k\right|^r\right] \leq \mathbb{E}\left[\left|\sum_{k=1}^n Y_k\right|^r\right] + \sum_{k=1}^n \mathbb{E}\left[|Z_k|^r\right]$$

where the second sum also got split up term by term with the c_r inequality. The small parts are bounded, $|\sum Y_k| \leq nM$, and the big parts are identically distributed, so with (7.12),

$$\mathbb{E}\left[|S_n|^r\right] \leq (nM)^r + n\,\mathbb{E}\left[|Z_1|^r\right] \leq (nM)^r + n\epsilon$$

If we divide both sides by n, then

$$\mathbb{E}\left[\left|\frac{S_n}{n^{1/r}}\right|^r\right] = \frac{\mathbb{E}\left[|S_n|^r\right]}{n} \leq n^{r-1}M^r + \epsilon \tag{7.13}$$

In the limit that $n \to \infty$, since $r \in (0,1)$, the power $r-1$ is negative, so the first term vanishes. What we have is

$$\limsup_{n\to\infty} \mathbb{E}\left[\left|\frac{S_n}{n^{1/r}}\right|^r\right] \leq \epsilon$$

And since ϵ can be made arbitrarily small, we have that $\frac{S_n}{n^{1/r}}$ converges to 0 in L_r, which implies convergence in probability. Notice how the proof mirrors the intuition: the bounded part contributes like n (way less than $n^{1/r}$), and the heavy part is controlled by the tail of $\mathbb{E}|X|^r$.

7.1.7 Application: Empirical Distributions

Here we will present an application of the law of large numbers through **empirical distributions**. Suppose that we have i.i.d. samples $\{X_1, X_2, ..., X_n\}$ and we don't know what their distribution is. This is not unheard of; in real life, when we collect data, we often have no idea what the underlying distribution is. However, we can still form a distribution from the data, and this distribution is called the empirical distribution (because it is a distribution found empirically from the data, duh), defined as

$$F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbb{I}_{\{X_k \leq x\}} \tag{7.14}$$

It's the fraction of the data at or below x, a staircase that jumps by $1/n$ at each data point. Who cares? Well, we are going to make the claim that even though we don't know what the underlying distribution is,

$$F_n(x) \xrightarrow{\mathbb{P}} F(x) \quad \text{for every } x \tag{7.15}$$

and this is a powerful result! What this means is that even though we have no information on the underlying distribution of the X_k, if the sample size is large enough, the empirical distribution function converges to the true distribution. How can we see this? For a fixed x, the indicators $\mathbb{I}_{\{X_k \leq x\}}$ are i.i.d. Bernoulli random variables (functions of independent things are independent), with mean

$$\mathbb{E}[\mathbb{I}_{\{X_1 \leq x\}}] = \mathbb{P}(X_1 \leq x) = F(x)$$

then the weak law tells us that

$$F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbb{I}_{\{X_k \leq x\}} \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbb{I}_{\{X_1 \leq x\}}] = F(x)$$

which is exactly the distribution function. In fact much more is true. The **Glivenko-Cantelli theorem** (sometimes called the fundamental theorem of statistics) says the convergence is uniform in x and almost sure, $\sup_x |F_n(x) - F(x)| \to 0$ a.s., and you can see it happening in figure 7.4. Every statistic that's a nice function of the data (the sample mean, the sample median, sample quantiles) is a function of F_n, and this is why they all converge to their population versions.

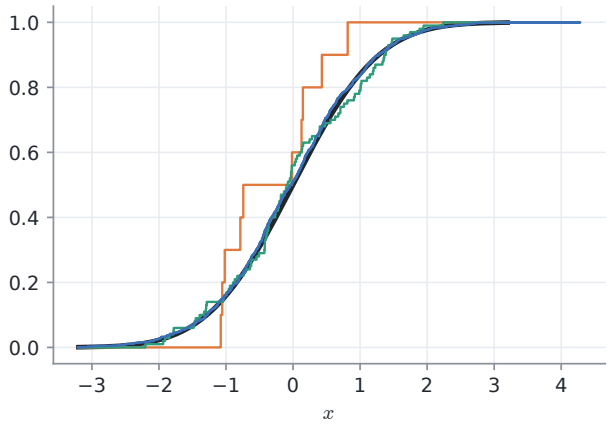


Figure 7.4: Empirical distribution functions of $n = 10$ (orange), 100 (green) and 1000 (blue) standard normal samples, against the true CDF (black).

7.1.8 Application: Monte Carlo

The other application that I can't skip is **Monte Carlo integration**, which is the law of large numbers used as a calculator. Suppose we want an integral $I = \int_0^1 g(x)\,dx$ that we can't do by hand. If U is uniform on $[0, 1]$, then $\mathbb{E}[g(U)] = I$, so with i.i.d. uniforms,

$$\hat{I}_n = \frac{1}{n}\sum_{k=1}^{n} g(U_k) \longrightarrow I$$

The classic toy example is computing π: throw points uniformly at the unit square and count the fraction that land inside the quarter disk, which has area $\pi/4$ (figure 7.5). Equation (7.3) tells us the typical error is $\sigma/\sqrt{n}$, so to get one more digit of accuracy you need 100 times as many samples. That's terrible for computing π. But here's the remarkable part:

the error rate $1/\sqrt{n}$ doesn't care about the dimension of the integral. A grid method in d dimensions needs m^d points for resolution $1/m$, which is hopeless for $d = 100$, while Monte Carlo still converges like $1/\sqrt{n}$. That's why every high dimensional integral in physics (the partition function of a lattice model, a path integral) gets done this way.

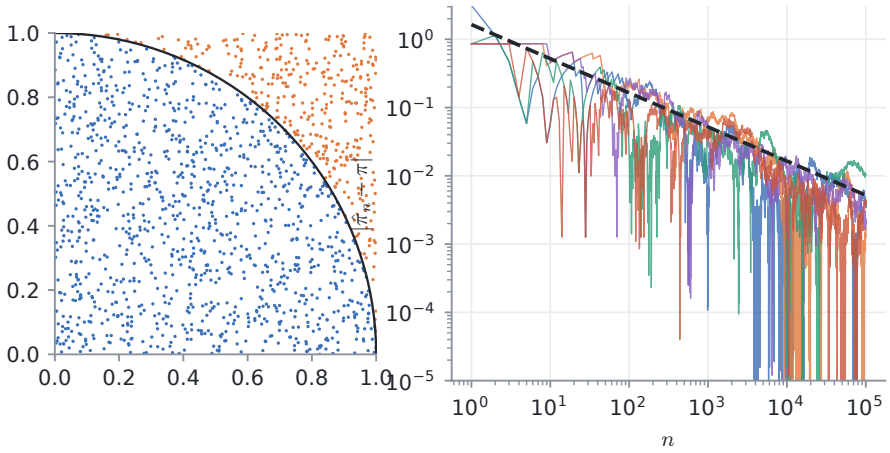


Figure 7.5: Left: estimating π by throwing 1500 uniform points at the unit square; 4 times the fraction inside the quarter circle (blue) is close to π. Right: the error $|\hat{\pi}_n - \pi|$ for 5 independent runs, on log-log axes, which decays like $1/\sqrt{n}$ (dashed).

7.2 The Strong Law Of Large Numbers

Now that we have some exposition on the weak law of large numbers, it would be appropriate now to turn our heads to the beast that is the strong law of large numbers

$$\mathbb{E}|X_1| < \infty \quad \Longrightarrow \quad \bar{X}_n \xrightarrow{\text{a.s.}} \mu \tag{7.16}$$

What does "strong" add? Go back to figure 7.1. The weak law is a statement about vertical slices of that picture: pick a large n, and most of the paths are inside the band at that time. But it doesn't rule out a path that wanders out of the band every once in a while, forever, as long as at any given time only a small fraction of the paths are doing it (like the typewriter sequence of chapter 6). The strong law is a statement about each path: with probability 1, the path you are watching eventually enters the band and never leaves again. If you think about how statistics is actually done (one person, one data set, collecting more and more data), the strong law is the one you want, because it's a statement about *your* sequence of estimates.

7.2.1 The Easy Case: Four Moments

Just like with the weak law, there's a cheap derivation if we're willing to assume more. Remember from chapter 6 that convergence in probability upgrades to almost sure convergence if the probabilities are summable, by Borel-Cantelli. Chebyshev gave us $\mathbb{P}(|\bar{X}_n - \mu| > \epsilon) \leq \sigma^2/(n\epsilon^2)$, and $\sum 1/n$ diverges, so it's not quite enough. But we can use a higher moment. Suppose $\mathbb{E}[X_1^4] < \infty$, and let $Y_k = X_k - \mu$, which have mean zero. Expand the fourth power of the sum,

$$\mathbb{E}\left[\left(\sum_{k=1}^{n} Y_k\right)^4\right] = \sum_{i,j,k,l} \mathbb{E}\left[Y_i Y_j Y_k Y_l\right]$$

By independence, any term where some index appears exactly once has a factor of $\mathbb{E}[Y] = 0$ and dies. The survivors are $i = j = k = l$ (n terms, each $\mathbb{E}[Y^4]$), and two distinct indices each appearing twice ($3n(n-1)$ terms, each $\mathbb{E}[Y^2]^2 = \sigma^4$; the 3 is the number of ways to pair up four slots). So

$$\mathbb{E}\left[(S_n - n\mu)^4\right] = n\,\mathbb{E}[Y^4] + 3n(n-1)\sigma^4 \leq Cn^2$$

for some constant C. Now Markov's inequality with the fourth power gives

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \epsilon\right) = \mathbb{P}\left(|S_n - n\mu|^4 > n^4\epsilon^4\right) \leq \frac{Cn^2}{n^4\epsilon^4} = \frac{C}{\epsilon^4 n^2}$$

which is summable! By the first Borel-Cantelli lemma, $|\bar{X}_n - \mu| > \epsilon$ happens only finitely often, almost surely, and that's almost sure convergence (by the characterization in chapter 6). The whole difference between weak and strong here was getting the probability of a bad time to decay like $1/n^2$ instead of $1/n$, so that the total expected number of bad times is finite. The rest of this section is about getting the same conclusion with only a finite mean, and for that we need some preliminaries.

7.2.2 The Random Kronecker Lemma

The first of these is a result from analysis, the **Kronecker lemma** (or sometimes the deterministic Kronecker lemma). It states that if you have a sequence of real numbers $\{x_n\}$ and an increasing (to infinity) sequence of positive numbers $\{a_n\}$, then

$$\sum_{n=1}^{\infty} \frac{x_n}{a_n} \quad \text{converges} \quad \Longrightarrow \quad \frac{1}{a_n}\sum_{k=1}^{n} x_k \to 0 \tag{7.17}$$

Why is this true? Let $b_n = \sum_{k \leq n} x_k/a_k$ be the partial sums, which converge to some b. Then $x_k = a_k(b_k - b_{k-1})$, and summing by parts (the discrete version of integration by

parts),

$$\frac{1}{a_n}\sum_{k=1}^{n} x_k = \frac{1}{a_n}\sum_{k=1}^{n} a_k(b_k - b_{k-1}) = b_n - \frac{1}{a_n}\sum_{k=1}^{n-1}(a_{k+1} - a_k)b_k$$

The last sum is a weighted average of $b_1, \ldots, b_{n-1}$ (the weights $a_{k+1} - a_k$ are positive and add up to about a_n), and since the $b_k \to b$, so does their weighted average. So the whole thing goes to $b - b = 0$. With $a_n = n$, this says that if $\sum x_n/n$ converges, then the average of the x_n goes to zero. That's exactly the shape of a law of large numbers!

The modification that we are going to make is called the **random Kronecker lemma**, and it states something similar. Suppose we have $\{X_n\}$ a sequence of random variables and $\{a_n\}$ a positive, increasing (to infinity) sequence of real numbers. Then

$$\sum_{k=1}^{\infty} \frac{X_k}{a_k} \text{ converges a.s.} \Rightarrow \frac{1}{a_n}\sum_{k=1}^{n} X_k \xrightarrow[n\to\infty]{\text{a.s}} 0$$

A quick look into the future shows that if we can show $\sum_{k=1}^{\infty} \frac{X_k - \mu}{k}$ converges a.s., then the strong law will follow. To prove the random Kronecker lemma, it's actually really straightforward. Let

$$A = \left\{\omega \in \Omega : \sum_{k=1}^{\infty} \frac{X_k(\omega)}{a_k} \text{ converges}\right\}, \qquad B = \left\{\omega \in \Omega : \frac{1}{a_n}\sum_{k=1}^{n} X_k(\omega) \underset{n\to\infty}{\to} 0\right\}$$

By assumption, we know that $\mathbb{P}(A) = 1$. For each $\omega \in A$, the numbers $X_k(\omega)$ are just a deterministic sequence, so the deterministic Kronecker lemma says $\omega \in B$. That is, $A \subseteq B$. By monotonicity, $\mathbb{P}(A) \leq \mathbb{P}(B)$. But we know that $\mathbb{P}(A) = 1$, and thus $\mathbb{P}(B) = 1$. Almost sure statements are just deterministic statements that hold for almost every ω, and that's the entire trick.

7.2.3 Kolmogorov's Convergence Criterion

So now we need to know when a random series converges almost surely. To put it bluntly, the **Kolmogorov convergence criterion** is just a condition for when an infinite series of independent random variables converges almost surely. To state it, suppose you have $\{X_n\}$ a sequence of independent random variables with finite variances. If the infinite series of variances is finite, then the sum of the centered random variables converges almost surely

$$\sum_{n=1}^{\infty} \text{Var}\left[X_n\right] < \infty \Rightarrow \sum_{n=1}^{\infty} \left(X_n - \mathbb{E}\left[X_n\right]\right) \text{ converges a.s.} \tag{7.18}$$

We also see that as a consequence of the above, if the infinite sum of expectations converges as well, then the series $\sum X_n$ itself converges almost surely. In order to prove the above, we are going to rely on the Kolmogorov inequality from chapter 4

$$\mathbb{P}\left(\max_{1\leq k\leq n}|S_k|\geq t\right)\leq\frac{1}{t^2}\sum_{i=1}^{n}\mathrm{Var}[X_i]$$

for sums of independent mean zero random variables. Without loss of generality, let the X_n have mean zero (otherwise subtract the means), so we want to show that the partial sums S_n converge almost surely. We prove this by showing that they are a Cauchy sequence. If we let $\epsilon>0$ and $n<m$, the increments S_k-S_n for $n<k\leq m$ are themselves partial sums of the independent mean zero variables $X_{n+1},X_{n+2},\ldots$, so the Kolmogorov inequality applies to them

$$\begin{aligned}\mathbb{P}\left(\max_{n<k\leq m}|S_k-S_n|>\varepsilon\right)&=\mathbb{P}\left(\max_{n<k\leq m}\left|\sum_{j=n+1}^{k}X_j\right|>\varepsilon\right)\\&\leq\frac{1}{\varepsilon^2}\sum_{k=n+1}^{m}\mathrm{Var}\left[X_k\right]\leq\frac{1}{\varepsilon^2}\sum_{k=n+1}^{\infty}\mathrm{Var}\left[X_k\right]\end{aligned}$$

If we send $m\to\infty$, the events on the left increase, so by continuity from below,

$$\mathbb{P}\left(\sup_{k>n}|S_k-S_n|>\varepsilon\right)\leq\frac{1}{\varepsilon^2}\sum_{k=n+1}^{\infty}\mathrm{Var}\left[X_k\right]\underset{n\to\infty}{\longrightarrow}0$$

since the right hand side is the tail of a convergent series. This is exactly the situation we had with the random harmonic series in chapter 4, and the same argument finishes it: the maximal oscillation $D=\lim_n\sup_{k,l>n}|S_k-S_l|$ is at most $2\sup_{k>n}|S_k-S_n|$ for every n, so $\mathbb{P}(D>2\epsilon)=0$ for every ϵ, so the partial sums are Cauchy almost surely, so they converge almost surely.

One thing that we will make note of is that if the sequence of random variables is uniformly bounded and independent, then the converse holds (if the centered series converges a.s., the variances are summable). We will not prove this, but will take it as is.

Problem 1.

Let $\{X_n:n\geq1\}$ be a sequence of i.i.d. symmetric coin flips, $\mathbb{P}(X_1=1)=\mathbb{P}(X_1=$

$-1) = 1/2$. What does the Kolmogorov convergence criterion tell us about the series

$$\sum_{n=1}^{\infty} \frac{X_n}{n} = \sum_{n=1}^{\infty} \frac{\pm 1}{n}$$

and what does the Kronecker lemma then say about $\bar{X}_n$?

Solution: Since the random variable is symmetric, it has mean 0, and thus the variance is

$$\text{Var}[X_n] = \mathbb{E}[X_n^2] = 1 \cdot \mathbb{P}(X_n = 1) + 1 \cdot \mathbb{P}(X_n = -1) = 1$$

The variance of the scaled random variable is

$$\text{Var}\left[\frac{X_n}{n}\right] = \frac{1}{n^2}\,\text{Var}[X_n] = \frac{1}{n^2}$$

This tells us that

$$\sum_{n=1}^{\infty} \text{Var}\left[\frac{X_n}{n}\right] = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

and since this converges, the criterion tells us that $\sum_n X_n/n$ converges a.s., which is what we showed by hand in chapter 4. Then the Kronecker lemma with $a_n = n$ says $\frac{1}{n}\sum_{k\leq n} X_k \to 0$ almost surely, which is the strong law for fair coin flips: the fraction of heads converges to 1/2 almost surely.

I think the above example is cute. We know from calculus 2 the conditions for when an alternating series converges. In this case, instead of the sign alternating deterministically (every other), the sign alternates randomly, and we still get convergence!

7.2.4 The Three Series Theorem

The Kolmogorov convergence criterion is pretty dandy, but it relies on the existence of variances, and even more strictly, that the infinite series of the variances converges. Can we say something about the convergence of a random series without assuming anything about variances? Suppose we have a series of independent random variables

$$\sum_{n=1}^{\infty} X_n$$

and obviously we want to know whether or not this converges almost surely. Let's play around and see what we find, using the truncation idea again. Let us define a truncated

random variable

$$Y_k = X_k \mathbb{I}_{\{|X_k| \leq A\}}$$

for some $A > 0$. If Y_k and X_k are convergence equivalent, then $\sum_k Y_k$ converges a.s. exactly when $\sum_k X_k$ does. When are they convergence equivalent? They differ exactly when $|X_k| > A$, so we need

$$\sum_{k=1}^{\infty} \mathbb{P}(X_k \neq Y_k) = \boxed{\sum_{k=1}^{\infty} \mathbb{P}(|X_k| > A) < \infty}$$

This gives us the first of the 3 series, and it tells us that in order for the infinite sum of X_k to converge, large jumps have to stop happening (sometimes called tail control or "no large jumps"). It's also clearly necessary: if the series converges, its terms must go to zero, so $|X_k| > A$ can only happen finitely often, and by the second Borel-Cantelli lemma (the X_k are independent) that forces the sum of probabilities to be finite.

This is great because now, if the tails of X_k die, then it is convergence equivalent to Y_k, and if the series of Y_k converges, then so must the series of X_k. However, this is not all (otherwise the title of this section would just be the 1 series theorem). We have to know when the sum of the Y_k converges. But the Y_k are bounded, so they have variances, and we can use the Kolmogorov convergence criterion

$$\sum_{k=1}^{\infty} \operatorname{Var}[Y_k] < \infty \Rightarrow \sum_{k=1}^{\infty} (Y_k - \mathbb{E}[Y_k]) \text{ converges a.s.} \tag{7.19}$$

If, in addition, the series of the means converges (to a finite number, positive or negative),

$$\sum_{k=1}^{\infty} \mathbb{E}[Y_k] \text{ converges}$$

then we can just add it back into the centered series, since it's a deterministic number, and we get that $\sum_k Y_k$ converges a.s. And from there, we reveal the remaining 2 series in this theorem. We summarize our findings here: let $A > 0$, suppose $\{X_n : n \geq 1\}$ are independent random variables and set $Y_k = X_k \mathbb{I}_{\{|X_k| \leq A\}}$. Then $\sum_{k=1}^{\infty} X_k$ converges a.s. if and only if

(1) $\sum_{k=1}^{\infty} \mathbb{P}(|X_k| > A) < \infty$,
(2) $\sum_{k=1}^{\infty} \mathbb{E}[Y_k]$ converges,
(3) $\sum_{k=1}^{\infty} \operatorname{Var}[Y_k] < \infty$.

This is Kolmogorov's **three series theorem**. We derived the "if" direction, and we saw that the first series is necessary. The "only if" for the other two takes more work (the standard

trick is symmetrization: subtract an independent copy of the whole sequence to kill the means, and then use the converse of the Kolmogorov criterion for bounded variables), and since we won't need it, I'll leave it to the references [4, 2]. The nice thing is that the theorem reduces the convergence of a random series to the convergence of three deterministic series: no big jumps, no drift, and not too much wiggling.

7.2.5 Proving The Strong Law

After all this work, we are finally in the business of proving the strong law of large numbers. Let us state the theorem here. Let $\{X_n\}$ be a sequence of i.i.d. random variables.

(1) If $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$, then $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.
(2) If $\bar{X}_n \xrightarrow{\text{a.s.}} c$ for some $c \in \mathbb{R}$, then $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = c$.
(3) If $\mathbb{E}[|X_1|] = \infty$, then $\limsup_{n\to\infty} |\bar{X}_n| = +\infty$ almost surely.

We see that we have many more conclusions than the weak law of large numbers. Parts 2 and 3 say that the finite mean isn't just a convenient assumption, it's exactly the right one. This will be a long journey, but I hope that it will be worth it. Here's the plan:

- truncate at level n, which by the convergence equivalence calculation changes nothing;
- show the truncated variables have summable variances (after dividing by n);
- use Kolmogorov's criterion and Kronecker's lemma to get the strong law for the truncated variables;
- and finally fix up the means, which the truncation shifted slightly.

Begin by defining $Y_n = X_n \mathbb{I}_{\{|X_n| \le n\}}$. We showed earlier that if $\mathbb{E}|X_1| < \infty$, then Y_n and X_n are convergence equivalent ($\sum_n \mathbb{P}(|X_1| > n) \le \mathbb{E}|X_1|$), so the averages of Y_n and of X_n differ by something that goes to zero almost surely. So it's enough to prove the strong law for the Y_n. The Y_n are independent, but they are not identically distributed anymore, and that's fine. The key estimate is the next problem.

Problem 2.

Prove that if $\mathbb{E}[|X_1|] < \infty$ then $\sum_{n=1}^{\infty} \text{Var}\left[\frac{Y_n}{n}\right] < \infty$.
Solution: We have that

$$\sum_{n=1}^{\infty} \text{Var}\left[\frac{Y_n}{n}\right] \le \sum_{n=1}^{\infty} \frac{1}{n^2}\mathbb{E}\left[Y_n^2\right] = \sum_{n=1}^{\infty} \frac{1}{n^2}\mathbb{E}\left[X_1^2 \mathbb{I}_{\{|X_1| \le n\}}\right]$$

where we used that X_n has the same distribution as X_1. Everything inside is

non-negative, so the monotone convergence theorem lets us move the sum into the expectation

$$= \mathbb{E}\left[X_1^2 \sum_{n=1}^{\infty} \frac{\mathbb{I}_{\{|X_1|\le n\}}}{n^2}\right] = \mathbb{E}\left[X_1^2 \sum_{n\ge|X_1|,\, n\ge 1} \frac{1}{n^2}\right]$$

It's tempting to throw away the indicator and bound the sum by $\pi^2/6$, but that leaves $\mathbb{E}[X_1^2]$, which might be infinite. The indicator is the whole point: it says the sum starts at $n \approx |X_1|$, and the tail of $\sum 1/n^2$ starting at m is about $1/m$. Precisely, comparing with an integral,

$$\sum_{n\ge m} \frac{1}{n^2} \le \frac{1}{m^2} + \int_m^{\infty} \frac{dx}{x^2} = \frac{1}{m^2} + \frac{1}{m} \le \frac{2}{m}$$

For $|x| > 1$, the sum starts at $m = \lceil |x| \rceil \ge |x|$, so it's at most $2/|x|$. For $0 < |x| \le 1$ it starts at $m = 1$ and is $\pi^2/6 \le 2 \le 2/|x|$. Either way, $x^2 \sum_{n\ge|x|} n^{-2} \le x^2 \cdot \frac{2}{|x|} = 2|x|$. Therefore

$$\sum_{n=1}^{\infty} \operatorname{Var}\left[\frac{Y_n}{n}\right] \le 2\,\mathbb{E}|X_1| < \infty$$

The truncation at level n is doing exactly the right thing: it makes the second moments finite, but only just barely, $\mathbb{E}[Y_n^2]$ grows at most like n, which is slow enough to be killed by the $1/n^2$.

Now we can put the pieces together.

- By problem 2 and the Kolmogorov convergence criterion (7.18) applied to the independent random variables Y_k/k,

$$\sum_{k=1}^{\infty} \frac{Y_k - \mathbb{E}[Y_k]}{k} \quad \text{converges a.s.}$$

- By the random Kronecker lemma with $a_n = n$,

$$\frac{1}{n}\sum_{k=1}^{n} (Y_k - \mathbb{E}[Y_k]) \xrightarrow{\text{a.s.}} 0$$

- The means: $\mathbb{E}[Y_k] = \mathbb{E}[X_1 \mathbb{I}_{\{|X_1|\le k\}}] \to \mathbb{E}[X_1] = \mu$ as $k \to \infty$, by dominated convergence (dominated by $|X_1|$). A sequence of numbers that converges to μ has averages that converge to μ (Cesàro), so $\frac{1}{n}\sum_{k\le n} \mathbb{E}[Y_k] \to \mu$. Adding this to the previous line,

$$\frac{1}{n}\sum_{k=1}^{n} Y_k \xrightarrow{\text{a.s.}} \mu$$

- Finally, by convergence equivalence, $X_k = Y_k$ for all but finitely many k, almost surely, so $\frac{1}{n}\sum_{k\le n}(X_k - Y_k) \to 0$ a.s. and
$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu$$

And that's the strong law of large numbers, with nothing more than a finite mean!

7.2.6 Why The Mean Is Necessary

The converse parts 2 and 3 are the reason I like this theorem so much, and they're surprisingly easy with the second Borel-Cantelli lemma. Suppose $\mathbb{E}|X_1| = \infty$, and pick any $A > 0$. Using the right half of (7.6) on $|X_1|/A$,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(|X_n| > An\right) = \sum_{n=1}^{\infty} \mathbb{P}\left(\frac{|X_1|}{A} > n\right) \ge \mathbb{E}\left[\frac{|X_1|}{A}\right] - 1 = \infty$$

The X_n are independent, so by the second Borel-Cantelli lemma, $|X_n| > An$ happens infinitely often, almost surely. In words: with infinite mean, every so often a single sample is bigger than A times the number of samples so far, which is enough to single handedly move the average by A (that's what you see in figure 7.3). To make it precise, write the newest sample in terms of the averages,

$$\frac{X_n}{n} = \bar{X}_n - \frac{n-1}{n}\bar{X}_{n-1}$$

If $|X_n|/n > A$, then at least one of $|\bar{X}_n|$ or $|\bar{X}_{n-1}|$ has to be bigger than $A/2$. This happens infinitely often, so $\limsup|\bar{X}_n| \ge A/2$ almost surely, and since A was arbitrary, $\limsup|\bar{X}_n| = \infty$. That's part 3. For part 2, if $\bar{X}_n \to c$ almost surely, then the right hand side of the display goes to $c - c = 0$, so $X_n/n \to 0$, which means $|X_n| > n$ only finitely often, and by what we just showed that's impossible unless $\mathbb{E}|X_1| < \infty$. Then part 1 says the limit is μ, so $c = \mu$.

This also gives a nice example of the difference between the two laws. We noticed at the end of the weak law by truncation that we only really used $x\,\mathbb{P}(|X_1| > x) \to 0$, and there are symmetric random variables with $\mathbb{P}(|X_1| > x) \approx \frac{1}{x\log x}$, for which $x\,\mathbb{P}(|X_1| > x) \approx 1/\log x \to 0$ but $\mathbb{E}|X_1| = \infty$ (since $\int \frac{dx}{x\log x}$ diverges). For these, the weak law holds, $\bar{X}_n \to 0$ in probability, but by part 3 the strong law fails spectacularly, with $\limsup|\bar{X}_n| = \infty$. At any given time the average is almost certainly near zero, but every path gets kicked arbitrarily far away infinitely often, it just happens at rarer and rarer times.

7.3 What The Law Of Large Numbers Says

We'll end with a few consequences, because the strong law is one of those theorems that quietly sits underneath a lot of things.

7.3.1 The Gambler Revisited

Remember the gambling problem from chapter 3? Each round, your balance gets multiplied by $M = 1.8$ or $M = 0.5$ with equal probability. The expected multiplier is 1.15, so the expected balance after n rounds is $10(1.15)^n$, which goes to infinity. But balances multiply, so take a log: $\log(\text{balance}_n) = \log 10 + \sum_{k=1}^{n} \log M_k$, a sum of i.i.d. terms. By the strong law,

$$\frac{1}{n} \log\left(\text{balance}_n\right) \xrightarrow{\text{a.s.}} \mathbb{E}[\log M] = \frac{1}{2}\left(\log 1.8 + \log 0.5\right) = \frac{1}{2} \log 0.9 \approx -0.053$$

So with probability 1, your balance behaves like $10\, e^{-0.053n}$ and goes to zero! The expectation and the typical behavior point in opposite directions (figure 7.6), and Jensen's inequality from chapter 4 explains why: $\mathbb{E}[\log M] \leq \log \mathbb{E}[M]$. The expectation is dominated by astronomically unlikely winning streaks, and the law of large numbers tells you what actually happens to *you*. This is the idea behind the Kelly criterion in betting and finance: if you are going to play a multiplicative game many times, maximize $\mathbb{E}[\log(\text{wealth})]$, not $\mathbb{E}[\text{wealth}]$. In physics language, the expectation is not self-averaging here, but the log is.

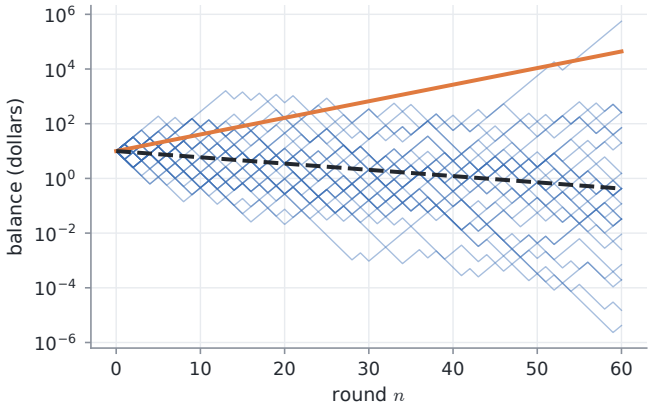


Figure 7.6: 30 players of the 1.8 or 0.5 game, balance on a log scale (thin blue). The expected balance $10(1.15)^n$ (orange) grows exponentially, but almost every player follows the typical rate $10\, e^{n\mathbb{E}[\log M]}$ (dashed) down to zero.

7.3.2 Normal Numbers

Pick a number ω uniformly at random from $[0, 1]$ and write it in binary, $\omega = 0.d_1 d_2 d_3 \ldots$. Under the Lebesgue measure, the digits d_k are i.i.d. fair coin flips (the first digit tells you which half you're in, the second which quarter, and each split is even). The strong law says that the fraction of 1's among the first n digits goes to $1/2$, for almost every ω. Applying it to blocks of digits, every finite pattern of k digits shows up with frequency 2^{-k}. Such numbers are called **normal**, and this is Borel's theorem: almost every number is normal. The funny thing is that it's very hard to prove that any *specific* number (like $\sqrt{2}$ or π) is normal. Almost all of them are, and we can't put our finger on a single one, which is a recurring theme with measure theory.

7.3.3 Entropy And Typical Sequences

Here's one for the physicists. Let $X_1, X_2, \ldots$ be i.i.d. draws from a discrete distribution $p(x)$, and look at the probability of the whole sequence you observed, $p(X_1, \ldots, X_n) = \prod_k p(X_k)$. Taking logs turns it into a sum of i.i.d. terms, so by the strong law

$$-\frac{1}{n}\log p(X_1, \ldots, X_n) = -\frac{1}{n}\sum_{k=1}^{n} \log p(X_k) \xrightarrow{\text{a.s.}} -\mathbb{E}\left[\log p(X_1)\right] = -\sum_x p(x)\log p(x) = H$$

which is the Shannon entropy. This says that, with probability going to 1, the sequence you see has probability about e^{-nH}. All the probability is carried by a "typical set" of roughly e^{nH} sequences, each about equally likely, and everything else is negligible. This is the **asymptotic equipartition property**, and it's the reason that entropy is the compression limit in information theory. It's also Boltzmann's $S = k \log W$ in disguise: the entropy counts (the log of) the number of typical microstates, and in a macroscopic system they're all that ever happens.

7.3.4 How Fast?

The law of large numbers tells us that $\bar{X}_n - \mu \to 0$, but it says nothing about how big the error is for a given n. Chebyshev's inequality gave us a hint: the error is typically of size $\sigma/\sqrt{n}$. What if we zoom in by exactly that factor and look at $\sqrt{n}(\bar{X}_n - \mu)/\sigma$? The law of large numbers says the thing we're zooming in on goes to zero, and the zoom factor goes to infinity, so it's not at all clear what happens to the product. The answer is the subject of the next chapter, and it's one of the most beautiful results in all of mathematics.

7.4 Problems

1. Let $X_1, X_2, \dots$ be random variables with $\mathbb{E}[X_k] = \mu$, $\mathrm{Var}[X_k] \le C$ for all k, and $\mathrm{Cov}(X_j, X_k) = 0$ for $j \neq k$. Show that $\bar{X}_n \to \mu$ in probability. Where exactly did the two line derivation use independence?

2. Let $U_1, U_2, \dots$ be i.i.d. uniform on $[0,1]$. Find the almost sure limit of the geometric mean $(U_1 U_2 \cdots U_n)^{1/n}$. Compare it with $\mathbb{E}[U_1]$.

3. (Computational) Use Monte Carlo to estimate $\int_0^1 e^{-x^2} dx$ with $n = 10, 10^2, \dots, 10^6$ samples. Plot the absolute error against n on a log-log plot and compare with $\sigma/\sqrt{n}$, where $\sigma^2 = \mathrm{Var}[e^{-U^2}]$.

4. Show that if $\mathbb{E}|X_1| < \infty$, then $\bar{X}_n \to \mu$ in L_1 as well. (Hint: show that $\{\bar{X}_n\}$ is uniformly integrable, and use chapter 6.)

5. (Kelly betting) You can bet any fraction $f \in [0,1]$ of your wealth on a coin that lands heads with probability $p > 1/2$. If it lands heads you win the amount you bet, otherwise you lose it. Show that your wealth grows like $e^{n g(f)}$ almost surely, find $g(f)$, and show that the optimal fraction is $f^* = 2p - 1$. What happens if you bet everything every time?

6. Let $X_1, X_2, \dots$ be independent with $\mathbb{P}(X_n = n^\alpha) = \mathbb{P}(X_n = -n^\alpha) = 1/2$. For which α does $\sum_n X_n / n$ converge almost surely? For which α does $\bar{X}_n \to 0$ almost surely?

7. Let X be symmetric with $\mathbb{P}(|X| > x) = \frac{e}{x \log x}$ for $x \ge e$. Show that $\mathbb{E}|X| = \infty$ but $x\,\mathbb{P}(|X| > x) \to 0$. Conclude that i.i.d. copies of X satisfy the weak law with $\mu = 0$, but not the strong law.

8. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_1^2] < \infty$. Show that the sample variance $\frac{1}{n}\sum_{k=1}^n (X_k - \bar{X}_n)^2$ converges almost surely to $\mathrm{Var}[X_1]$.

8

Central Limit Theorems

"Everything is Gaussian if you squint hard enough."

– Hieu

We ended the last chapter with a question. The law of large numbers says that $\bar{X}_n - \mu \to 0$, and Chebyshev's inequality hinted that the error is typically of size $\sigma/\sqrt{n}$. So what happens if we zoom in on the error by exactly that factor? The answer is that, no matter what distribution we started with (as long as it has a finite variance), what we see under the microscope is always the same shape: the Gaussian bell curve. This is the **central limit theorem**, and it's the crowning achievement of all of probability and statistics. It's the reason the normal distribution is everywhere, it's the reason error bars work, and it's the reason that the macroscopic world is so boring compared to the microscopic one.

Like the last chapter, I want to understand where this comes from more than I want to prove it in its most general form. We'll derive it with characteristic functions, then look at it from a couple of other angles (cumulants, and a renormalization group picture that I think is the real reason it's true), then ask how good the approximation is, and then what happens when the assumptions break. Throughout, $X_1, X_2, \ldots$ are i.i.d. with mean μ and variance σ^2, $S_n = \sum_{k \le n} X_k$, and the **standardized sum** is

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}\left(\bar{X}_n - \mu\right)}{\sigma} \tag{8.1}$$

which has mean 0 and variance 1 for every n. The theorem says

$$Z_n \rightsquigarrow N(0,1), \qquad \text{that is,} \qquad \lim_{n\to\infty} \mathbb{P}\left(Z_n \le z\right) = \Phi(z) = \int_{-\infty}^{z} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx \text{ for every } z \tag{8.2}$$

8.1 Zooming In On The Law Of Large Numbers

Why $\sqrt{n}$ and not some other power? The sum S_n has mean $n\mu$ and variance $n\sigma^2$, so it lives at distance about $n\mu$ from the origin, give or take $\sigma\sqrt{n}$. The fluctuations grow, but more slowly than the mean, because independent errors partially cancel. That's the whole content of "variances add". Now think about what different zoom levels show us.

- Divide the centered sum $S_n - n\mu$ by n, and the spread is $\sigma/\sqrt{n} \to 0$. We see a point mass at zero. That's the law of large numbers.
- Divide by $n^{1/2}$ and the spread stays exactly σ. We see a distribution that neither collapses nor blows up. That's the central limit theorem.
- Divide by anything smaller, say $n^{1/4}$, and the spread is $\sigma n^{1/4} \to \infty$. Everything smears out to nothing.

So $\sqrt{n}$ is the only interesting scale. The surprise isn't the scale, it's that the limiting *shape* doesn't depend on what we started with. Figure 8.1 shows it for exponential random variables, which are about as un-Gaussian as a distribution with a finite variance can be: they're one-sided and very skewed. After adding 30 of them, you'd be hard pressed to tell the standardized sum from a normal. If you go back to figure 5.1 in chapter 5, you'll see the same thing happening for uniforms even faster.

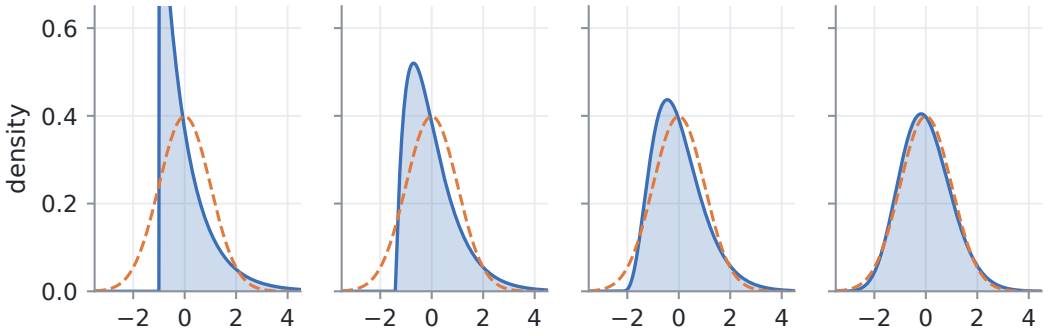


Figure 8.1: The exact density of the standardized sum Z_n of n exponential random variables (blue), against the standard normal density (dashed), for $n = 1, 2, 5, 30$ from left to right.

8.2 The Derivation

The tool is the characteristic function, for the same reason as in the last chapter: sums of independent things become products, and products of many things near 1 become exponentials. By standardizing each term, $Y_k = (X_k - \mu)/\sigma$, we can assume mean 0 and variance 1, so $Z_n = \frac{1}{\sqrt{n}} \sum_{k \le n} Y_k$. Let $\varphi(t) = \mathbb{E}[e^{itY_1}]$. Then by independence and the scaling property,

$$\varphi_{Z_n}(t) = \left[\varphi\left(\frac{t}{\sqrt{n}} \right) \right]^n \tag{8.3}$$

Again we only need φ near $t = 0$, and this time we need it to second order. Since $\mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^2] = 1$, we'd expect

$$\varphi(s) = \mathbb{E}\left[1 + isY - \frac{s^2 Y^2}{2} + \cdots \right] = 1 - \frac{s^2}{2} + o(s^2) \qquad \text{as } s \to 0 \tag{8.4}$$

Then

$$\varphi_{Z_n}(t) = \left[1 - \frac{t^2}{2n} + o\left(\frac{1}{n} \right) \right]^n \underset{n \to \infty}{\longrightarrow} e^{-t^2/2} \tag{8.5}$$

and $e^{-t^2/2}$ is the characteristic function of the standard normal (problem 1 below). It's continuous at zero, so the Lévy continuity theorem from chapter 6 says $Z_n \rightsquigarrow N(0,1)$. That's the central limit theorem!

There's exactly one step that deserves suspicion, which is (8.4), because we only know that the second moment of Y exists, and the "$\cdots$" involves higher powers of Y that might have infinite expectations. Here's how to make it honest. Taylor's theorem with remainder (applied to cos and sin) gives two bounds for the error after the quadratic term,

$$\left| e^{ix} - \left(1 + ix - \frac{x^2}{2} \right) \right| \le \min\left(\frac{|x|^3}{6},\ x^2 \right)$$

The first one is the usual remainder from the third order term. The second one is cruder (the error after the linear term is at most $x^2/2$, and we're adding another $x^2/2$), but it only needs two powers of x. Putting $x = sY$ and taking expectations,

$$\left| \varphi(s) - \left(1 - \frac{s^2}{2} \right) \right| \le s^2\, \mathbb{E}\left[\min\left(\frac{|s||Y|^3}{6},\ Y^2 \right) \right]$$

As $s \to 0$, the thing inside the expectation goes to 0 for every ω (the first argument does), and it's always at most Y^2, which is integrable. So by dominated convergence the expectation goes to 0, which means the error is $o(s^2)$. The finite variance is used exactly once, as the dominating function. Finally, the limit $(1 + c_n/n)^n \to e^c$ whenever $c_n \to c$ (take logs and

use $\log(1+u) = u + O(u^2)$), and that's true for complex c_n too.

Problem 1.

Show that the characteristic function of a standard normal Y is $\varphi(t) = e^{-t^2/2}$.
Solution: We could complete the square in the exponent with a complex shift, but then we'd have to justify moving the contour. Here's a cleaner way. Differentiate under the integral (fine, since $|ixe^{itx}| \le |x|$ is integrable against the Gaussian)

$$\varphi'(t) = \int_{-\infty}^{\infty} ix\, e^{itx} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx$$

Now notice that $xe^{-x^2/2} = -\frac{d}{dx}e^{-x^2/2}$, and integrate by parts (the boundary terms vanish)

$$\varphi'(t) = \int_{-\infty}^{\infty} ie^{itx} \frac{d}{dx}\left(-\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right) dx = \int_{-\infty}^{\infty} i \cdot it\, e^{itx} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx = -t\,\varphi(t)$$

So φ solves the differential equation $\varphi' = -t\varphi$ with $\varphi(0) = 1$, whose only solution is

$$\boxed{\varphi(t) = e^{-t^2/2}}$$

The Gaussian is its own Fourier transform (up to scaling), which is a big part of why it's the answer to so many questions.

Look at how little went into this. We used independence (to turn the sum into a product), identical distribution (so all the factors are the same), and the first two moments (to get the quadratic behavior near $t = 0$). The shape of the original distribution only lives in the higher order terms of φ, and those get crushed in the limit. The next two sections are two ways of seeing exactly how they get crushed.

8.3 Why The Gaussian?

8.3.1 Cumulants

Take the log of the characteristic function. We saw in chapter 5 that the log of a moment generating function is the cumulant generating function $K(t) = \sum_m \kappa_m t^m/m!$, and that cumulants of independent random variables add. So the cumulants of the standardized sum

are easy to write down. Adding n copies multiplies every cumulant by n, and dividing by $\sqrt{n}$ multiplies the m-th cumulant by $n^{-m/2}$ (it's homogeneous of degree m). So

$$\kappa_m(Z_n) = n \cdot \frac{\kappa_m(Y)}{n^{m/2}} = \frac{\kappa_m(Y)}{n^{m/2-1}} \tag{8.6}$$

For $m = 1$, this is $\sqrt{n}\,\kappa_1(Y) = 0$, because we centered. For $m = 2$, it's $\kappa_2(Y) = 1$, independent of n. And for every $m \geq 3$, it goes to zero: the skewness ($m = 3$) dies like $1/\sqrt{n}$, the excess kurtosis ($m = 4$) like $1/n$, and so on (figure 8.2). In the limit, the only cumulant left standing is the variance. And the Gaussian is precisely the distribution whose only nonzero cumulants are the mean and the variance, since $\log e^{-t^2/2} = -t^2/2$ is a quadratic and nothing else. (In fact, there's a theorem of Marcinkiewicz that says if the cumulant generating function is a polynomial, it has degree at most 2, so the Gaussian is the *only* distribution with finitely many nonzero cumulants.) Compare this with the law of large numbers, where we divided by n instead of $\sqrt{n}$ and every cumulant except the mean died.

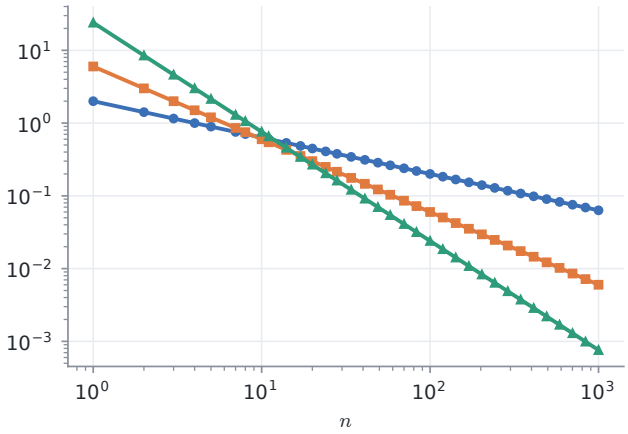


Figure 8.2: Sums of n exponentials: the skewness (blue), excess kurtosis (orange) and fifth cumulant (green) of the standardized sum, on log-log axes. They decay like $n^{-1/2}$, n^{-1} and $n^{-3/2}$, while the variance stays at 1.

8.3.2 A Renormalization Group Picture

Here's the way I actually think about it, and if you've seen statistical field theory, it will look familiar. Consider the following operation on distributions: take two independent copies X and X' of a mean zero random variable, add them, and rescale so the variance is unchanged,

$$X \longmapsto \frac{X + X'}{\sqrt{2}}$$

Doing this k times gives exactly the standardized sum of 2^k copies, so the central limit theorem is the statement that iterating this map drives every distribution (with finite variance) to the Gaussian. The Gaussian is a **fixed point**: if X and X' are independent $N(0,1)$, then $(X+X')/\sqrt{2}$ is $N(0,1)$ again (we showed this by convolution in chapter 5). What happens to a distribution that's *near* the fixed point? Each cumulant transforms on its own, $\kappa_m \mapsto 2 \cdot 2^{-m/2}\kappa_m = 2^{1-m/2}\kappa_m$. In the language of the renormalization group:

- κ_1 gets multiplied by $\sqrt{2}$ each step. It's a *relevant* direction; any mean that we don't subtract off grows and runs away. That's why we center.
- κ_2 gets multiplied by 1. It's *marginal*; it labels a whole line of fixed points, the Gaussians of different variances.
- κ_m for $m \geq 3$ gets multiplied by $2^{1-m/2} < 1$. These are all *irrelevant*, and they shrink to nothing under repeated coarse graining.

So the Gaussian is the universal attractor, and the details of the microscopic distribution (everything beyond the first two cumulants) are irrelevant at large scales. This is *universality*, the same phenomenon that makes the critical exponents of a magnet independent of the details of its lattice. And it immediately suggests a question: are there other fixed points? If κ_2 is infinite, the rescaling by $\sqrt{2}$ is the wrong one, and there's a whole family of other fixed points that we'll meet at the end of the chapter.

8.4 The de Moivre-Laplace Theorem

Historically, the first central limit theorem was for coin flips, and it's worth seeing it directly because it shows where the Gaussian shape comes from without any Fourier analysis. Let S_n be Binomial(n,p) and write $q = 1-p$. We want to look at $\mathbb{P}(S_n = k)$ for k near the mean, $k = np + x\sqrt{npq}$, with x of order 1. The tool is **Stirling's formula**, $\log m! = m\log m - m + \frac{1}{2}\log(2\pi m) + o(1)$. Applying it to all three factorials in $\binom{n}{k}$ and collecting terms (the $-m$'s cancel because $k + (n-k) = n$),

$$\log\left[\binom{n}{k}p^k q^{n-k}\right] = -k\log\frac{k}{np} - (n-k)\log\frac{n-k}{nq} - \frac{1}{2}\log\left(2\pi\frac{k(n-k)}{n}\right) + o(1)$$

The first two terms are $-n$ times a relative entropy (a Kullback-Leibler divergence) between the fraction of heads k/n and p, which is zero when $k = np$ and positive otherwise. Write $k = np + \delta$ and $n - k = nq - \delta$, where $\delta = x\sqrt{npq}$ is small compared to n. Expanding $(np+\delta)\log(1+\frac{\delta}{np}) = \delta + \frac{\delta^2}{2np} + O(\delta^3/n^2)$, and similarly for the other term with $-\delta$, the

linear terms cancel and the quadratic terms add,

$$-k\log\frac{k}{np} - (n-k)\log\frac{n-k}{nq} = -\frac{\delta^2}{2n}\left(\frac{1}{p}+\frac{1}{q}\right) + O\left(\frac{\delta^3}{n^2}\right)$$
$$= -\frac{\delta^2}{2npq} + O\left(\frac{1}{\sqrt{n}}\right) = -\frac{x^2}{2} + O\left(\frac{1}{\sqrt{n}}\right)$$

The last term is $-\frac{1}{2}\log(2\pi npq) + o(1)$, since $k(n-k)/n \approx npq$. Exponentiating,

$$\mathbb{P}(S_n = k) \approx \frac{1}{\sqrt{2\pi npq}} \exp\left[-\frac{(k-np)^2}{2npq}\right] \tag{8.7}$$

which is a normal density with mean np and variance npq, evaluated at k. This is the **local** form of the central limit theorem: not only do probabilities of intervals converge, the individual probabilities follow the bell curve (figure 8.3). And it shows that the Gaussian comes from the fact that the entropy is smooth with a maximum at the mean. A smooth function looks like a parabola near its maximum, and $e^{-\text{parabola}}$ is a Gaussian. It's the same reason that the Boltzmann distribution of macroscopic fluctuations in thermodynamics is Gaussian.

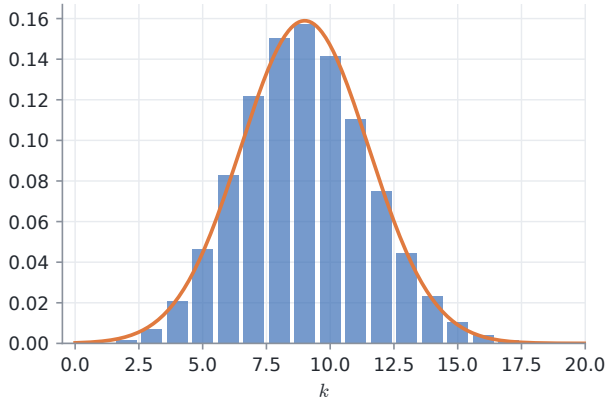


Figure 8.3: The Binomial$(30, 0.3)$ probabilities (bars) against the normal density with the same mean and variance (orange).

Problem 2.

Use the central limit theorem to guess Stirling's formula.

Solution: Here's a cute one. Let X_k be i.i.d. Poisson(1), so S_n is Poisson(n), with mean n and variance n. The probability that the sum lands exactly on its mean is

$$\mathbb{P}(S_n = n) = \frac{e^{-n} n^n}{n!}$$

On the other hand, S_n is integer valued, so $\{S_n = n\} = \{n - \frac{1}{2} < S_n \leq n + \frac{1}{2}\}$, and in standardized units that's an interval of width $1/\sqrt{n}$ around 0. The central limit theorem (in its local form) says the probability of a tiny interval is about its width times the normal density there

$$\mathbb{P}(S_n = n) \approx \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{2\pi}} e^0 = \frac{1}{\sqrt{2\pi n}}$$

Setting the two equal and solving for $n!$,

$$\boxed{n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n}$$

which is Stirling's formula, factor of $\sqrt{2\pi}$ and all. (This isn't a proof, since we need the local version of the theorem to justify the tiny interval, but it's the right answer, and that's not a coincidence.)

8.5 How Good Is The Approximation?

The central limit theorem is a limit statement. It says nothing about how big n has to be. In practice, we always use it at some finite n ("$n \geq 30$" is the rule of thumb they teach you in intro stats), so it would be nice to know the error.

8.5.1 Berry-Esseen

The cleanest answer is the **Berry-Esseen theorem**: if $\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$, then

$$\sup_z |\mathbb{P}(Z_n \leq z) - \Phi(z)| \leq \frac{C\rho}{\sigma^3 \sqrt{n}} \tag{8.8}$$

with a universal constant C (the best known value is below 0.48). So the error is of order $1/\sqrt{n}$, and the constant depends on how lopsided the distribution is, through the third absolute moment. You can see the $1/\sqrt{n}$ in figure 8.4. Notice that fair coin flips, which are perfectly symmetric, still only get $1/\sqrt{n}$. That's because the binomial is a lattice distribution: its CDF is a staircase with steps of height about $1/\sqrt{n}$, and no smooth curve

can be closer than half a step to a staircase. For a symmetric distribution with a density (like the uniform) the error is actually of order $1/n$, which is why the uniform sums in chapter 5 looked Gaussian so fast.

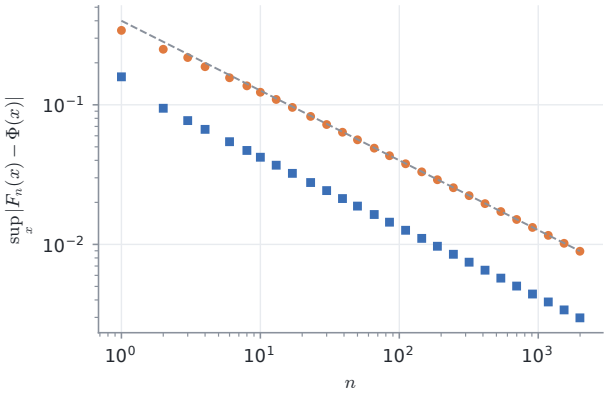


Figure 8.4: The largest distance $\sup_x |F_n(x) - \Phi(x)|$ between the CDF of Z_n and the normal CDF, for sums of fair coin flips (orange) and of exponentials (blue), on log-log axes. Both decay like $1/\sqrt{n}$ (dashed), as Berry-Esseen predicts.

8.5.2 The Skewness Correction

Can we do better than just bounding the error? We can actually compute the leading correction, by keeping one more term in the cumulant expansion. With $\gamma = \kappa_3(Y)$ the skewness of a single term,

$$\log \varphi_{Z_n}(t) = -\frac{t^2}{2} + \frac{\gamma}{6\sqrt{n}}(it)^3 + O\left(\frac{1}{n}\right) \quad \Longrightarrow \quad \varphi_{Z_n}(t) \approx e^{-t^2/2}\left(1 + \frac{\gamma}{6\sqrt{n}}(it)^3\right)$$

Now we invert the Fourier transform. Multiplying the characteristic function by (it) corresponds to differentiating the density and flipping the sign, $-\frac{d}{dz}$ (differentiate the inversion formula (5.4) with respect to z). And the derivatives of the Gaussian density $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$ are polynomials times ϕ, the **Hermite polynomials**: $-\phi' = z\phi$, $\phi'' = (z^2 - 1)\phi$, and $-\phi''' = (z^3 - 3z)\phi$. So the density of Z_n is approximately

$$f_{Z_n}(z) \approx \phi(z)\left[1 + \frac{\gamma}{6\sqrt{n}}\left(z^3 - 3z\right)\right] \tag{8.9}$$

This is the first term of the **Edgeworth expansion**. The correction is odd in z, so it tilts the bell curve: for positive skewness it moves probability from the left shoulder to the right tail and pushes the peak to the left. Figure 8.5 shows how much of the error it captures, already at $n = 5$. Keep going and you get a correction at order $1/n$ involving the

kurtosis and γ^2, and so on, which is the cumulant decay of the previous section turned into a practical tool.

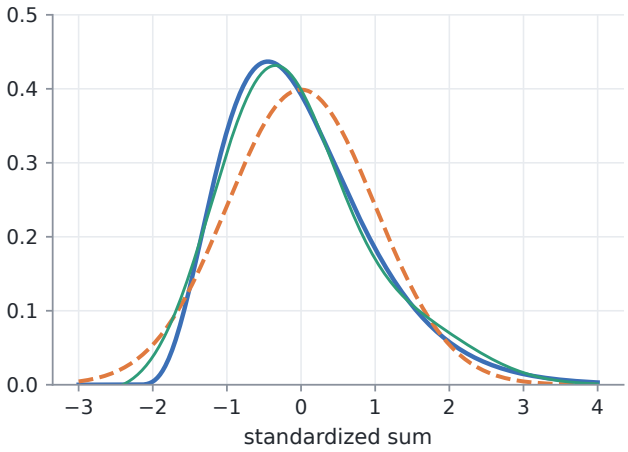


Figure 8.5: The exact density of the standardized sum of 5 exponentials, which has skewness $2/\sqrt{5}$ (blue), the normal approximation (dashed orange), and the normal with the first Edgeworth correction (green).

Problem 3.

You roll 100 fair dice. Estimate the probability that the total is at least 370.

Solution: A single die has mean 3.5 and variance $\frac{1}{6}(1+4+9+16+25+36)-3.5^2 = \frac{35}{12}$. So the total S has mean 350 and variance $100 \cdot \frac{35}{12} \approx 291.7$, a standard deviation of about 17.08. The total is an integer, so "at least 370" is the same as "more than 369.5" (this is the **continuity correction**: when approximating a lattice distribution by a continuous one, split the difference between neighboring values). Standardizing,

$$\mathbb{P}(S \geq 370) = \mathbb{P}\left(Z > \frac{369.5 - 350}{17.08}\right) \approx 1 - \Phi(1.142) \approx 0.127$$

The exact answer (by convolving the die distribution with itself 100 times, which a computer will happily do) is 0.1269. Not bad! Without the continuity correction we'd get $1 - \Phi(1.171) \approx 0.121$, which is noticeably worse. The die is symmetric, so there's no skewness correction, and the error is of order $1/n$.

8.6 Beyond Identical Distributions

In real life the things we add up are rarely identically distributed. The total error in a measurement comes from many different sources, a stock price moves for many different reasons. Does the central limit theorem still hold? Suppose $X_1, X_2, \ldots$ are independent, with means μ_k and variances σ_k^2, and let $s_n^2 = \sum_{k \le n} \sigma_k^2$ be the variance of the sum. The question is whether $(S_n - \sum \mu_k)/s_n \rightsquigarrow N(0,1)$.

It can't always be true. If X_1 is, say, uniform with variance 1 and all the other X_k are zero, then the "sum" is just X_1, which is not normal. More subtly, if the variances grow so fast that the last term always carries most of the total variance, the sum looks like its last term. The derivation needed every factor $\varphi(t/\sqrt{n})$ to be close to 1, so that the log could be expanded, and that requires each individual term to be small compared to the whole. The right condition is the **Lindeberg condition**: for every $\epsilon > 0$,

$$\frac{1}{s_n^2} \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mu_k)^2 \, \mathbb{I}_{\{|X_k - \mu_k| > \epsilon s_n\}}\right] \underset{n\to\infty}{\longrightarrow} 0 \tag{8.10}$$

In words: the fraction of the total variance that comes from terms that are individually a noticeable fraction of the whole spread goes to zero. No single summand gets to matter. The **Lindeberg-Feller theorem** says this is enough for the central limit theorem, and (as long as no single variance dominates s_n^2) it's also necessary. The derivation is the same as before, done term by term: $\log \varphi_{S_n/s_n}(t) = \sum_k \log \varphi_k(t/s_n)$, each term is $-\frac{\sigma_k^2 t^2}{2 s_n^2}$ plus an error, the quadratic parts add up to $-t^2/2$, and the Lindeberg condition is exactly what makes the errors add up to something that goes to zero (it's the dominated convergence step from before, but uniform over the summands). A handy sufficient condition is **Lyapunov's**: $\frac{1}{s_n^3} \sum_k \mathbb{E}|X_k - \mu_k|^3 \to 0$. For instance, if the X_k are uniformly bounded, $|X_k| \le M$, then the Lindeberg condition holds as soon as $s_n \to \infty$, because eventually $\epsilon s_n > 2M$ and every indicator is zero.

Problem 4.

Let $U_1, U_2, \ldots$ be i.i.d. uniform on $[0,1]$, and say that a **record** happens at time k if U_k is bigger than all of $U_1, \ldots, U_{k-1}$. Let R_n be the number of records up to time n. Show that $\frac{R_n - \log n}{\sqrt{\log n}} \rightsquigarrow N(0,1)$.

Solution: Let I_k be the indicator that time k is a record. By symmetry, each of $U_1, \ldots, U_k$ is equally likely to be the largest of the first k, so $\mathbb{P}(I_k = 1) = 1/k$. It's a classic (and surprising) fact that the I_k are independent: whether U_k beats

its predecessors says nothing about the relative order of the predecessors among themselves. So $R_n = \sum_{k \le n} I_k$ is a sum of independent Bernoulli$(1/k)$ variables, which are definitely not identically distributed. The mean and variance are

$$\mathbb{E}[R_n] = \sum_{k=1}^{n} \frac{1}{k} \approx \log n, \qquad s_n^2 = \sum_{k=1}^{n} \frac{1}{k}\left(1 - \frac{1}{k}\right) \approx \log n$$

The summands are bounded by 1 and $s_n \to \infty$ (slowly, but it does), so the Lindeberg condition holds, and

$$\frac{R_n - \sum_{k \le n} 1/k}{s_n} \rightsquigarrow N(0,1)$$

The difference between $\sum 1/k$ and $\log n$ is bounded (it tends to Euler's constant), as is the difference between s_n^2 and $\log n$, so after dividing by $\sqrt{\log n}$ they don't matter (Slutsky), and we get the claim. In a million samples you should expect about $\log 10^6 \approx 14$ records, give or take about 4.

8.7 When The Variance Is Infinite

Every derivation in this chapter used the finite variance in an essential way. What happens without it? We already know the answer for the Cauchy distribution from the last chapter: $\varphi(t) = e^{-|t|}$, so $\varphi(t/n)^n = e^{-|t|}$, and the sum of n Cauchys divided by n (not $\sqrt{n}$!) is exactly Cauchy again. So the Cauchy is a fixed point too, of the map $X \mapsto (X + X')/2$ instead of $(X + X')/\sqrt{2}$.

That's the general pattern. Suppose the tails decay like a power law, $\mathbb{P}(|X| > x) \approx cx^{-\alpha}$ for some $0 < \alpha < 2$, so the variance is infinite (and for $\alpha \le 1$ the mean is too). Then the near-zero behavior of the characteristic function is not quadratic but $\varphi(t) \approx 1 - c'|t|^\alpha$ (heavy tails make φ non-smooth at the origin, which is the Fourier dual of the statement that the moments don't exist). The scale that keeps $n\,(1 - \varphi(t/a_n))$ finite is $a_n = n^{1/\alpha}$, and then

$$\left[\varphi\left(\frac{t}{n^{1/\alpha}}\right)\right]^n \approx \left(1 - \frac{c'|t|^\alpha}{n}\right)^n \longrightarrow e^{-c'|t|^\alpha}$$

(after centering, if $1 < \alpha < 2$). These limits are the **stable distributions**, and this is the **generalized central limit theorem**: the only possible limits of normalized sums of i.i.d. random variables are the stable laws, with the Gaussian ($\alpha = 2$) as the one special case that has a finite variance. You might remember problem 9 of chapter 5, which asked for which α the function $e^{-c|t|^\alpha}$ is a characteristic function. The answer, $0 < \alpha \le 2$, is exactly the family of fixed points. In the renormalization group language, these are the other universality

classes, and which one you flow to is decided entirely by the power in the tail.

The scaling $n^{1/\alpha}$ grows faster than $\sqrt{n}$, because the sum is dominated by its few largest terms instead of by many small cancelling ones. It's the same intuition as the Marcinkiewicz-Zygmund law of the last chapter. Figure 8.6 shows what this looks like for a random walk: with Gaussian steps you get the familiar jittery path, and with Cauchy steps you get long stretches of jitter punctuated by enormous jumps. These are called Lévy flights, and people have argued that foraging animals and financial markets move like this.

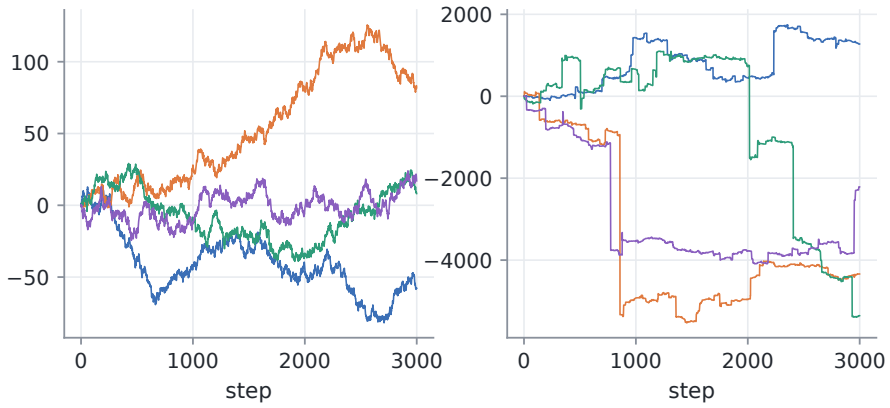


Figure 8.6: Random walks with standard normal steps (left, Brownian motion) and standard Cauchy steps (right, Lévy flights). Notice the scales on the vertical axes: after 3000 steps, the Gaussian walks are of order $\sqrt{3000} \approx 55$, while the Cauchy walks are of order 3000 and dominated by a few huge jumps.

8.8 The Central Limit Theorem In Statistics

I said at the beginning that this is the theorem that makes statistics possible, so let's end with how it gets used.

8.8.1 Confidence Intervals

The law of large numbers says $\bar{X}_n$ is close to μ. The central limit theorem says how close, in a usable way:

$$\mathbb{P}\left(\left|\bar{X}_n - \mu\right| \leq \frac{1.96\,\sigma}{\sqrt{n}}\right) \approx \mathbb{P}\left(|Z| \leq 1.96\right) = 0.95$$

So $\bar{X}_n \pm 1.96\sigma/\sqrt{n}$ is an (approximate) 95% confidence interval for μ, whatever the distribution of the data. Of course we don't know σ either. But the sample standard deviation $\hat{\sigma}_n \to \sigma$ almost surely by the law of large numbers (problem 8 of chapter 7), and Slutsky's

theorem from chapter 6 says that replacing σ by something that converges to it doesn't change the limit: $\sqrt{n}(\bar{X}_n - \mu)/\hat{\sigma}_n \rightsquigarrow N(0,1)$ as well. That's the justification for every error bar you've ever seen. For example, a poll estimating a proportion near $1/2$ has $\sigma \leq 1/2$, so a margin of error of $\pm 3\%$ at 95% confidence needs $1.96 \cdot \frac{0.5}{\sqrt{n}} = 0.03$, or about $n \approx 1068$ people, regardless of whether the country has a million people or a billion. The population size doesn't appear anywhere; only the sample size does.

8.8.2 The Delta Method

Often we care about a function of the mean rather than the mean itself, like $1/\mu$ for a rate, or $\log \mu$. If g is differentiable at μ, then a first order Taylor expansion gives $g(\bar{X}_n) - g(\mu) \approx g'(\mu)(\bar{X}_n - \mu)$, and the error is of smaller order than $\bar{X}_n - \mu$, which is itself $O(1/\sqrt{n})$. So multiplying by $\sqrt{n}$ and using Slutsky to throw away the remainder,

$$\sqrt{n}\left(g(\bar{X}_n) - g(\mu)\right) \rightsquigarrow N\left(0,\, g'(\mu)^2\sigma^2\right) \tag{8.11}$$

This is the **delta method**, and it's the reason that smooth functions of averages are also approximately normal, with the error bars rescaled by the slope. (If $g'(\mu) = 0$, the first order term vanishes and you have to go to second order, where the answer is a χ^2 instead. That's problem 5 below.) In the multivariate version, with the multivariate central limit theorem $\sqrt{n}(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \rightsquigarrow N(0, \Sigma)$ (which follows from the one-dimensional one by applying it to every linear combination $\boldsymbol{t} \cdot \boldsymbol{X}$, the **Cramér-Wold device**), the delta method is how the asymptotic normality of maximum likelihood estimators is derived.

8.8.3 Diffusion

And one for the physicists. A particle in a fluid gets kicked by molecules many times a second, and each kick is a small independent displacement with mean zero and variance ℓ^2. After time t there have been t/τ kicks, so by the central limit theorem the displacement is approximately $N(0, \ell^2 t/\tau)$, with a density

$$p(x,t) = \frac{1}{\sqrt{4\pi D t}} e^{-x^2/4Dt}, \qquad D = \frac{\ell^2}{2\tau}$$

which is the solution of the diffusion equation $\partial_t p = D\partial_x^2 p$ starting from a point. The mean squared displacement grows linearly in time, $\langle x^2 \rangle = 2Dt$, which is the $\sqrt{n}$ scaling in disguise. This is Einstein's 1905 explanation of Brownian motion, and taking the limit of finer and finer kicks properly (keeping track of the whole path, not just the endpoint) is how you build Brownian motion itself, which is where a second course in probability usually starts. So I think that's a good place to stop.

8.9 Problems

1. Let $X_\lambda \sim \text{Poisson}(\lambda)$. Using characteristic functions, show that $(X_\lambda - \lambda)/\sqrt{\lambda} \rightsquigarrow N(0,1)$ as $\lambda \to \infty$. Why is this a central limit theorem even though there's no sum in sight?

2. Let $Q_n \sim \chi^2_n$. Show that $(Q_n - n)/\sqrt{2n} \rightsquigarrow N(0,1)$, and use the cumulants to find the skewness of Q_n. Use this to explain the shapes in figure 5.2.

3. A fair coin is flipped 10,000 times. Use the central limit theorem (with a continuity correction) to approximate the probability of getting between 4,900 and 5,100 heads, inclusive. How many flips would you need for the fraction of heads to be within 0.001 of $1/2$ with probability 0.99?

4. Let $X_1, X_2, \ldots$ be i.i.d. uniform on $[-1, 1]$. Find the exact characteristic function of Z_n, expand $\log \varphi_{Z_n}(t)$ to order t^4, and show that the first correction to the normal approximation is of order $1/n$ rather than $1/\sqrt{n}$.

5. Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance σ^2. Find the limiting distribution of $\sqrt{n}(\bar{X}_n^2 - \mu^2)$ when $\mu \neq 0$. When $\mu = 0$, show that $n\bar{X}_n^2/\sigma^2 \rightsquigarrow \chi^2_1$ instead.

6. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_k = k) = \mathbb{P}(X_k = -k) = 1/2$. Find s_n^2, and check whether the Lindeberg condition holds. Does S_n/s_n converge to a normal?

7. Let $X_1, X_2, \ldots$ be i.i.d. with density $f(x) = |x|^{-3}$ for $|x| \geq 1$. Show that the variance is infinite, but that $S_n/\sqrt{n \log n} \rightsquigarrow N(0,1)$. (Hint: truncate at $\sqrt{n \log n}$ and use the Lindeberg-Feller theorem on the truncated variables. This is the borderline case $\alpha = 2$, and the Gaussian survives with a logarithmic correction to the scaling.)

8. (Computational) Build a Galton board: simulate 10^5 balls each falling through 20 rows of pegs, bouncing left or right with probability $1/2$ at each peg, and plot the histogram of final positions against the de Moivre-Laplace approximation. Then make the board biased (probability 0.8 of bouncing right) and see how the agreement changes with the number of rows.

Bibliography

[1] Patrick Billingsley. *Probability and Measure*. Wiley, anniversary edition, 2012.

[2] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5th edition, 2019.

[3] William Feller. *An Introduction to Probability Theory and Its Applications, Volume II*. Wiley, 2nd edition, 1971.

[4] Allan Gut. *Probability: A Graduate Course*. Springer, 2nd edition, 2013.

[5] Jeffrey S. Rosenthal. *A First Look at Rigorous Probability Theory*. World Scientific, 2nd edition, 2006.

[6] Larry Wasserman. *All of Statistics: A Concise Course in Statistical Inference*. Springer, 2004.